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Energy Performance Optimization of Integrated Passive Envelope for Mid-Rise Office Buildings in Tropical Climate

Dimas Maulana Fasya

Department of Architecture
The Graduate School
Pusan National University

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has been approved by the committee members.

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Chair	Changbae Park	______________
Member	Yujin Nam	______________
Member	Seonghwan Yoon	______________

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List of Abbreviations

ACH	:	Air Change per Hour
AFN	:	Airflow Network
ASR	:	Aerodynamic Short-Circuit Ratio
ASHRAE	:	American Society of Heating, Refrigerating, and Air conditioning Engineers
BAU	:	Business-as-Usual
BCA	:	Building Construction Authority
BES	:	Building Energy Simulation
BMKG	:	*Badan Meteorologi, Klimatologi, dan Geofisika* (Agency for Meteorology, Climatology, and Geophysics)
CFD	:	Computational Fluid Dynamics
CTF	:	Conduction Transfer Function
DA	:	Daylight Autonomy
DOAS	:	Dedicated Outdoor Air System
DSF	:	Double-Skin Façade
EPW	:	EnergyPlus Weather Data
EUI	:	Energy Use Intensity
GBCI	:	Green Building Council Indonesia
GHG	:	Greenhouse Gas
GIZ	:	*Deutsche Gesellschaft für Internationale Zusammenarbeit* (German Society for International Cooperation)
GSA	:	Global Sensitivity Analysis
HVAC	:	Heating, Ventilation, and Air Conditioning
MCDA	:	Multi-Criteria Decision Analysis
MOO	:	Multi-Objective Optimization
OT	:	Operative Temperature
PMV	:	Predicted Mean Vote
PPD	:	Predicted Percentage of Dissatisfied
RAC	:	Refrigeration and Air Conditioning
RUEN	:	*Rencana Umum Energi Nasional* (National Energy General Plan)
sDA	:	Spatial Daylight Autonomy
SHGC	:	Solar Heat Gain Coefficient
SNI	:	*Standar Nasional Indonesia* (Indonesia National Standard)
TARP	:	Thermal Analysis Research Program
THADES	:	Thermal-Hygric Adaptive Decoupled Envelope System
TST	:	Thermal Stack Tower
UDI	:	Useful Daylight Illuminance
UHI	:	Urban Heat Island
VLT	:	Visual Light Transmittance
VT	:	Visible Transmittance
WWR	:	Window-to-Wall Ratio

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Abstract

The rapid urbanization in tropical regions, particularly in Jakarta, necessitates a fundamental paradigm shift from active cooling dependence toward passive architectural defence mechanism. Although passive design is frequently extolled as the pinnacle of sustainable solutions, the thermodynamic feasibility of purely passive strategies in mitigating extreme ambient humidity remains a subject of critical debate. This study investigates the energy performance boundaries of an integrated passive envelope system applied to mid-rise office buildings. The proposed system integrates three distinct technologies: a deep-cavity double-skin façade (DSF) to intercept solar heat, an aerodynamic windcatcher – empirically recharacterized herein as a thermal stack tower (TST) – to induce buoyancy-driven exhaust ventilation, and optimized external louvers to regulate radiant heat gain. Rather than merely quantifying cooling load mitigation, this study establishes the hard thermodynamic boundaries where passive sensible cooling optimization inadvertently triggers latent (dehumidification) energy penalties and visual-thermal trade-offs.

Methodologically, this research employs a rigorous two-phase parametric optimization framework coupled with an airflow network (AFN) diagnostic model. This approach represents an advancement beyond conventional thermal simulations by addressing the intricate relationship between buoyancy-driven ventilation and dynamic moisture (latent heat) intrusion. This study evaluated a range of design scenarios, from optimal to worst-case configurations, with the intention of isolating the impact of geometric variables, such as cavity depth and louver angles, on

sensible and latent heat transfer mechanisms. A novel three-pathway heat gain decomposition further revealed that whilst the louver system (case C2) achieves marginally superior solar interception (18.07 vs. 19.23 kWh/m^2/year), the DSF (case A10) demonstrates decisive superiority through its conductive heat flux pathway (21.25 vs. 32.75 kWh/m^2/year) – resolving the counterintuitive paradox between the psychrometric solar shading mandate and DSF's aggregate cooling superiority. The results obtained establish a thermodynamic limit within the parametric and climatic domain investigated. The optimal configuration (scenario 1), characterized by a 1.5 m cavity depth and high-value glazing thermal properties, successfully intercepted solar radiation. This resulted in a substantial 30.3% reduction in annual sensible cooling energy demand, lowering the cooling EUI from 146.51 kWh/m^2/year (baseline model) to 102.14 kWh/m^2/year. When accounting for the concurrent artificial lighting penalty, the total EUI reached 110.14 kWh/m^2/year – representing a 26.47% reduction from the baseline total EUI of 149.80 kWh/m^2/year. Conversely, the worst-case configuration (scenario 3) paradoxically achieved a marginally lower cooling EUI of 95.25 kWh/m^2/year, highlighting a severe visual-thermal trade-off where apparent cooling conservation is achieved through extreme shading that aggressively penalizes the building's artificial lighting energy demand. Moreover, the optimal cooling efficiency in scenario 1 exposed a critical latent energy paradox; the porous geometry required to facilitate natural ventilation drew significant tropical moisture into the occupied zone, forcing the mechanical system to expend additional energy for dehumidification.

Furthermore, the system encountered a profound "aerodynamic short-circuiting" and "thermodynamic imbalance" phenomenon, where the strong stack effect generated by the heated façade (operating at a massive exhaust rate of -1.50 m^3/s) created a parasitic vacuum that compromised the intended wind-driven ventilation efficacy. While the bottom ventilation in scenario 1 delivered a deceptively active internal circulation driven by DSF cavity recirculation (0.392 ach), the actual fresh outdoor air captured by the TST was severely limited to a mere 0.073 ach — representing a net aerodynamic improvement of only 0.029 ach above the unmodified building baseline of 0.0439 ach. This aerodynamic failure translates to an 89.3% fresh air delivery deficit against the ASHRAE Standard 62.1 minimum requirement, wherein 81.4% of total internal air movement constitutes parasitic recirculation rather than genuine outdoor air exchange. Because the façade aggressively monopolized the aerodynamic flow, the building failed to autonomously utilize outdoor air to flush internal heat and moisture. The findings

of this research provide empirical evidence that in humid equatorial climates, optimization reaches a “hard” thermodynamic boundary where maximizing passive sensible cooling inversely exacerbates hidden energy penalties from artificial lighting and mechanical dehumidification. This study addresses a significant gap in the field of tropical building science by shifting the discourse from aggregate sensible cooling savings to the granular diagnosis of total energy equilibrium. This prompts a reconsideration of the commonly accepted notion that a purely passive design constitutes a universal solution for the tropics. Hence, the findings substantiate a “decoupled hybrid” framework – whereby passive envelopes manage sensible loads while targeted mechanical systems handle latent loads and daytime ventilation – as the sole viable pathway to resolve the conflict between sensible heat rejection and total energy efficiency in tropical climates.

Keywords: Thermodynamic limits, integrated passive envelopes, energy performance optimization, latent energy penalty, visual-thermal thermal comfort.

1. Introduction

1.1. Research Background

Across tropical regions, rapid urbanization is propelling an unsustainable spike in energy consumption strongly correlated with economic growth and environmental degradation (Anser et al., 2022; Zhang et al., 2024). This crisis is predominantly driven by the refrigeration and air conditioning (RAC) sector. The building sector stands as the primary driver to this issue, accounting for approximately 40% of the global energy demand and 30% of global CO_2 emissions (Xing et al., 2011). This escalating energy demand is further fuelled by rapid population growth and the urban heat island (UHI) effect, a phenomenon exacerbating urban environmental temperatures (Santamouris, 2020) and annual temperature increases ranging from 2 to 2.5˚C (Pachauri et al., 2015).

This energy challenge manifests acutely within the Indonesian archipelago. In 2015, the data reported by the Emissions Database for Global Atmospheric Research indicated that the RAC sector alone generated 77.31 Mt CO_2-eq greenhouse gas (GHG) emissions, representing approximately 15.4% of the nation's total GHG output, a figure comparable to regional trends observed throughout Southeast Asia (Janssens-Maenhout et al., 2017). Global warming and surging demand are projected to propel Indonesia's GHG emissions to 218 Mt CO_2-eq by 2050 (GIZ, 2017)[1] as illustrated in Figure 1. Consequently, contemporary building designs must pivot towards integrating passive cooling strategies to achieve zero-energy buildings (Nejat et al., 2015), which have been proven to curtail energy demand, particularly during peak hours (Khani et al., 2017). This approach aligns with national policies, such as the National Energy General Plan (*Rencana Umum Energi Nasional*) (RUEN) (Presidential Regulation No. 22, 2017)[2] and the commitment to reduce carbon footprints by 2060 (Law (UU) of the Republic of Indonesia No. 16, 2016)[3], while

1) For a more comprehensive overview, please refer to the Refrigeration and Air Conditioning Greenhouse Gas Inventory for Indonesia by *Deutsche Gesellschaft für Internationale Zusammenarbeit* (GIZ) GmbH, in cooperation with the Directorate General of New Renewable Energy and Energy Conservation, The Republic of Indonesia.

2) For more comprehensive information, please refer to Presidential Regulation No. 22 of 2017, accessible via the following website: https://climate-laws.org/; Accessed: October 2nd, 2025

3) For a more detailed overview, please refer to the Law (*Undang-Undang*) of the Republic of Indonesia Number 16 of 2016, accessible via the following link: https://peraturan.bpk.go.id/; Accessed: September 29th, 2025

simultaneously adhering to the Paris Agreement's goal of achieving net-zero emissions by 2050 (Murali et al., 2025).

To address these decarbonization imperatives through architectural intervention, a number of passive architectural technologies have been developed, including the double-skin façade (DSF), windcatchers, and louvers. Extant literature has widely validated the functionality of each technology for addressing building performance challenges. For instance, research has revealed that the DSF is particularly effective in mitigating excessive solar heat gain, where optimizing the cavity space depth can significantly influence sensible heat decoupling and buoyancy-driven aerodynamic performance (Aziiz et al., 2018; Yasa, 2015). Similarly, the efficacy of louvers has been thoroughly documented, with research demonstrating their capacity to attain comfortable levels of useful daylight illuminance (UDI) while yielding substantial energy savings of up to 31% compared to conventional buildings (Suryandono et al., 2021). In the realm of ventilation, an experimental analysis confirmed the viability of windcatcher technology in facilitating effective airflow in Indonesia. The study found that a hybrid longitudinal windcatcher could achieve internal wind speeds of up to 0.3 m/s even under low ambient wind speeds of 3 m/s (Satwiko and Tuhari, 2016).

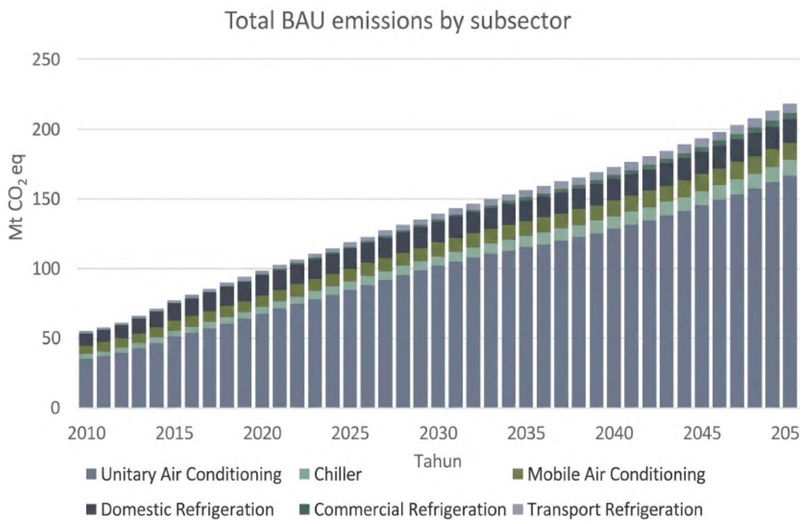


Figure 1. The growth of the RAC sector in Indonesia and its emissions until the year 2050 (GIZ, 2017)

Nevertheless, applying these strategies in the context of Jakarta's hot-humid climate introduces considerable complexity. The effectiveness of these individual passive elements is often compromised by the necessity to manage not only sensible thermal loads but also massive moisture intrusion and artificial lighting demands simultaneously. Persistent high relative humidity (RH) values, which frequently average 80% annually (BMKG, 2024)[4], generate substantial latent cooling loads that are often not sufficiently addressed by simple shading or single pass ventilation strategies. The issue of balancing energy efficiency in naturally ventilated tropical buildings poses a significant challenge, as passive systems that rely on extreme shading and airtightness for sensible heat management inherently trigger severe visual penalties-drastically increasing artificial lighting demand. As demonstrated in the research (Bernal et al., 2024a), windcatchers have been shown to be effective in increasing air exchange rates. However, it should be noted that the inclusion of outdoor air in tropical climates concomitantly introduces substantial latent loads (moisture). Consequently, the mechanical energy required to mitigate this moisture intrusion offsets the savings generated from passive cooling. Accordingly, the design imperative undergoes a transition from the minimisation of the sensible cooling load in isolation to a holistic balance encompassing solar heat rejection, daylight admission, and latent energy penalties. This requirement underscores the urgent need to transition from single-technology studies to an integrated approach capable of managing these competing demands.

Despite extensive research conducted on these individual passive cooling technologies, a significant research gap exists in terms of the integrated performance of combined DSF-windcatcher-louver system, particularly within the hot-humid climate of Jakarta. The majority of extant studies have evaluated single technologies in isolation, neglecting the complex interplay inherent in a holistic systems approach (Khani et al., 2022; Kwong et al., 2014). Moreover, there is a paucity of holistic assessments that comprehensively quantify the critical trade-offs between sensible cooling efficiency, artificial lighting energy penalties, and mechanical dehumidification loads for hot-humid climates. This fragmentation hinders the ability to predict low-energy building performance, particularly in cases where seamless subsystem operation is crucial (Hong et al., 2018). To address this challenge, it is necessary to implement a comprehensive optimization framework that rigorously evaluates

4) *Badan Meteorologi, Klimatologi, dan Geofisika* (Meteorology, Climatology, dan Geophysical Agency). (2024) *Catatan Iklim dan Kualitas Udara Indonesia* (Indonesia's Climate and Air Quality Records). https://iklim.bmkg.go.id; Accessed: November 11th, 2025

these components as an integrated passive envelope system.

Furthermore, the implementation of these passive strategies in tropical climates is constrained by critical thermodynamic considerations regarding latent heat loads. A recent systematic review revealed that while natural ventilation has been identified as a pivotal component for decarbonization, it is increasingly vulnerable to extreme ambient conditions, necessitating a trade-off between sensible heat rejection and latent energy penalties (Hitchin, 2021). This challenge has been further compounded by the integration of these systems. Quantitative analyses suggested that the integration of shading within ventilated cavities introduces complex performance boundaries, where maximizing sensible cooling through physical obstruction often disrupt fluid dynamic behaviour and triggers severe secondary energy demands (Jankovic and Goia, 2021). Significantly, the integration of windcatchers with auxiliary cooling remains a pivotal challenge in non-arid regions (Heidari et al., 2024). Consequently, a substantial knowledge gap persists regarding the precise thermodynamic limits where passive sensible cooling efficiency is compromised by massive artificial lighting demands and mechanical dehumidification loads.

This study addresses these deficiencies by developing a performance-based optimization framework for an integrated system combining a DSF, a windcatcher, and louvers. Departing from single-component assessments, this research employs a robust two-phase simulation workflow coupled with multi-criteria decision analysis (MCDA), to navigate the complex trade-offs between conflicting performance objectives. The objective of this investigation is to quantify the synergistic effects on total energy use intensity (EUI), with a particular focus on mapping the visual-thermal trade-offs and the latent energy penalties associated with tropical moisture intrusion. Furthermore, a parametric sensitivity analysis is integrated to identify the most influential geometric variables dictating the building's energy behaviour, thereby preventing over-engineering and ensuring robustness against climatic uncertainties. Rather than merely seeking a theoretical optimum, this research intends to rigorously define the exact "tipping point" or thermodynamic equilibrium of passive systems in tropical climates. This study specifically quantifies the threshold where extreme shading and ventilation fail, necessitating the transition from purely passive designs to active mechanical dehumidification. This approach provides a validated pathway for decarbonizing the tropical building sector by achieving true, holistic energy optimization.

1.2. Research Purposes

The fundamental objective of this research is to rigorously delineate the thermodynamic performance boundaries of integrated passive envelope systems within high-enthalpy tropical climates, specifically in the context of Jakarta. Contemporary architecture discourse has increasingly advocated for passive design as a primary decarbonization strategy to mitigate the UHI effect (Long et al., 2025; Putra et al., 2021, 2022). However, existing implementations often fail to address the critical dichotomy between sensible cooling efficiency and hidden secondary penalties. In tropical regions where ambient humidity frequently exceeds 80%, the prevailing design paradigms, which predominantly focus on minimizing solar heat gain, often inadvertently compromise daylight autonomy (DA) (Bahdad et al., 2024; Lim and Heng, 2016) and trigger massive mechanical dehumidification (Ding et al., 2022). Consequently, this study aims to move beyond single-component analysis by developing a holistic optimization framework that integrates a DSF, aerodynamic windcatchers, and shading louvers. This study intends to establish a quantitative method for determining the precise tipping point at which extreme passive interventions become counterproductive to total energy reduction, thereby establishing the ultimate limit for purely passive operations. Specifically, this study addresses the following central research question: To what thermodynamic extent can an integrated passive envelope – comprising a deep-cavity DSF, aerodynamic windcatcher (Thermal Stack Tower (TST) recharacterization), and permeable louver system – mitigate sensible cooling loads in Jakarta's equatorial climate before reaching aerodynamic and hygric performance boundaries.

To answer this question and achieve the overarching objective, the research employs a two-phase parametric optimization method coupled with airflow network (AFN) diagnostics. This methodological choice is critical; conventional thermal simulations often assume fixed air change rates, thereby masking the complex interplay between buoyancy-driven ventilation and hydrostatic pressure equilibrium. The utilization of AFN in this study aims to elucidate the "choked flow" phenomenon, which occurs when the aerodynamic conflict between the façade's stack effect and the roof's wind pressure creates a parasitic vacuum, paralyzing the building's natural ventilation capacity. This approach facilitates a detailed evaluation of the influence of geometric variations in the building envelope on the trade-off between EUI, artificial lighting demands (visual penalty), and mechanical dehumidification requirements (latent penalty).

To ensure the robustness of the optimization process, this research isolates and permutes key geometric variables that govern heat and mass transfer. Through a rigorous parametric sensitivity analysis, the geometric parameters employed in this study serve to define which architectural elements dictate the most significant shifts in total energy consumption. The following table (Table 1) delineates the scope of the parametric variables investigated. The purpose of selecting these specific ranges is to simulate a comprehensive spectrum of design scenarios, ranging from the “Baseline” (business-as-usual (BAU)) to the “Optimal” configuration, and contrasting them with “Worst-Case” scenarios to validate the energy sensitivity of the system and prove the necessity of a balanced optical-thermal equilibrium.

Table 1. Range of geometric parameters for passive optimization

Component	Design Parameter	Min. Value	Max. Value	Unit	Physical Objective
DSF	Cavity Depth	0.25	1.5	m	Enhance stack effect
	Glazing (SHGC)	0.174	0.219	-	Limit solar transmittance
	Glazing (U-Value)	0.345	2.8618	W/m^{2-K}	Minimize conductive heat
Windcatcher	Tower Shape	Short-rectangular	Long-rectangular	-	Maximize inlet pressure
	Tower Height	2	9	m	Increase driving force
Louver	Louver Gap	30	270	mm	Prevent airflow choking
	Width of Slats	30	340	mm	Define shading cutoff
	Louver Angle	30	60	^{o} (degree)	Block direct radiation
	Distance from the Window	30	300	mm	Enable convective flushing

Note: This table summarizes the primary decision variables used in the parametric evolutionary algorithm to determine the optimal trade-off solutions.

By systematically analysing the permutations of these variables, this research aims to construct a pareto-front that visualises the conflicting relationship between energy conservation and physiological safety. Preliminary hypotheses suggest that while optimizing geometric configuration, particularly cavity depth and louver angles can drastically reduce the sensible cooling load (lowering EUI), it may simultaneously elevate the artificial lighting demand due to extreme shading and explode the mechanical dehumidification load owing to tropical moisture intrusion, a limitation that has been increasingly questioned in recent tropical studies (Verma et al., 2023). Therefore, a core purpose of this analysis is to identify the specific design configurations where this optical-thermal trade-off reaches its limit, and where the "sensible-latent decoupling" becomes necessary to prevent total energy spikes.

Moreover, this study intends to challenge the "passive-only" idealism that is often prescribed for the tropics. By benchmarking performance against rigorous international energy standards (ASHRAE Standard 55 and 62.1, and ISO 7730) while adhering to regional tropical building codes (Green Mark, Green Building Council Indonesia (GBCI), and *Standar Nasional Indonesia* (Indonesian National Standard) (SNI)), the research aims to demonstrate that meeting stringent low-energy building targets in Jakarta's climate requires a paradigm shift from purely passive systems to "hybrid decoupled" frameworks. The analysis will quantify the exact aerodynamic shortfall (in ACH) inherent in optimized passive designs, providing empirical evidence that justifies the integration of auxiliary mechanical dehumidification to handle the massive latent loads that passive envelopes physically cannot resolve. Ultimately, the results of this research are intended to formulate a validated design guideline for mid-rise office buildings, offering architects a strategic roadmap to balance the urgent need for decarbonization with the non-negotiable requirement for holistic energy equilibrium, ensuring that passive sensible cooling does not inadvertently backfire into massive latent and visual energy penalties.

1.3. Research Hypothesis

Synthesizing the theoretical constraints of high-enthalpy thermodynamics with the mechanics of buoyancy-driven ventilation, the following hypotheses are formulated in this study to interrogate the energy performance boundaries of the integrated passive envelope system:

1. Non-Linear Synergistic Efficiency

The integrated application of a double-skin façade (DSF), aerodynamic windcatcher, and optimized louvers is hypothesized to yield a non-linear reduction in EUI that exceeds the aggregate efficiency of individual components. However, this synergy is thermodynamically bounded by diminishing returns where extreme shading and aerodynamic resistance are introduced to the building envelope.

2. The Visual-Thermal Inverse Correlation

Optimization targeting minimal sensible cooling loads will demonstrate a statistically significant inverse correlation with DA. Specifically, the aggressive solar shading required for maximum thermal buffering is predicted to trigger a severe visual penalty, proportionally exploding the building's artificial lighting energy demand.

3. Ther Latent Energy Penalty in Passive Ventilation

The passive envelope is predicted to successfully suppress sensible heat gains. However, it is posited that maximizing natural ventilation inevitably introduces massive unmanaged latent heat loads (tropical moisture) into the occupied zone. This forces the mechanical system to expend secondary energy for active dehumidification, a penalty that mathematically offsets the initial sensible cooling savings.

4. Aerodynamic Short-Circuiting of Airflow Drivers

Geometric configurations of the envelope, specifically the cavity depth and the slat angles of the louvers, are postulated to function as the principal determinants of aerodynamic pressure differentials by modulating the thermal gradient within the cavity space. It is asserted that intense solar heating of the façade generates a localized stack effects so powerful that it creates a parasitic vacuum. This internal suction overpowers the roof-level windcatcher, precipitating as aerodynamic "choked flow" phenomenon where the building fails to autonomously harvest fresh outdoor air.

5. The Hard Thermodynamic Limit

Within the conditions studied, a thermodynamic boundary exists where purely passive strategies become insufficient to balance the conflicting demands of sensible heat rejection, daylight admission, and moisture control, thereby necessitating a transition to a "hybrid decoupled" framework where sensible loads are managed passively while latent loads require targeted auxiliary mechanical intervention to achieve true, holistic energy optimization.

1.4. Research Contributions

This study transcends the conventional optimization of singular building components by systematically isolating the "hard" thermodynamic boundaries inherent to passive cooling in high-humidity climates. The resulting findings offer critical implications that extend beyond theoretical discourse, providing tangible frameworks for scientific advancement, regulatory formulation, and architectural practice.

1. Advancement in Tropical Building Science

This research addresses a significant gap in the field by shifting the academic discourse from aggregate sensible cooling energy savings to the granular diagnosis of "latent and visual barriers," thereby challenging the prevailing orthodoxy that purely passive design is a universal solution for tropical regions. This study provides a comprehensive overview of the critical thermodynamic rationale underpinning the transition to a "hybrid decoupled" framework. It further delineates the precise aerodynamic thresholds where passive strategies trigger severe secondary energy demands. Furthermore, it quantifies the boundary condition at which sensible cooling efficiency begins to inversely correlate with artificial lighting needs and mechanical dehumidification loads, thus providing a roadmap for future mixed-mode simulation studies.

2. Evidence-Based Policy Formulation for Jakarta

This study provides a robust empirical basis for recalibrating regional building codes, specifically SNI and GBCI standards, to prioritize strict mechanical dehumidification baselines alongside natural ventilation strategies. By measuring the latent energy spikes linked to unmanaged humidity in passively ventilated envelopes, this study supports the needs for enforcing decoupled-system readiness in mid-rise office developments. This contribution directly facilitates the alignment of technical building standards with *Rencana Umum Energi Nasional* (National Energy General Plan) (RUEN) and 2060 Net-Zero targets, thereby ensuring that aggressive decarbonization mandates do not inadvertently backfire into massive operational energy deficits.

3. Decision-Support Framework for Architectural Practice

For practitioners, this study presents a replicable optimization workflow that leverages MCDA to navigate the complex trade-offs between façade geometry, sensible cooling performance, and DA. Rather than being dependent on intuition, it provides architects with a quantitative basis on which to justify the integration of complex assemblies, such as deep-cavity DSFs and aerodynamic windcatchers, against economic and spatial constraints. This research identifies a matrix of “pareto-optimal” design configurations that have been rigorously stress-tested to withstand Jakarta’s climatic extremities and prevent aerodynamic short-circuiting. This, in turn, serves to reduce uncertainty in the decision-making process during the early stages of design.

4. Methodological Innovations in Building Diagnostics

This study introduces two novel diagnostic frameworks that represent a significant advance in the methodological frontier of tropical building performance research. Firstly, the aerodynamic short-circuit ratio is established as an AFN-derived quantitative metric – defined as the proportion of total internal building air circulation attributable to recirculated rather than genuine outdoor air – providing a precise diagnostic instrument for fresh air delivery failure in integrated passive envelope systems, a metric not previously applied in tropical DSF research. Secondly, this study presents a three-pathway heat gain decomposition framework that disaggregates the annual sensible cooling load into solar radiative, conductive, and internal sensible constituents – validated through arithmetic closure against EnergyPlus zone-level outputs – proving the replicable analytical instrument for resolving counterintuitive performance paradoxes in integrated passive envelope systems operating under high-enthalpy equatorial conditions.

1.5. Research Roadmap

To navigate the complex interdependence between thermodynamic constraints and architectural optimization, this study operationalizes a structured, four-stage investigative framework derived from a two-phase simulation workflow. The trajectory is initiated with the calibration of the baseline energy model against Jakarta’s climatic boundary conditions, thereby establishing the validity of the simulation. This is followed by a component-level parametric exploration, where the envelope elements, a DSF, windcatchers, and louvers, are optimized independently. In contradistinction to linear assessments, this roadmap employs sensitivity analysis as a filtration mechanism to isolate geometric variables that most significantly dictate the building's total energy behaviour, prior to synthesising them into holistic scenarios.

The subsequent phase subjects these integrated scenarios to a MCDA filter, with the aim of identifying pareto-optimal configurations that balance conflicting energy objectives – namely, maximizing sensible cooling efficiency while mitigating artificial lighting demands. This analytical progression culminates in a mass-balance diagnostic phase, where AFN analytics interrogate the system's limits against massive latent load intrusion and aerodynamic stagnation. This final stage is critical for delineating the "choked flow" phenomenon and defining the hard thermodynamic boundaries where passive operations fail to minimize the total EUI. Consequently, this systematic transition from isolated component sensitivity to an integrated multi-objective diagnostic ensures that the subsequent methodology directly addresses the conflicting thermodynamic nature of tropical passive design. Table 2 outlines the strategic milestones and anticipated outputs for each investigative phase.

Table 2. Strategic research roadmap and milestones

Phase	Research Stage	Key Activity and Methodological Focus	Strategic Output / Milestone
I	**Diagnostic and Validation**	Calibration of EnergyPlus / AFN models against Jakarta's EPW data and ASHRAE-compliant geometry calibrated to local operational profiles	Validated BAU model and identification of baseline energy and aerodynamic performance gaps
II	**Component-Level Optimization**	Independent parametric permutation and sensitivity analysis (Range (R) index) to isolate high-leverage geometric design variables	Identification of dominant total energy performance drivers and optimal configuration for each subsystem
III	**Holistic Integration**	Synthesis of components	Selection of the "global

	and MCDA	into consolidates scenarios (optimal, moderate, and worst-case) and MCDA filtering	optimum" scenario and quantification of synergistic trade-offs (sensible cooling vs. visual and latent energy penalties)
IV	**Thermodynamic Diagnostic**	AFN-based forensic analysis of aerodynamic stagnation and psychrometric overlay to detect massive moisture intrusion	Empirical definitions of thermodynamic limits and justification for the "hybrid decoupled" transition to achieve total energy equilibrium

1.6. Structure of the Thesis

The articulation of this research is systematically structured into five cohesive chapters, designed to guide the analytical trajectory from the theoretical identification of energy barriers to the empirical definition of the thermodynamic limits in tropical passive envelope design. The structure of the thesis is as follows:

1. Introduction

The research context is established by foregrounding the conflict between regional decarbonization mandates and the severe aerodynamic and moisture constraints inherent to the equatorial climate of Jakarta. It delineates the research gap, articulates the "thermodynamic limit" hypothesis and outlines the strategic roadmap for investigating integrated passive envelope systems to achieve total energy equilibrium.

2. Critical Literature Review

A critical examination of the extant corpus of knowledge pertaining to passive cooling technologies – specifically DSF, windcatchers, and louvers system – in hot-humid climates. Through systematic deconstruction of prior studies this chapter exposes the scarcity of holistic assessments addressing secondary energy penalties and the "sensible latent" decoupling problem, thereby establishing the empirical and methodological justification for the multi-criteria diagnostic framework adopted in this research.

3. Parametric Thermodynamic Assessment of Individual Passive Strategies

This chapter delineates the integrated simulation architecture, detailing the coupling of EnergyPlus with AFN diagnostic, the calibration of the baseline building model against Jakarta-Kemayoran station climate data, and the psychrometric characterization of seasonal thermodynamic operating condition. It further specifies the parametric boundaries, sensitivity analysis protocols, and MCDA weighting criteria governing the phase 1 component-level assessment, the empirical results of which are presented as the concluding section of this chapter.

4. Integrated Passive Envelope Simulation

This section details the integrated phase 2 performance assessment, in which the component-level optima identified in chapter 3 are systematically combined into three scenarios – optimal, moderate, and worst-case – to evaluate the coupled aerodynamic and hygric synergies of the integrated passive system. This chapter builds upon the diagnostic findings of the integrated assessment to establish the thermodynamic boundaries of the passive envelope within the studied tropical archetype. It then goes on to quantify the aerodynamic short-circuiting phenomenon and latent energy mandate. Collectively, these define the hard limits of purely passive design in Jakarta's equatorial climate.

5. Conclusion

In this chapter, the findings from the empirical research are consolidated into a structured set of conclusions and actionable recommendations. This chapter commences with a synthesis of the key thermodynamic outcomes derived from both simulation phases. This is followed by the formulation of a validated hybrid decoupled design framework that positions the passive envelope as a dedicated sensible thermal buffer while delegating latent load management and aerodynamic fresh air delivery to targeted mechanical supplementation. Subsequent to this, evidenced-based recommendations are advanced for the revision of regional building codes – specifically the SNI and GBCI GREENSHIP frameworks – to enforce performance-verified aerodynamic delivery standards and active dehumidification baselines as preconditions for natural ventilation credit. Conclusively, it acknowledges the limitations of the current simulation framework and identifies potential future research directions, including transient occupancy profiling and life-cycle cost analysis of the proposed hybrid decoupled approach

1.7. The Typological Focus: Mid-Rise Office Buildings in Jakarta

The strategic emphasis on mid-rise office buildings is predicated on converging architectural, economic, and thermodynamic determinants. Firstly, the architectural typology's inherent compatibility with passive strategies contributes to its prominence in urban development. A mid-rise structure is generally defined as a building between four and ten stories (Mid Rise Building Guidelines, 2019), and this definition is universally accepted in various urban design guidelines that emphasize sustainable density and low-carbon operations as a core principle (Toronto Mid-Rise Building Design Guidelines, 2024). This typology is of particular importance in achieving sustainable density in urban cores. Office buildings, irrespective of their height, are thermodynamically conducive to advanced energy analysis, as their environmental impact is predominantly influenced by the operational (use) phase (Esteghamati et al., 2022), underscoring the imperative for early design to prioritize operational efficiency.

As the nation's capital and a major business centre, Jakarta possesses a significant and complex office stock. As of the first half of 2024, the office market was marked by significant supply and a highly competitive environment (Savills Indonesia Research & Consultancy, 2024; Colliers, 2024) (see Figure 2). This competitive environment prompts tenants to prioritize high-quality space and sustainability initiatives to maintain market competitiveness (Knight Frank, 2024). Landlords are progressively acknowledging the imperative to incorporate environmental, social, and governance practices and green building concepts to appeal to and retain tenants (Colliers, 2024). This transition has led to a discernible market trend in which stringent energy efficiency and reduced operational expenditures derived from high-performance facades have become non-negotiable performance metrics for new mid-rise office developments (Jakarta CBD office market overview, 2023).

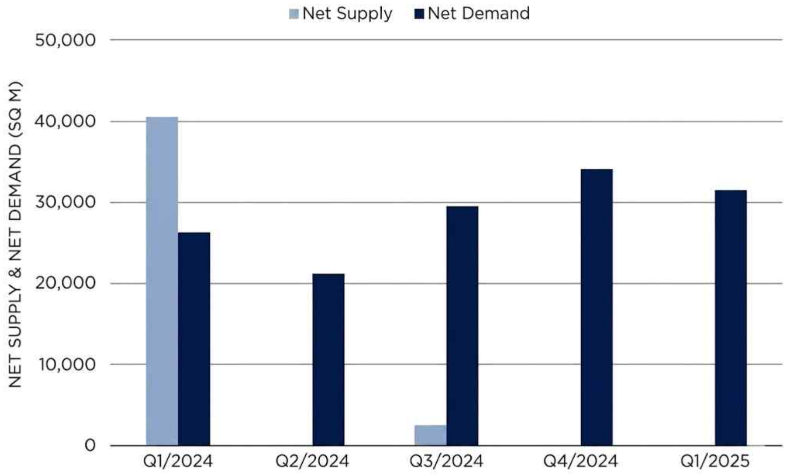


Figure 2. The pronounced supply-demand disparity in Jakarta's office market, where resilient net demand significantly outstrips the insufficient net supply (Savills Indonesia Research & Consultancy, 2024)

The preponderance of energy consumption in the building sector for the hot-humid climate of Jakarta is predominantly driven by the cooling load. Research focusing on office buildings in South Jakarta has demonstrated that the air conditioning system is responsible for over 53% of the total annual energy consumption, thereby establishing the building envelope as the primary thermodynamic boundary for initial energy intervention (Farizal et al., 2024). This assertion is further substantiated by simulation studies, which demonstrated that the reference energy use intensity (EUI) for typical Jakarta Office buildings falls within the range of 200 to 226 kWh/m^2/year. These findings strongly emphasize the necessity for robust design strategies (Fajar and Tokimatsu, 2024). Additionally, while the implementation of nearly zero energy building goals mandates energy efficiency as a conceptual cornerstone for sustainability (Hafez et al., 2023), research indicated that the most economically feasible path to achieving deep energy savings avoids expensive active measures like massive solar panel installation. Instead, it prioritizes load-reduction efficiency gains from optimized passive façade designs and targeted mechanical systems (Farizal et al., 2024). This finding necessitates an architectural approach that prioritizes maximal thermal performance through passive integration rather than relying solely on costly renewable energy offsets.

Consequently, the mid-rise typology, which is often constructed more efficiently at a relatively lower cost than high-rises (Podesto, 2015), represents the optimal thermodynamic scale for implementing the integrated passive envelope system proposed in this study. The adopted six-story mid-rise archetype, which complies with the ASHRAE Standard 90.1 (2019) prototype for medium offices, ensures that the research findings are directly applicable to the most relevant and dominant segment of the tropical urban building sector in Jakarta. This focus enables a precise assessment of the synergistic benefits derived from the integrated DSF and windcatchers. These aerodynamic components function most effectively at mid-rise heights, where buoyancy-driven pressure differentials are sufficient to induce vertical flow without being overwhelmed by the extreme wind shear forces and turbulence typical of high-rise structures. Simultaneously, this scale allows for the integration of deep external louvers without necessitating the heavy structural reinforcement required in taller typologies to withstand high-velocity wind loads. This aerodynamic and structural distinction clearly differentiates the feasibility of this study from research focused solely on high-rise structures.

2. Critical Literature Review

2.1. Performance of DSF in Tropical Climate

A DSF is a building envelope system consisting of two layers separated by a ventilated cavity space that serves as a thermal buffer at a specific depth (Al-awag and Wahab, 2022). The system performance is primarily determined by several key parameters, including the cavity geometry, ventilation mode (natural, forced, or hybrid), thermal and optical properties of the materials, and ventilation operating strategy (Wong et al., 2008). This operational versatility is empirically evident in recent studies verifying that DSF configurations can be engineered for opposing climatic objectives, ranging from passive preheating in cold regions (Hou et al., 2021) to heat dissipation in warm climates. Early simulation studies in cooling-dominated climates have established that specific geometric configurations, particularly cavity width and exterior opening dimensions, are decisive factors in minimizing annual cooling loads (Torres et al., 2007). This thermal barrier mechanism is the foundation of the system, as conceptually illustrated in Figure 3. The DSF is regarded as a core green architecture technology that offers significant energy-saving advantages over conventional facades, particularly in subtropical regions (Khoshbakht et al., 2017).

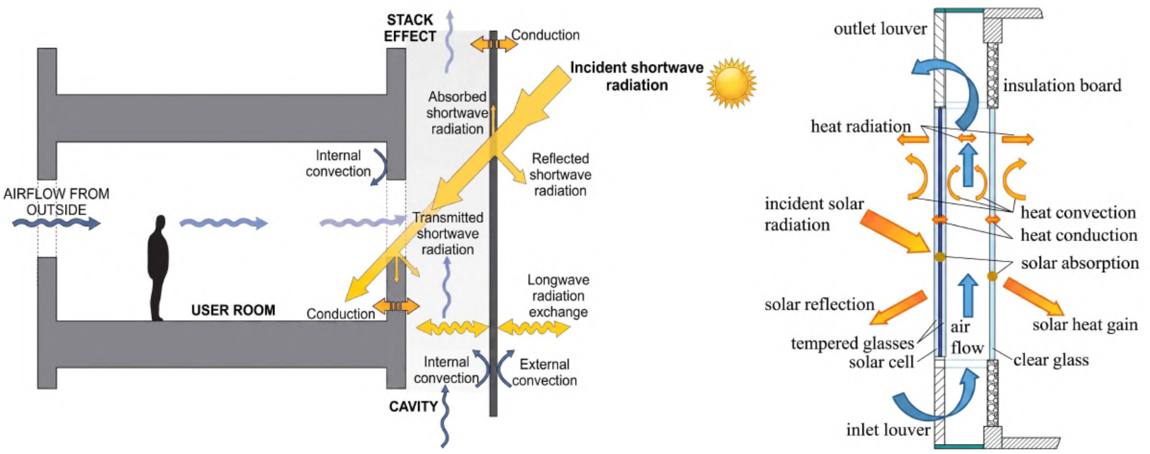


Figure 3. The detail of the double skin facade (DSF) of an office building illustrates the heat transfer mechanism and airflow (Barbosa et al., 2015)

Despite the relative scarcity of data and research on the development of this technology in tropical regions (Aziiz et al., 2018; Pomponi et al., 2017), several studies have yielded promising results. The implementation of DSFs in tropical climates has been shown to reduce the overall heat gain and energy consumption of buildings. For instance, studies on DSF using low-E triple glazing with a low solar heat gain coefficient (SHGC) and U-Value have led to significantly lower total annual electricity demands (Ardiani and Koerniawan, 2017; Mulyadi et al., 2011). This potential is further corroborated by regulatory assessments in similar hot-humid context, such as Malaysia, where naturally ventilated DSFs were found to significantly lower the overall thermal transfer value (OTTV) compared to traditional single-skin envelopes, thereby aligning with stringent energy efficiency standards (Ayegbusi et al., 2017). Similarly, research conducted in Taiwan’s humid climate indicated that a wider cavity space (e.g., 1.2 m) is more effective in reducing energy requirements and enhancing thermal insulation because of enhanced natural ventilation through the buoyancy effect (Dewi et al., 2013). The effectiveness of a DSF in reducing the solar heat gain and sensible cooling load is consistent across various studies.

The core function of the DSF cavity in hot-humid environments is to facilitate buoyancy-driven airflow, which removes heat buildup and mitigates excessive solar heat gain (Kang et al., 2019). Consequently, optimization efforts frequently concentrate on parameter analysis to enhance this stack effect, confirming that wider cavity spaces (i.e., 1.5 m) are often the most effective in tropical regions for maximizing sensible heat extraction and overall natural ventilation efficiency (Aziiz et al., 2018). More recent optimization workflows leveraging computational fluid dynamics (CFD) have refined these geometric parameters, demonstrating that optimizing vent design alongside cavity depth can exponentially increase ventilation ratios and improve the volumetric airflow rates significantly beyond base-case scenarios (Ma’bdeh et al., 2025). Conversely, suboptimal configurations, often characterized by shallow cavity depths, have been shown to result in reduced air exchange rates and lower overall energy efficiency.

The selection of glazing material is highly critical, as optimal DSF performance is contingent upon a delicate balance between thermal and optical properties. Furthermore, the thermal response of the façade is intrinsically linked to material properties; research in tropical educational buildings confirmed that the thermal lag and decrement factor are heavily influenced by the specific heat capacity and conductivity of the chosen skin materials (Barbosa and Alberto, 2019). This necessitates the incorporation of low-SHGC components to mitigate excessive radiant heat transfer, while simultaneously maintaining adequate visual light transmission (VLT) to minimize reliance on artificial lighting (Baetens et al., 2010; Tong et al., 2021). Achieving the multi-criteria targets of reducing both the cooling and lighting loads is contingent upon this dual requirement. This emphasis on DSF as a principal strategy for mitigating solar heat gain and preventing the visual-thermal trade-offs is imperative to accomplishing sustainable building performance objectives (Pelletier et al., 2023).

While the capacity of the DSFs to curtail sensible cooling loads has been comprehensively documented, their holistic impact on total operational energy in hot-humid climates remains highly contested. Research on an office building model in a tropical climate (Rio de Janeiro) found that a DSF equipped with shading devices could significantly intercept sensible heat transfer into the occupied zone (Barbosa et al., 2015). A direct measurement study on Kuala Lumpur also revealed that the integration of DSF with a low-E film reduced heat gain and facilitated enhanced thermal resistance (Qahtan, 2019). Despite these encouraging findings, a substantial research gap remains. Most studies on DSF, particularly those assessing cooling performance, tend to prioritize sensible temperature metrics while disregarding the pivotal role of RH in hot-humid climates. Although a recent review highlighted the potential of a DSF to reduce air temperature, it also noted an increase in RH by up to 5% (Lim and Ismail, 2023). This observation is critical because elevated moisture levels introduced by passive airflow directly dictate the auxiliary energy required for mechanical dehumidification, a penalty that mathematically offset the initial sensible cooling savings. Therefore, while the extant literature corroborates the potential of DSFs to enhance sensible building performance, there is a pressing need for research that holistically evaluates the coupled thermodynamic effects of DSFs on sensible energy consumption, artificial lighting demand, and latent heat accumulation. Most studies have either focused on single-performance metrics or were conducted in temperate climates (Jomehzadeh et al., 2017). This study aims to fill this critical gap by assessing the complex interaction of DSF elements with other passive technologies to achieve a total operational energy equilibrium within an integrated passive cooling system.

2.2. Windcatcher Technologies

Windcatchers, also known as wind scoops, represent a historical architectural element. They operate on the principle of induced natural ventilation to harness external winds and fresh air while simultaneously promoting the expulsion of internal hot air (Jomehzadeh et al., 2017). The performance of this system is contingent on two primary driving forces: the wind force, which prevails under high-wind-speed conditions, and the buoyancy effect, which dominates in tranquil environments (Ma et al., 2024). Furthermore, experimental assessments have elucidated that integrating evaporative cooling modules, such as wetted pads, within the system can significantly augment the sensible cooling load and induce airflow even under near-zero wind velocity conditions (Noroozi and Veneris, 2018). As illustrated in Figure 4, contemporary windcatcher mechanisms frequently incorporate features such as dampers and diffusers to optimize the filtration and directional flow of air entering the building. Consequently, windcatchers are regarded as a highly promising technology for delivering low energy sensible cooling solutions in passive cooling systems (Li et al., 2025).

However, a critical review of the literature reveals a conspicuous geographical bias within the extant research, which predominantly focuses on arid or semi-arid climates, such as in Iran, and several countries in the Middle East (Alrebei et al., 2022; Foroughi et al., 2024; Li et al., 2025; Movahed, 2016; Saif et al., 2021), as well as in several northern and central European countries, such as Austria (Shayegani et al., 2024), and in North America: for example, Mexico (Reyes et al., 2015). This discourse has recently expanded to Western Europe, where assessments across 15 capitals have highlighted the system’s underutilized passive cooling potential in educational facilities despite the temperate climatic context (Nejat et al., 2025). Yet, a knowledge deficit results in a significant gap in understanding their efficacy and geometrical optimization in humid tropical environments, such as the Southeast Asian region (Nejat et al., 2019), where elevated relative RH poses distinctive thermodynamic challenges, specifically regarding latent energy penalties.

Nevertheless, investigations on windcatchers in tropical climates have yielded encouraging, albeit limited, results. For example, studies undertaken in Malaysia substantiated the premise that augmented tower elevation concurrently enhances windcatcher performance (Gharakhani et al., 2017), which is consistent with previous discoveries pertaining to this configuration (Ghadiri et al., 2014). Moreover, computational fluid dynamics (CFD) modeling conducted on a tropical edifice in Panama indicated that windcatchers constitute an efficacious ventilatory methodology. This finding is evidenced by an increase in air change per hour (ACH) of up to 29.2% and sensible heat dissipation (Bernal et al., 2024b). These advantages are further substantiated by the findings that windcatchers provide enhanced airflow compared with conventional windows and that their geometric configuration is adaptable to various architectural structures (Reyes et al., 2015; Sangdeh and Nasrollahi, 2022). For instance, CFD optimization applied to modern residential typologies revealed that specific orientation adjustments, specifically a 30° rotation relative to prevailing winds, can yield superior velocity performance compared to standard cardinal alignments (Mamari et al., 2025). A study conducted in Yogyakarta, Indonesia revealed that a hybrid longitudinal windcatcher could attain optimal internal wind speeds of up to 0.3 m/s under low ambient wind velocities of 3 m/s (Satwiko and Tuhari, 2016).

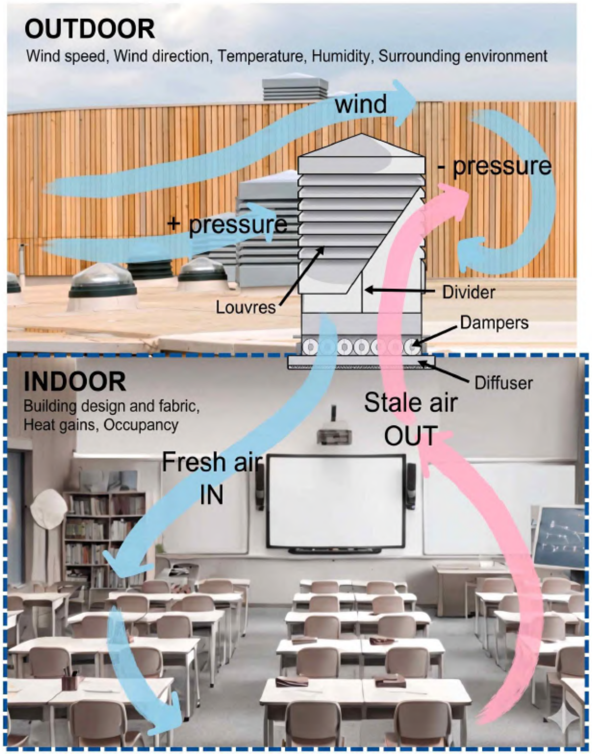


Figure 4. Contemporary windcatcher operating system in a building, and the parameters that influence it (Li et al., 2025)

The validation of these optimized geometries is increasingly reliant on advanced simulation methodologies, such as CFD simulation. CFD simulations are used to support the development of precise windcatcher shape topology (Katona et al., 2022). The advancement of meticulous geometrical design guidelines for windcatchers is imperative, requiring a comprehensive roadmap to standardise the diverse geometrical classifications impacting aerodynamic efficiency (Mayhoub et al., 2025). This optimization is of paramount importance because the windcatcher's functions extend beyond mere thermal mitigation, serving as a primary mechanism for driving volumetric airflow to passively extract internal heat gains, a critical factor for minimizing reliance on mechanical cooling (Khalid et al., 2025). Attaining a high ACH is not solely a thermal strategy, but rather a fundamental prerequisite for a functionally efficient passive building envelope.

Beyond isolated sensible efficiency, the aerodynamic integration of windcatchers is a critical metric. Their capacity to augment airflow rates significantly and facilitate consistent air exchange is vital for flushing out structural heat accumulation (Nejat et al., 2021). However, this mechanism introduces a severe thermodynamic paradox in the tropics that is frequently disregarded in standalone performance studies. For instance, while a study conducted in Kuwait demonstrated the efficacy of this technology in maximizing air exchange without latent consequences (Saif et al., 2021), replicating this immense volume of unconditioned outdoor air in Jakarta would mathematically explode the auxiliary energy required for mechanical dehumidification. Crucially, the extant literature fails to assess synergistic performance of integrated systems holistically. Most CFD and experimental models have addressed the performance of windcatchers only in isolation, neglecting to evaluate their coupled aerodynamic interaction with other passive technologies, such as a DSF and louvers. Therefore, this study fills a critical gap by utilizing advanced simulations to quantify the complex aerodynamic conflicts – such as parasitic vacuum effects – between the geometric and operational parameters of a windcatcher, a DSF, and louvers. Consequently, this research first isolates the aerodynamic parameters of windcatcher, before specifically assesses their combined influence on sensible cooling efficiency, auxiliary mechanical ventilation demands, mechanical dehumidification penalties, and total operational EUI in a hot-humid environment.

2.3. Louvers System

Louvers are a prevalent architectural element that are employed for sun shading, ventilation modulation, and facade integration, particularly in tropical climates (Iqbal et al., 2025). The performance of the system is predominantly influenced by its geometric configuration, which encompasses the slat angle, depth, and gap. These parameters, as illustrated in Figure 5 (Riviere and Damour, 2023), play pivotal roles in the regulation of solar radiation and daylight distribution (Dinapradipta et al., 2024; Ma et al., 2024). The strategic implementation of louvers has been shown to yield substantial advantages, including enhanced energy efficiency and optimized solar-optical balances (Ma et al., 2024).

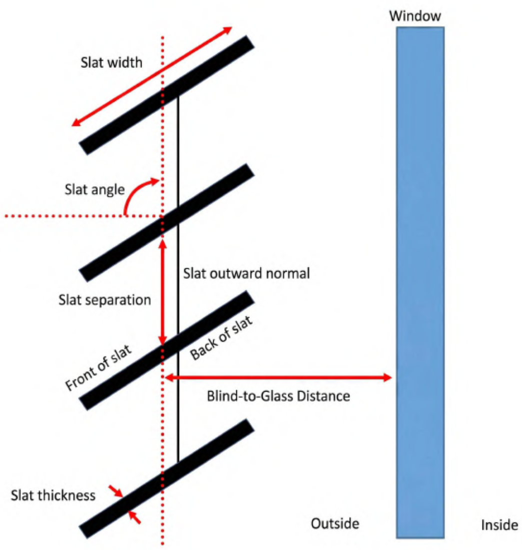


Figure 5. The configuration and parameters in louvers (Riviere and Damour, 2023)

An extensive corpus of existing scholarship provides compelling empirical data demonstrating the efficacy of louvers in mitigating solar heat influx and optimizing energy utilization. Investigations undertaken in tropical and humid environments, as exemplified by Nigeria, have substantiated the significant contribution of external louvers to the attenuation of solar irradiance. Specifically, the findings indicated that louvers can diminish solar radiation by 56.55%, thereby exerting a salutary influence on the building’s sensible cooling demands (Olaniyan et al., 2023). This reduction in solar heat gain directly corresponds to a significant decrease in the energy requirements. For instance, one study indicated that a specific louver geometry could reduce the cooling energy consumption by up to 21% compared with conventional buildings (Suryandono et al., 2020). This magnitude of efficiency is further corroborated by investigations into building orientation, which identified that external shading on south-facing envelopes, the orientation subject to maximum heat flux in the tropics, can achieve comparable energy savings of approximately 19 – 21% (Dutta et al., 2017). Concurrent research on a south-facing façade demonstrated that adjusting the slat inclination could result in a substantial reduction in energy consumption (Shaeri et al., 2022). The application of multi-objective optimization (MOO) has further underscored the capacity of louvers to curtail thermal energy expenditure while enhancing daylight performance (Khidmat et al., 2022).

The MOO approach has become the standard methodology for addressing inherent parameter conflict. Recent research in hot climates (e.g., Qom, Iran) has used MOO to evaluate shading strategies, such as louvers, against multiple conflicting performance metrics simultaneously: sensible cooling efficiency, daylight performance, thermal comfort, and operational cost (Abedini et al., 2025). This advanced analysis is necessary because louver geometry, such as slat angle and depth, fundamentally impacts thermal envelope resistance and daylight availability (Alafchi et al., 2025). Furthermore, studies focusing on tropical climates now use parametric optimization to evaluate dynamic louvers integrated into high-performance window system, i.e., DSF-IGUs, to maximize daylight availability without incurring severe sensible heat penalties (Bahdad et al., 2024). This advanced focus on daylighting metrics highlights the complexity of performance trade-offs.

Despite the evident advantages in terms of sensible energy efficiency, a critical trade-off emerges between solar gain management and secondary energy penalties (Shaeri et al., 2022). Although louvers are efficient in mitigating solar radiation, their deployment has been associated with a curtailment of natural light ingress, thereby potentially augmenting the reliance on artificial illumination, and consequently, escalating the overall total EUI (Bramiana et al., 2025). This study underscores the critical importance of refining louver configurations, including parameters such as their inclination and proximal spacing from fenestration, to establish an optimal equilibrium between the attenuation of radiant heat and the maximization of daylight penetration (Kirimtat et al., 2016; Nicoletti et al., 2023).

Conversely, a substantial research gap persists in studies that systematically assess this trade-off when coupled with complex aerodynamic ventilation strategies. The extant body of research on this aspect has primarily focused on the examination of louvers under isolated conditions or within a restricted set of optical-thermal metrics. Because external louvers physically alter the porosity of the facade, the majority of extant studies neglect to consider the aerodynamic flow resistance imposed on the building's natural ventilation and buoyancy-driven heat exhaust. The distinguishing feature of the present study is its comprehensive framework, which evaluates the interaction between the geometric parameters of louvers (e.g., angle and distance) and the reciprocal dynamics of the operational parameters of the windcatcher and the system configuration of the DSF. This comprehensive approach directly addresses the current gap by precisely evaluating their combined influence on aerodynamic flow resistance, secondary artificial lighting demands, mechanical dehumidification loads, and total operational energy demand in a hot-humid environment.

2.4. Jakarta Climate Context

Jakarta, which is a major coastal metropolis located in the lowland coastal area of northern Java (Martinez and Masron, 2020), is characterized by a tropical monsoon climate. This classification is consistent with high temperatures and humidity throughout the year owing to its proximity to the equator (Samodra and Yoon, 2015). The climatic conditions of the city are defined by two distinct seasons, namely a dry season and rainy season, which significantly impact architectural design (Maheng et al., 2023). The geographical position of Jakarta is visually represented in Figure 6.

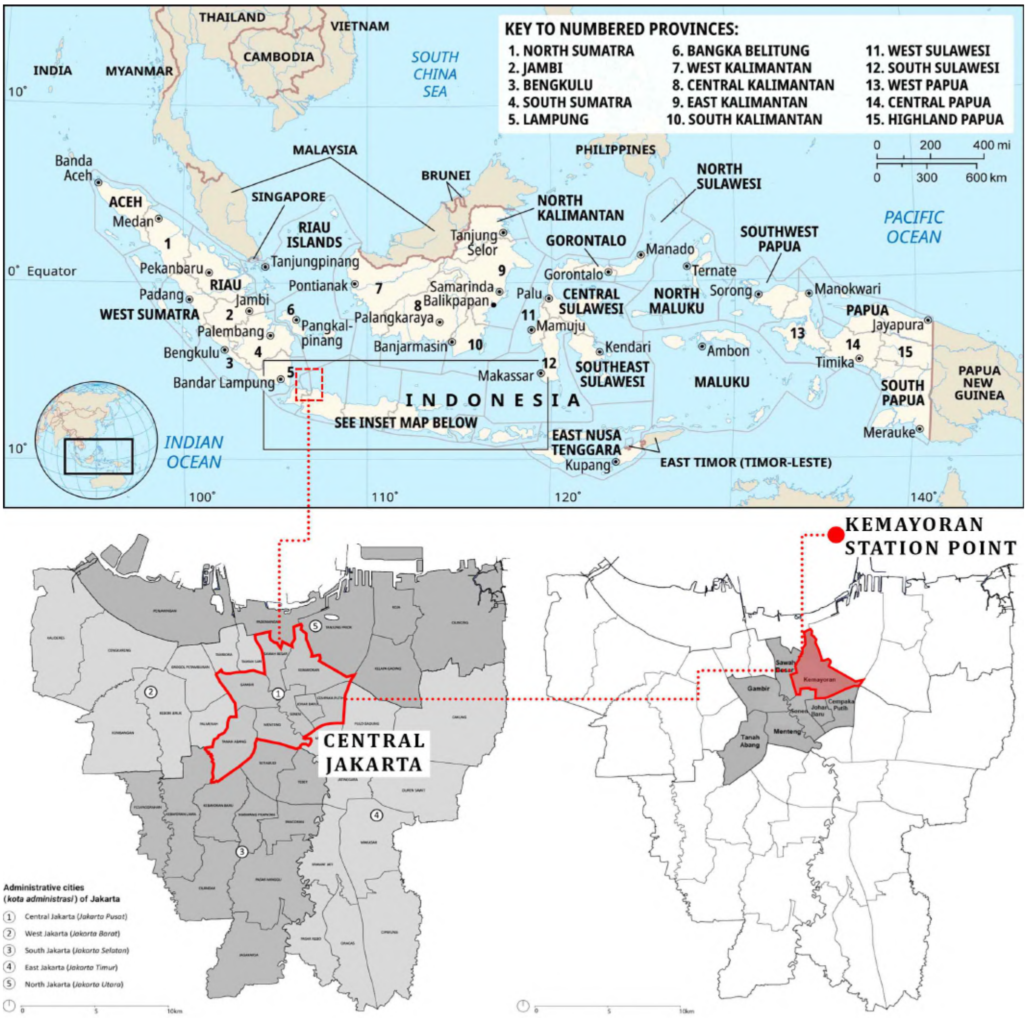


Figure 6. Map Jakarta with the KMY (Kemayoran Station) point representing Jakarta's climate in the DesignBuilder software (Source: www.britannica.com; Accessed: December 9th, 2025)

However, the climatic challenges faced by Jakarta are not solely meteorological in nature, but are also significantly exacerbated by the city's status as a high-density metropolis. This accelerated urbanization has led to a well-documented urban heat island (UHI) effect, a phenomenon of growing concern in tropical regions that are highly vulnerable to global warming (Lefevre et al., 2025). This phenomenon has been quantitatively validated in the Jakarta metropolitan area, where research has unequivocally associated the escalating intensity and extent of the UHI with substantial land use alterations and accelerated urban expansion (Putra et al., 2021). This phenomenon is attributed to the excessive use of building materials with high thermal mass, such as concrete and brick, which absorb solar radiation during the day and release it slowly at night. This mechanism directly contributes to severe sensible heat accumulation and increased operational cooling energy expenditure within the dense urban fabric (Wonorahardjo et al., 2020).

The thermal profile analysis is strictly based on annual EnergyPlus weather data (EPW) for the Jakarta-Kemayoran station[5)] utilized within Rhinoceros + Ladybug. This analysis indicates highly challenging, high-enthalpy conditions throughout the year (see Figure 7). The annual temperature range remains consistently high, spanning from 27.88 to 35.98 ˚C. This minor variation indicates an exceptionally low diurnal temperature range thereby effectively preventing nocturnal passive cooling. Furthermore, the RH data are substantial, ranging annually from 56.80% to 92.80% with a mean average of 80%. This confirms the presence of substantial latent cooling loads that persist throughout the year, creating a severe thermodynamic barrier whereby introducing unconditioned outdoor air would result in a significant increase in the demand for mechanical dehumidification.

With regard to wind availability, the EPW analysis indicates a mean annual velocity of 0.80 m/s, with predominant directions from west-southwest, northwest, and east. The wind profile analysis for mid-rise building design indicates a significant increase in wind velocity with height, suggesting strong wind energy availability for elevated structures. The windcatcher system has been developed to capitalise on precisely this roof-level dynamic pressure: its inlet geometry captures the high-velocity, steady flow at building crown and modulates it into a controlled aerodynamic pressure differential that drives consistent volumetric delivery to the occupied zone. This effectively compensates for the lower-level wind scarcity induced by urban roughness without generating disruptive indoor turbulence (Li et al., 2025; Mayhoub et al., 2025). This pressure-modulating capacity renders the high-velocity EPW profile conducive to windcatcher integration, as the system is engineered to operate across a broad wind speed spectrum – including conditions that nominally exceed the 0 to 5 m/s optimal range identified for conventional configurations (Alhraki and Sultan Qurraie, 2024).

The year-round high-temperature conditions that define Jakarta’s climate present significant challenges to the energy sector of the city, particularly regarding its high energy demand. Consequently, simple or stand-alone passive design techniques are insufficient to

5) Accessible via the following link: https://www.ladybug.tools/epwmap/; Accessed: May 30th, 2025

prevent severe secondary energy penalties without dedicated strategies to address the high radiant gains and latent loads. Therefore, the present investigation is warranted to explore how a tailored integrated passive envelope system can mitigate these demands by critically managing solar gains and optimizing buoyancy-driven heat exhaust in this challenging tropical environment without compromising the total operational EUI.

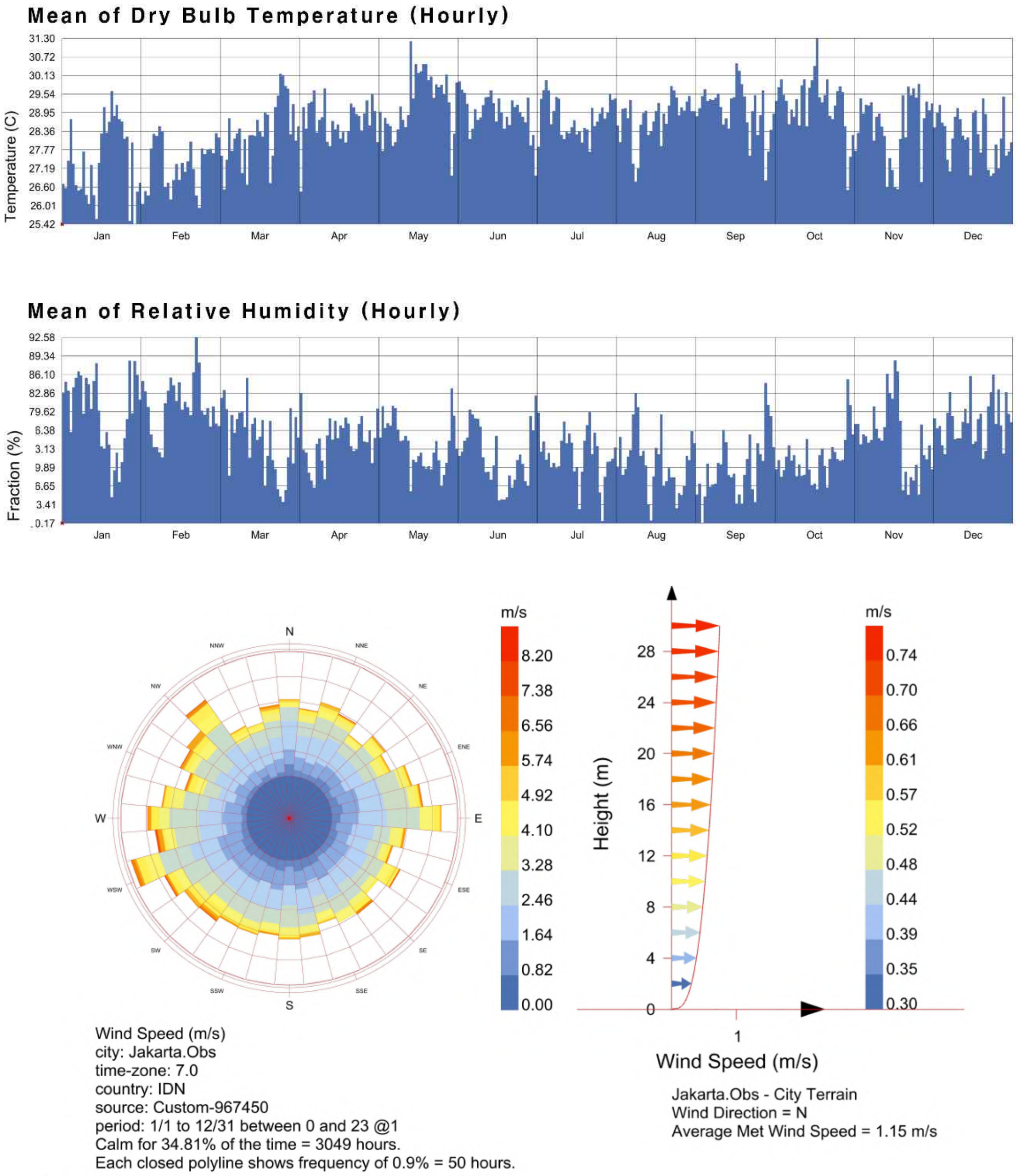


Figure 7. The chart relation to annual average of temperature, RH, and ambient wind speed

2.5. Synthesis of Critical Findings and Study Justification

Passive cooling technologies, including DSF, windcatchers, and louvers, were critically reviewed. This review confirms the individual potential of these technologies for enhancing building performance, including cooling loads and thermal performance in tropical or hot-humid climates. Table 3 summarizes the key findings, methodologies, and limitations of prior studies.

Table 3. Summary of prior research on integrated passive cooling system component technologies (DSF, windcatcher, and louvers) in tropical or hot-humid climates and other climate conditions

Author	Technology Focus	Location	Methodology	Key Variables	Main Findings	Limitation Identified
Prior research on DSF						
(Wong et al., 2008)	DSF	Singapore	Aipak CFD simulation	Facade orientation; Cavity gap (0.3 m, 0.6 m, 1.2 m); Galzing materials	South-facing DSF achieved best thermal performance (80% acceptability limit)	Simulation only; Energy efficiency was not evaluated
(Mulyadi et al., 2011)	DSF	Makassar, Indonesia	FORTRAN simulation	DSF geometry (cavity space depth: 1 m); Ventilation characteristics	Temperature reduction (solar heat gain reduction); Energy reduction	Simulation only; RH contril was not evaluated
(Dewi et al., 2013)	DSF	Taiwan	DesignBuilder + EnergyPlus simulation	Cavity space depth (0.3 m, 0.6 m, 0.9 m, and 1.2 m)	Cooling energy reduction 34.69%; Wider cavity space more effective in cooling energy reduction	Occupant thermal comfort was not evaluated
(Barbosa et al., 2015)	DSF	Rio de Janeiro, Brazil	IESVE simulation	Cavity space depth 1 m; North and south facade orientation; SSF vs. DSF	70% acceptable thermal comforts (occupied hours); Energy savings (DSF) 15%	No experimental validation; Results depend on weather data assumption
(Pomponi et al., 2017)	DSF	Rio de Janeiro, Brazil	IESVE and FLOVENT simulation	Building orientation (north, north-west, and north east)	61% occupied hours with comfort; Significant reduction in cooling loads	Simulation only; RH control was not evaluated
(Ayegbusi et al., 2017)	DSF	Kuala Lumpur, Malaysia	EnergyPlus	Facade orientations; Cavity space depth; Glazing material (inner and outer skin);	DSF reduced OTTV by 25-35%; Cooling load reduction reached 15-20%; Optimal cavity width was found to be between 0.6 m	This study relies solely on simulation results

				Natural ventilation strategies; Integration with internal shading device	and 1.0 m	
(Ardiani and Koerniawan, 2017)	DSF	Tangerang, Indonesia	Autodesk Green Building simulation	Perforated metal DSF vs. glass material DSF	Glass material DSF delivers the lowest energy demand	Thermal comfort was not evaluated; Ventilation not modelled
(Aziiz et al., 2018)	DSF	Bandung, Indonesia	ANSYS FLUENT simulation and experiment	Cavity space gap width (0.5 m, 1 m, and 1.5 m); Outer skin material (one-way galss)	0.5 m and 1.5 m air gap produced greatest reduction in indoor temperature	Specific material; Ony air gap and no additional factors; Energy consumption reduction was not evaluated
(Qahtan, 2019)	DSF	Kuala Lumpur, Malaysia	Experimental	Facade configuration (ventilated cavity); Measured parameters	Reduces heat gain-improving thermal comfort	No occupant level assessment
(Barbosa and Alberto, 2019)	DSF	Rio de Janeiro	IESVE simulation	External materials (clear glazing, perforated metallic sheet painted, and perforated galvanized metallic sheet)	Clear glazed DSF generated 15% higher airflow rates; Thermal comfort performance; The glazed model performed better during cold month	Geometry simplification; Weather data discrepancies (may not reflect the specific conditions); Focused solely on thermal performance
(Lim and Ismail, 2023)	DSF	Kuala Lumpur (Malaysia), Jakarta (Indonesia),	EnergyPlus simulation	SSF vs. DSF; Building Orientation (north, east, and west); envelope	DSF consistently outperforms SSF; Energy savings (more than 40%); DSF cavity space: lowering indoor temperature and	No experimental validation

		Phnom Penh (Cambodia), and Nha Trang (Vietnam)		properties (Low-E glass)	humidity	
Prior research on windcatcher						
(Calautit and Hughes, 2014)	Windcatcher	Ras Al Khaimah, UAE (Hot Dessert)	ANSYS FLUENT CFD study + Field measurements	Wind tower geometry (height, inlet/outlet size)	The windcatcher provides the required fresh-air rates; Cooling load reduction up to 30%	RH was not assessed
(Satwiko and Tuhari, 2016)	Windcatcher	Yogyakarta, Indonesia	CFD-ACEC simulation +	Windcatcher design (longitudinal type); inlet heights (1 m and 4 m)	CFD: wind speed of 1-5 m/s produced indoor air velocities of 0.19-1.01 m/s; Experiment: 3 m/s wind speeds	Design variations were limited; was not assessed
(Gharakhani et al., 2017)	Windcatcher	Johor, Malaysia	Experimental + CFD analysis	Windcatcher height (0.4 m, 0.8 m, 1.20 m)	Increasing height consistently improves ventilation performance; Thermal comfort implication	Steady-state assumption; Energy reduction was not assessed
(Saif et al., 2021)	Windcatcher	Ahmadi, Kuwait (hot dessert)	DesignBuilder simulation	Windcatcher geometry and dimensions	Windcatcher delivers a strong trade-off (thermal comfort)-substantially lower energy use (48% lesser tan baseline model)	Adaptive comfort model validity; No physical experiments
(Alrebei et al., 2022)	Windcatcher	Cardiff (UK), Doha (Qatar), and Amman	ANSYS CFX (CFD) simulation	Windcatcher geometry (size and shape); Orientation	Increase the actual-to-required ration by up to 9% thermal comfort improves	Simulation-only validation; Energy reduction impact

		(Jordan)				was not assessed
(Bernal et al., 2024)	Windcatcher	Panama	DesignBuilder simulation	Six windcatcher shapes and dimensions	Long-rectangular and semicircular shapes offered the highest air exchange; Short-rectangular delivered most balanced performance	Energy simulation was not evaluated; No experimental validation
(Bernal et al., 2024b)	Windcatcher	Panama	DesignBuilder (CFD)	Wind tower geometry (semicircular windcatcher)	Airflow improvement: 0.61 to 0.67 m/s (high-wind condition); 0.24 to 0.26 m/s (low-wind condition); Thermal comfort improvement; Energy implication	Steady-state assumption; Simulation only
(Shayegani et al., 2024)	Windcatcher	Vienna, Austria (Temperate)	DesignBuilder + EnergyPlus simulation	Windcatcher height (2.5 m); Inlet dimensions	The optimal configuration of windcatcher (2.5 m height and 0.9 x 1.4 m inlet) significantly reduces the cooling load	Simulation only without experimental validation
(Khalid et al., 2025)	Windcatcher	Muscat, Oman (Hot-humid)	STAR CCM (CFD) simulation	Windcatcher orientation	The windcatcher 30° from north toward the east produced the highest performance indicator and maximum average air velocity; This study indicates a need for dampers or control mechanism to close the windcatcher during high-speed winds	Climate specificity; Cooling sufficiency; Energy savings were not simulated
Prior research on louvers						
(Dutta et al., 2017)	Louver	Kolkata, India	TRNSYS 17 +	10-story commercial	Heat gain through south-facing	Scope of shading

		(Warm and Humid)	FORTRAN 77 + Autodesk Ecotect simulation	hospital building; Automated Movable Window Shading Device (venetian blinds) controlled by a Programmable Logic Controller (PLC) linked to the sun path; Windows orientations (south, east, west, and north)	windows was found to be 96% higher than north-facing windows; The highest saving energy occurred in June (14.9%); The shading device reduced the energy consumption for space cooling 18.7% and heat rejection by 16.4%	for movable shading; Autodesk Ecotect does not forecast the exact amount of solar irradiation owing to subjectivity in modelling surrounding objects, although it is useful for visualization
(Suryandono et al., 2020)	Louver	Jakarta, Indonesia	Rhinoceros + Grasshopper (Ladybug) simulation	L-shaped aluminium profiles (15 x 15 mm and 30 x 30 mm); Gap ratio (1:1, 1:2, 1:3); Building orientation	Cooling energy consumption by up to 18% compared with the baseline; Increase of 16.78% UDI; simple pay-back period (0.86 year)	Simulation only; limited set of louver size and gap ratio; no experimental validation
(Suryandono et al., 2021)	Louver	Jakarta, Indonesia	Rhinoceros + Grasshopper (Ladybug) simulation	L-shaped aluminium profiles (varying width, depth, and angle); Gap ratio; Building orientation	Energy savings (by up to 21%); Economic viability: simple pay-back period <1 year	No experimental validation; no lighting energy impact assessment
(Khidmat et al., 2022)	Louver	Jakarta (Indonesia), Birmingham (UK), and Sydney (Australia)	Rhinoceros + Grasshopper (ladybug and Honeybee) and Octopus simulation	Louvers geometry (overhang depth, blade space, and angle); Room orientation	UDI improvement (Jakarta +146%); Cooling energy reduction (Jakarta: 3.3%); Louvers dimensions and angles ffer the best trade-off (daylight quality and energy saving)	No experimental study

(Shaeri et al., 2022)	Louver	Bushehr, Shiraz, and Tabriz (Iran) (Hot arid)	DesignBuilder simulation	Louvers angle; Façade orientation (south, east, and west)	Significant reduction of cooling loads (up to 28.3%); Optimal louver angle depends on climate and façade orientation	No field validation
(Olaniyan et al., 2023)	Louver	Ogbomoso, Nigeria	DesignBuilder simulation	Five virtual models (overhangs, side-fins, and louvers); Material properties (glass and wall)	Louvers gave the best reduction in indoor solar gains (56.55%) and thermal benefits; substantial decrease in cooling load	No on-site validation
(Dinapradipta et al., 2024)	Louver	Surabaya, Indonesia	Radiance Desktop plug-in + Ecotect simulation and quasi-experimental modelling	Louver angles (30˚, 45˚, 60˚, and 90˚); Slat geometry (flat, one-bent, and two-bent)	Two-bent slat with a 90˚: provide the most even daylight distribution; Dynamic shading by bent can be effective to control daylight	Focus solely on daylighting; No energy savings impact assessment
(Bramiana et al., 2025)	Louver	Semarang, Indonesia	Autodesk Revit simulation	Lover window angles (0˚, 15˚, 30˚, and 45˚); Building orientation (west facing)	15 – 30 ˚: provides a balanced trade-off (daylight and acceptable glare); Contributing to energy savings	Simulation assumed static weather condition
(Ma et al., 2025)	Louver	Qingdao, China (Monsoon)	Rhinoceros + Grasshopper (Ladybug + Honeybee) simulation	Ventilation strategies (baseline + five optimized scenarios); Window type; Blind parameters	Energy reduction (34.4%); Tailoring ventilation schedules: Best trade-off (cooling and daylighting provision)	Simulation only; Study focuses on ventilation and blind parameters

A critical synthesis of the reviewed literature, visualized through the research landscape mapping (Figure 8) and multi-objective focus distribution (Figure 9), and the compiled limitations of prior studies (Table 3), elucidates a marked methodological predilection for numerical simulations over empirical fieldwork, alongside a distinct thematic compartmentalization across passive systems. While DSFs demonstrate a historically balanced investigation across the domains of energy, comfort, and ventilation, the analysis reveals a significant disparity in windcatcher and louver research. As evidenced in Figure 9, windcatcher studies are disproportionately skewed towards isolated aerodynamic performance – 7 of 9 windcatcher studies assess ventilation ACH exclusively, with zero studies assessing latent load RH or implications and only 1 of 29 total studies across all three technologies addressing this metric. This leaves their substantial impact on total operational energy and latent cooling penalties – specifically the recurring absence of RH control evaluations – substantially under investigated. Conversely, louver research is heavily dominated by daylighting and sensible heat reduction, frequently disregarding the physical aerodynamic flow resistance these devices impose on natural ventilation.

This observation is reinforced by the methodology distribution in Figure 8, which reveals that 23 of 29 studies (79.3%) rely exclusively on numerical simulation – building energy simulation (n = 15) and CFD (n = 8) – while experimental validation accounts for only 2 studies (6.9%), confirming a structural methodological saturation in singular-objective simulation approaches. This creates a noticeable asymmetry in which passive components are evaluated in isolation. This fails to take into account the severe secondary energy penalties, such as massive mechanical dehumidification demands and artificial lighting spikes, that are triggered when these systems interact in high-enthalpy tropical climates. Consequently, to address this scarcity and bridge the critical gap between isolated aerodynamic behaviour and holistic building thermodynamic, this study strategically discards isolated comfort metrics in favour of total operational energy parameters (sensible coolingefficiency, demand for auxiliary mechanical ventilation, and mechanical dehumidification loads) as the primary dependent variables. This selection entails a paradigm shift in investigative focus: moving away from theoretical, singular airflow rates towards a tangible, integrated aerodynamic and mass-balance analysis. By leveraging the established reliability of coupled numerical simulations, this study introduces a novel framework to quantify the complex aerodynamic conflicts – operationalized through a novel aerodynamic diagnostic metric and an integrated thermodynamic decomposition framework, both formally introduced in section 3 – and the optical-thermal trade-offs inherent in integrated passive envelopes. Ultimately, this research fills the extant literature gap by investigating and quantifying the “hard” thermodynamic limits – where purely passive ventilation fails, thereby providing empirical basis for evaluating for a “hybrid-decoupled” operational strategy in tropical mid-rise office buildings.

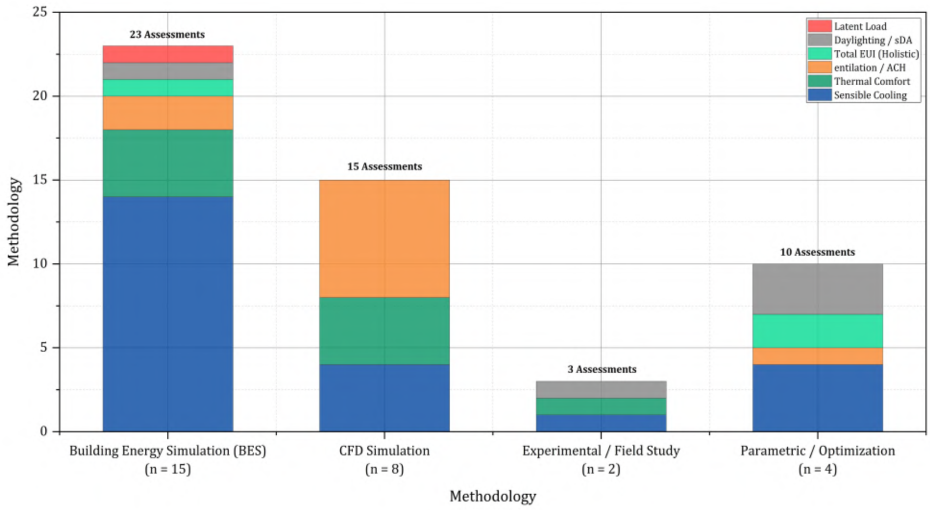


Figure 8. Research methodology distribution across reviewed studies – dominance of numerical simulation and singular-objective focus

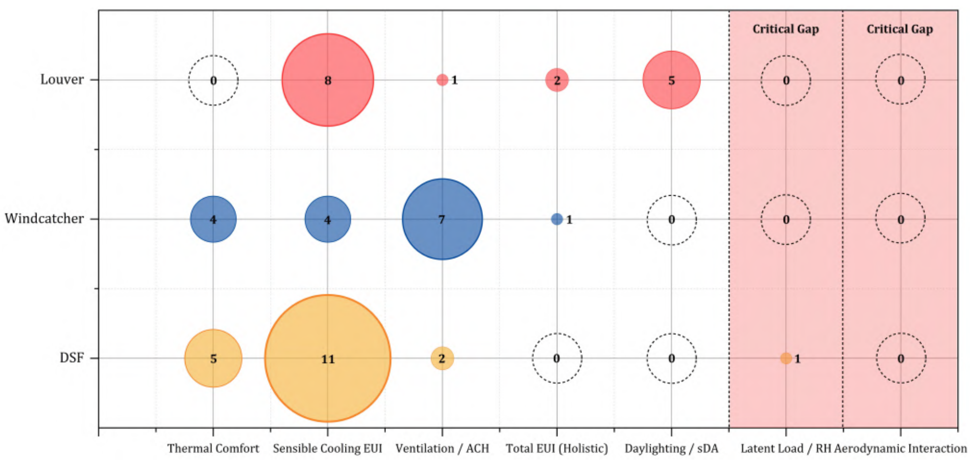


Figure 9. Research coverage gap matrix – performance metrics assessed per passive technology, revealing critical absence of latent load and aerodynamic interaction studies

3. Parametric Thermodynamic Assessment of Individual Passive Strategies

3.1. Simulation Framework and Thermodynamic Modelling Setup

This study employed a dynamic simulation framework to rigorously evaluate the thermodynamic performance boundaries and total operational energy equilibrium of integrated passive envelopes in Jakarta's high-enthalpy tropical climate. The analytical workflow integrated the DesignBuilder interface with the EnergyPlus to resolve complex heat and mass transfer phenomena. To address the transient heat flux through opaque envelopes, the simulation utilized the conduction transfer function (CTF) algorithm to decouple radiative and conductive heat gains. Simultaneously, the EnergyPlus daylighting module was employed to compute interior illuminance levels and the resulting secondary artificial lighting demands, establishing the quantitative basis for evaluating the visual-thermal trade-offs.

Crucially, the thermal model and optical models were fully coupled with the AFN to accurately capture buoyancy-driven ventilation, the aerodynamic flow resistance introduced by shading devices, and convective heat transfer within the DSF cavity and windcatcher. The AFN discretizes the building into a series of pressure nodes, where the mass flow rates are governed by the Bernoulli equation, utilizing surface-averaged wind pressure coefficients, which are derived from the wind's angle of incidence and the building's aspect ratio relative to the local wind velocity profile. This integration ensures the precise calculation of the specific convective heat transfer coefficient at internal surfaces and the psychrometric behaviour, which are pivotal for assessing the "latent barrier" and quantifying the exact mechanical dehumidification penalties in high-enthalpy humid environments. To ensure the reproducibility of the computational workflow and provide a clear inventory of the technological instrumentation employed, Table 5 details the specific software versions, simulation engines, and algorithmic modules utilized in this study. To synthesize the sequential logic of this multi-stage simulation, Figure 10 visualises the comprehensive methodological workflow, delineating the critical interface between parametric optimization, sensitivity assessment, and the thermodynamic diagnostic framework.

It is imperative to acknowledge that, whilst the EnergyPlus AFN constitutes the established computational standard for whole-building pressure-network analysis – and has been validated for DSF cavity airflow modelling in prior tropical studies (Lim and Ismail, 2023; Ayegbusi et al., 2017) – its pressure-node discretisation framework inherently simplifies three-dimensional flow phenomena. Specifically, the AFN does not resolve the Navier-Stokes equations at the intra-cavity scale, precluding the direct capture of vortex structures, turbulent jet separation, and localized recirculation cells that govern the near-opening aerodynamic behaviour of the DSF cavity. Consequently, the aerodynamic short-circuiting ratios and cavity flow directionality metrics reported in Section 4.3 and 4.4 represent network-averaged pressure equilibria rather than spatially resolved velocity fields. These results are interpreted as quantitative indicators of systematic aerodynamic behaviour, which is sufficient for the thermodynamic boundary analysis objectives of this study. However, the spatial resolution limitations of the data are identified as a primary candidate for future CFD based validation within the scope of subsequent research. The absence of physical experimental validation is acknowledged as a boundary condition of this study. Simulation credibility is established through three convergent proxies: EnergyPlus has been widely validated for DSF thermal simulation in tropical climates (e.g., Lim and Ismail, 2023; Ayegbusi et al., 2017); the Jakarta-Kemayoran TMYx EPW represents the highest-fidelity climatic dataset available for this location; and the three-pathway heat gain decomposition all 32 parametric configurations – collectively sufficient for the parametric boundary definition objectives of this study.

Whilst these internal proxies establish simulation stability, a formal methodological justification for the selection of AFN over CFD is warranted given the aerodynamic complexity of the integrated DSF-Windcatcher (later TST) system investigated. Table 4 presents a comparative capability assessment demonstrating that AFN constitutes the methodologically appropriate choice for the specific research objectives of this study.

Table 4. Comparative capability assessment of AFN and CFD relative to the research objectives of this study, justifying EnergyPlus-AFN as the methodologically appropriate simulation framework

Evaluation Criterion	AFN-This Study	CFD	Implication for This Research Objectives
Research purpose alignment	System-level pressure equilibria; thermodynamic boundary definition at building scale	Spatially resolved velocity and turbulence fields; zone-level flow characterisation	Boundary analysis requires network-level mass balance – spatial velocity maps provide unnecessary resolution at prohibitive computational cost
Temporal coverage	Full annual dynamic simulation (8,760 hours) feasible across all 32 configurations	Typically limited to steady-state or short transient periods owing to computational demand	Annual temporal coverage is necessary for seasonal hygric and aerodynamic boundary characterisation in a climate exhibiting monsoon variability
DSF multi-storey cavity	Validated against measurements for multi-storey naturally validated DSF using EnergyPlus AFN	Validated for single-cavity cross-section; full building-scale integration rarely demonstrated	Anđelković et al. (2016) established EnergyPlus-AFN fidelity for the exact system type used here, across three seasonal conditions
Aerodynamic short-circuit ration	Directly computable from AFN inter-zone node flow outputs – a primary novel metric of this study	Requires spatial integration of velocity fields across multi-zone boundaries – complex post-processing	The ASR diagnostic is only computationally accessible through AFN node-level flow accounting; CFD cannot produce this metric efficiently at building scale
Coupled hygric-thermal analysis	Integrated latent load and moisture mass balance natively within EnergyPlus thermal zone model	Requires separate humidity transport solver and co-simulation coupling protocol	The hygric boundary diagnosis requires tightly coupled thermal-moisture output that AFN-EnergyPlus provides natively
Validated simulation precedent	Anđelković et al. (2016); Catto Lucchino et al. (2021)	Whole-building energy balance integration requires co-simulation infrastructure	EnergyPlus-AFN is the community-validated standard for naturally ventilated multi-storey DSF simulation
Residual limitation	No spatially resolved	Computationally	CFD remains the

	recirculation calls within DSF near-opening zone	prohibitive for full annual parametric sweep across 32 configurations	appropriate future tool for spatial flow validation of the choked flow mechanism – identified as a primary direction for subsequent research

Furthermore, the EnergyPlus-AFN modelling approach for multi-storey naturally ventilated DSF systems has been independently validated against physical experimental measurements at a real office building by Anđelković et al. (2016), who recorded statistically acceptable agreement between simulated and measured DSF cavity air temperature and velocity across multiple seasonal thermal boundary conditions. A systematic comparison of four major BES platforms by Catto Lucchino et al. (2021) found that validation activities for DSF simulation remain scarce and tool selection is context-dependent; within this context, the Anđelković et al. (2016) validation constitutes a directly applicable precedent for the EnergyPlus-AFN configuration deployed in this study.

Table 5. Key geometric and operational specifications of the baseline mid-rise office model

Tools used	Used for
AutoCad and SketchUp	Development of the baseline geometry and detailed 3D architectural prototypes
DesignBuilder + EnergyPlus	Execution of dynamic thermal simulations, AFN modelling, and whole-building energy performance analysis
Rhinoceros + Ladybug and Climate Consultant 6.0	Environmental analysis (climate data interpretation)
Python (Pandas/NumPy) +	Automated data post-processing, numerical computation, and statistical sensitivity analysis (Range method)
OriginPro	Scientific data visualization, graphical representation of results, and simulation output organization

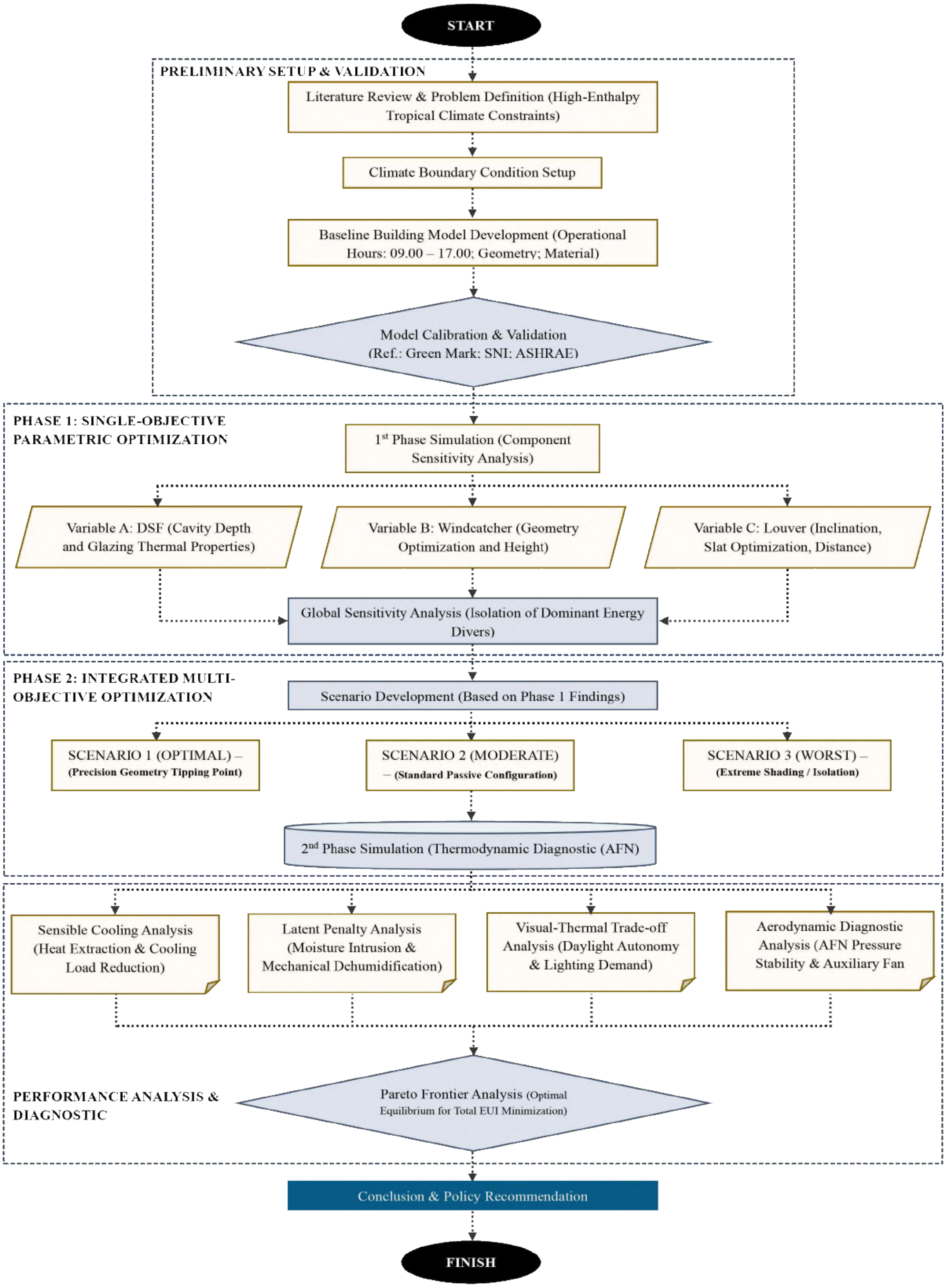


Figure 10. Methodological workflow illustrating the integration of parametric optimization, sensitivity analysis, and thermodynamic diagnostics

3.2. Climate Boundary and Model Calibration

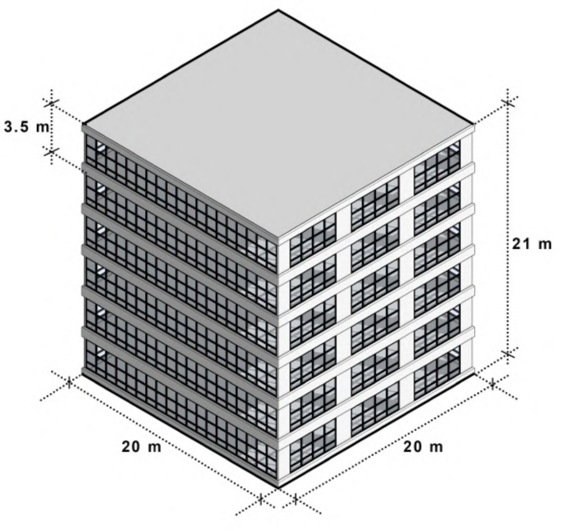
This simulation was subjected to data from International Weather for Energy Calculation via the EPW format for Jakarta-Kemayoran (6.15 ˚S, 106.85 ˚E), a region characterized by extreme tropical humidity, with dry-bulb temperatures ranging from 27.88 to 35.98 ˚C and a mean RH of 80%. This analysis covered a full meteorological year (8760 hours) to capture seasonal thermodynamic variations and impact of diffuse solar radiation. To reflect realistic office functionality, the building's operational schedule was strictly set from 9:00 a.m. to 5:00 p.m. on weekdays, in accordance with Green Mark guidelines (BCA, 2021) and Indonesia's Government Regulation (Government Regulation (PP) No. 35, 2021). The internal boundary conditions were set to a high-load scenario, with an occupant density of 0.188 people/m^2 on the ground floor, 0.138 people/m^2 on upper floors, and a baseline lighting power density of 11.8 W/m^2. These internal heat gains were configured to rigorously test the passive system's sensible heat removal capacity while simultaneously establishing a quantitative benchmark for subsequent artificial lighting penalty assessments (Thornton et al., 2009).

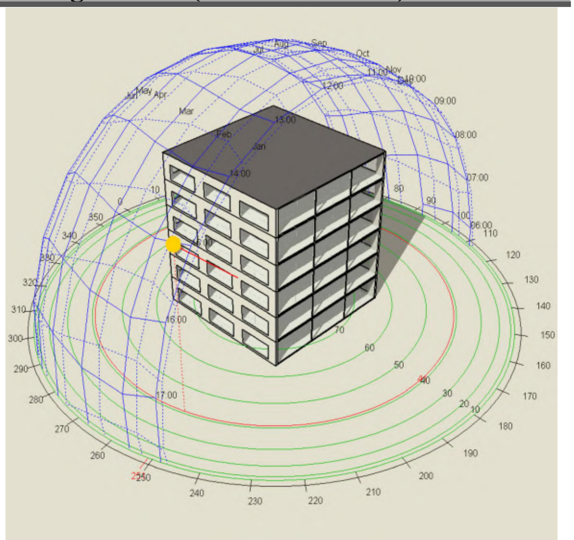
The baseline model for this study is a six-story mid-rise office building with a gross floor area of 2400 m^2 that is geometrically compliant with the ASHRAE 90.1 prototype (ASHRAE 90.1, 2019). The envelope has a window-to-wall ratio of 40% and the primary façade's principal axis is fixed to a southern orientation as a constant boundary condition throughout the simulation matrix. To ensure experimental reproducibility and strict adherence to standard office functionality, the precise geometric dimensions and operational setpoints governing this baseline model are itemized in Table 6. Moreover, given the study's emphasis on transient heat flux analysis, the constituent thermo-physical properties of the envelope material – specifically, thermal conductivity, density, and specific heat capacity – are explicitly defined in Table 7 to validate the inputs for the CTF algorithm. The baseline model served as the absolute control variable to quantify the initial total EUI, providing a definitive datum to measure the exact aerodynamic, visual, and latent energy trade-offs of the proposed passive interventions. The primacy of the mechanical cooling system in the baseline energy balance is corroborated by empirical measurement in Jakarta's commercial office sector, wherein the HVAC system constitutes the single largest category of building electricity consumption (Farizal et al., 2024), confirming that the modelled archetype replicates the thermodynamic operating conditions characteristic of the target building population.

Table 6. Key geometric and operational specifications of the baseline mid-rise office model

Category		Description
Building Type	:	Mid-Rise Office Building[6]
Location (Weather Data)	:	Jakarta (Jakarta – Kemayoran)[7]
Number of Floors	:	6 (Six) Floors
Gross Floor Area (per Floor Area)	:	400 m^2
Total Gross Floor Area	:	2,400 m^2
Building Surface	:	2,480 m^2
Floor-to-Floor Height	:	3.5 m
Total Building Height	:	21 m
Building Footprint	:	Rectangle (20 x 20 m)
Window-to-Wall Ratio (WWR)	:	40%
Facade Orientation	:	South
HVAC (Cooling System)	:	Fan Coil Unit (4 Pipe) with District Heating + Cooling
Building Operation Hours	:	09.00 a.m. to 17.00 p.m.
Population Density	:	1st Floor: 0.188 people/m^2; 2nd – 6th Floors: 0.138 people/m^2
Power Density	:	11.8 W/m^2

Illustration of Mid-Rise Offoce Building Model (Refernce Model)





6) For a more detailed overview, please refer to https://www.energycodes.gov/prototype-building-models; Accessed: April 30th, 2025

7) For a more detailed overview, please refer to https://climate.onebuilding.org/; Accessed: April 28th, 2025

Table 7. Thermo-physical properties of the building envelope materials

Material Case	Layers	Thickness (m)	Thermal Conductivity (W/m^{-k})	Density (kg/m^3)	Specific Heat (J/kg^{-k})
External Walls	Cement/Plaster/ Mortar-Plaster, Sand, Aggregate	0.02	0.82	1680	840
	Brick-Aerated	0.13	0.3	1000	840
	Cement/Plaster/ Mortar-Cement Screed	0.03	1.4	2100	650
Internal Walls	Cement/Plaster/ Mortar-Cement Plaster, Sand Aggregate	0.03	0.72	1860	840
	Brick-Aerated	0.13	0.3	1000	840
	Cement Sand Render	0.02	1.0	1800	1000
Roof	Bitumen, Felt/Sheet	0.02	0.23	1100	1000
	Concrete Cast-Lightweight	0.15	0.17	500	840
	Plaster Board	0.013	0.25	2800	896
Floor	Cast Concrete (Dense)	0.12	1.4	2100	840
	Cement/Plaster/ Mortar-Cement	0.02	0.72	1860	840
	Ceramic, Glazed	0.02	1.4	2500	840
Ground Floor	Cast Concrete (Dense)	0.12	1.4	2100	840
	Cement/Plaster/ Mortar-Cement	0.02	0.72	1860	840
	Ceramic, Glazed	0.02	1.4	2500	840
Windcatcher	Aluminium	0.003	160	2800	880

To characterise the thermodynamic operating envelope within which the passive systems must function, a seasonal psychrometric analysis was conducted using the Climate Consultant 6.0 bioclimatic tool. This analysis was referenced against ASHRAE Standard 55 comfort criteria and applied to the same EPW dataset for Jakarta-Kemayoran station (6.156˚ S, 106.8424˚E). The annual hourly data were disaggregated into two climatically distinct periods: the dry season (June through September; 2,928 operational hours) and the rainy season (October through May; 5,832 operational hours). These periods correspond to the southwest and northwest monsoon cycles that govern Jakarta's Aw tropical savanna classification (Maheng et al., 2023; Putra et al., 2021). This seasonal decomposition is of critical analytical significance because the two periods impose fundamentally different thermodynamic demands on the building envelope – a distinction that aggregate annual analyses systematically obscure.

The psychrometric distribution of the dry season (see Figure 11) demonstrates a tendency for data points to congregate in the elevated range of dry-bulb temperature (approximately 26 – 33 ˚C) with RH levels that persistently surpass 60%. This observation validates the notion that the diminished precipitation during this phase does not result in a concomitant decrease in latent load intensity. The Climate Consultant diagnostic identifies that 94.2% of dry season hours (2,759 of 2,928) require simultaneous mechanical cooling and dehumidification, whilst a further 5.8% (169 hours) demand standalone dehumidification in the absence of a concurrent sensible cooling requirement. It is imperative to note that not a single operational hour falls within the designated natural comfort zone. Furthermore, the suite of passive bioclimatic strategies – encompassing natural ventilation cooling, evaporative cooling, high thermal mass, and fan-forced ventilation -registered zero hours of effective applicability. The sole passive intervention with demonstrable reach is the use of solar shading of windows, which extends coverage to 33.2% of dry season hours (971 hours). However, this strategy addresses radiant heat interception exclusively and exerts no mitigating influence on the latent load mandate.

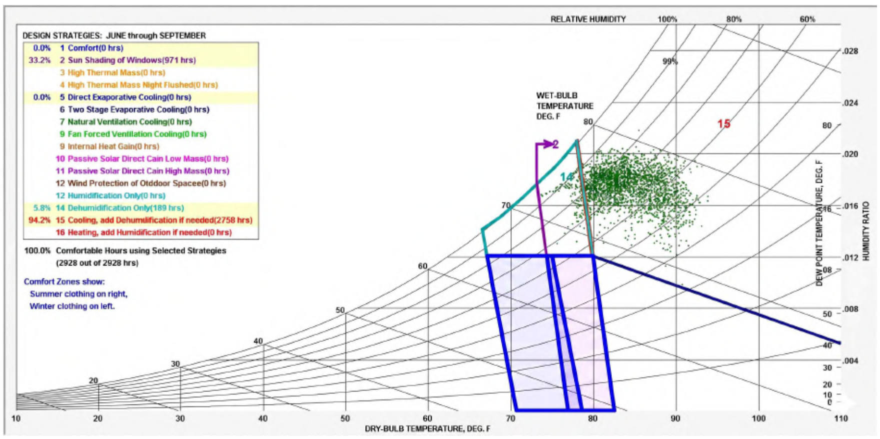


Figure 11. Psychrometric distribution of Jakarta-Kemayoran EPW data during the dry season (June – September, n = 2,928 hours)

The psychrometric distribution of the rainy season (see Figure 12) presents a thermodynamically more severe hygric condition. Although the sensible cooling fraction experiences a slight contradiction to 90.3% of rainy season hours (5,266 of 5,832), the standalone dehumidification fraction increases to 9.7% (566 hours) – a 3.35-fold increase relative to the dry season in dehumidification-only exposure. The observed seasonal inversion can be attributed to the elevated atmospheric moisture loading that is introduced by the northwest monsoon. This sustained moisture content is sufficient to maintain ambient humidity ratios above 0.016 kg_{water}/kg_{dry} air for the majority of rainy season hours. This is evidenced by the rightward displacement of the psychrometric data cluster relative to the dry season distribution. The passive strategy assessment corroborates the findings from the dry season, precisely indicating that natural ventilation, evaporative cooling, and all thermal mass strategies registered zero hours of effectiveness, whilst solar shading reached 31.1% of rainy season hours (1,813 hours) - marginally lower than the dry season figure, reflecting the reduced direct beam radiation intensity under monsoon overcast conditions. It is important to note that zero comfort hours can only be achieved through mechanical intervention. The aggregate implication of this seasonal psychrometric analysis constitutes the empirical climate mandate that frames the entire two-phase simulation framework of this study. Across the complete annual cycle of 8,760 hours, mechanical dehumidification is required without exception. In fact, 91.6% of all hours (8,025 hours) demand concurrent cooling and dehumidification, whilst the remaining 8.4% (735 hours) require standalone dehumidification. This distribution unequivocally established that latent load management is not a secondary consequence of passive envelope intervention, but rather a primary and non-negotiable thermodynamic constraint of Jakarta’s hot-humid climate. The finding directly informs the research hypothesis that purely passive strategies are insufficient to resolve the total energy burden of tropical mid-rise office buildings – a premise that the phase 1 parametric analysis and phase 2 integrated assessment presented in the subsequent chapters are designed to quantify, bound, and ultimately justify the transition to a hybrid decoupled operational framework.

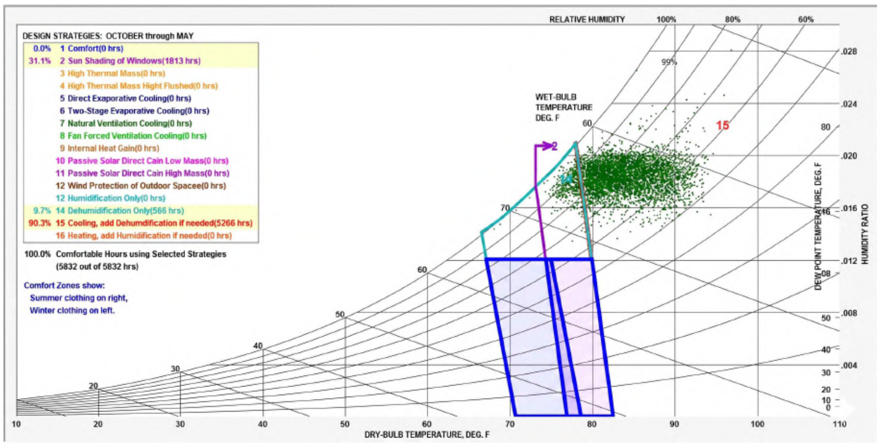


Figure 12. Psychrometric distribution of Jakarta-Kemayoran EPW data during the rainy season (October – May, n = 5,832 hours)

3.3. Integrated Passive Strategies and Parametric Variables

The proposed system integrates three passive architectural elements, a DSF, a roof-mounted windcatcher, and external louvers. These elements are strategically designed to achieve a total operational energy equilibrium by navigating the complex trade-offs between sensible heat rejection, massive latent load management, and DA. For the DSF, optimization centred on the thermodynamic behaviour within the cavity, with particular attention to convective heat transfer and stack effect efficiency (see Figure 13b). The thermal analysis research program (TARP) algorithm was utilized to calculate the variable surface convection coefficient (h_c), with dynamically adjusted based on the surface-to-air temperature differential. This approach enabled the accurate resolution of buoyancy-driven airflow loops and convective heat transfer inside the variable-depth DSF cavity (Lim and Ismail, 2023).

Concurrently, the windcatcher optimization targeted aerodynamic topology (tower shape) and tower elevation (see Figure 13a) to maximize wind pressure differential (C_p) and hydrostatic pressure column. This configuration is critical for ensuring a sufficient driving force during both wind-driven and buoyancy-dominant periods, specifically to counteract the parasitic vacuum effects generated by the heated facade (Li et al., 2025; Mayhoub et al., 2025). Simultaneously, the configuration of the external louvers was optimized based on the targeted solar geometry relation (see Figure 13c). Specific slat angles and dimensions were systematically permuted not only to intercept direct solar radiation for sensible cooling, but also to facilitate diffuse light transmission to prevent severe artificial lighting penalties (Bramiana et al., 2025; Dinapradipta et al., 2024; Suryandono et al., 2020). Furthermore, this geometrical permutation explicitly accounts for the physical aerodynamic flow resistance the louvers impose on the building's natural ventilation porosity. The specific independent variables and their respective parametric domains involved in this optimization process are detailed in Table 8.

Table 8. Parametric simulation matrix defining the geometric variables and thermodynamic functions

System Domain	Specific Design Parameters	Parametric Domain / Configuration	Physics-Based Function and Rationale
DSF (Case A)	Cavity Depth	• 0.25 m • 0.5 m • 1.0 m • 1.2 m • 1.5 m	Modulates convective heat transfer coefficients (h_c) via TARP algorithm and buoyancy-driven stack effect intensity to extract accumulated sensible heat
	Glazing Material Thermal Properties	• **Type 1** (SHGC: 0.174; U-Value: 0.345 W/m^2K) • **Type 2** (SHGC: 0.219; U-Value: 2.8618 W/m^2K)	Controls conductive heat flux (q^n) and radiant transmission through the inner boundary layer, directly influencing the core optical-thermal trade-off
Windcatcher (Case B)	Aerodynamic Topology (Windcatcher Tower Shape)	• Long-Rectangular (Footprint: 1 x 20 m) • Short-Rectangular (Footprint: 1 x 1.2 m)	Determines the effective wind capture area (A_{inlet}) and maximizes the wind pressure differential (C_p) to drive airflow and counteract parasitic vacuum effects from the heated facade
	Tower Elevation (Height)	• 2 m • 3 m	Maximizes the hydrostatic pressure

		• 5 m • 7 m • 9 m	column (ΔP_{stack}) to induce buoyancy-driven ventilation during low-wind periods, stabilizing the building's aerodynamic pressure
Louvers (Case C)	Louvers Gap and Width[8] of Slats	• Type 1 (Gap: 30 mm; Width of Slats: 30 mm) • Type 2 (Gap: 270 mm; Width of Slats: 340 mm)	Defines the solar cut-off angle to intercept direct beam radiation while managing diffuse light transmission, and dictates the physical aerodynamic flow resistance (porosity) imposed on the natural ventilation network
	Louvers Angle	• 30° • 45° • 60°	Optimizes the thermodynamic balance between shading coefficient (heat rejection) and visual permeability (view factor) to prevent severe artificial lighting penalties, and aerodynamic friction
	Louvers Distance from the Window	• 30 mm • 300 mm	

8) Slath width refers to horizontal chord depth – the dimension measured along the façade face – not structural blade thickness. This upper bound value was included to empirically define the maximum shading coefficient achievable within the parametric domain.

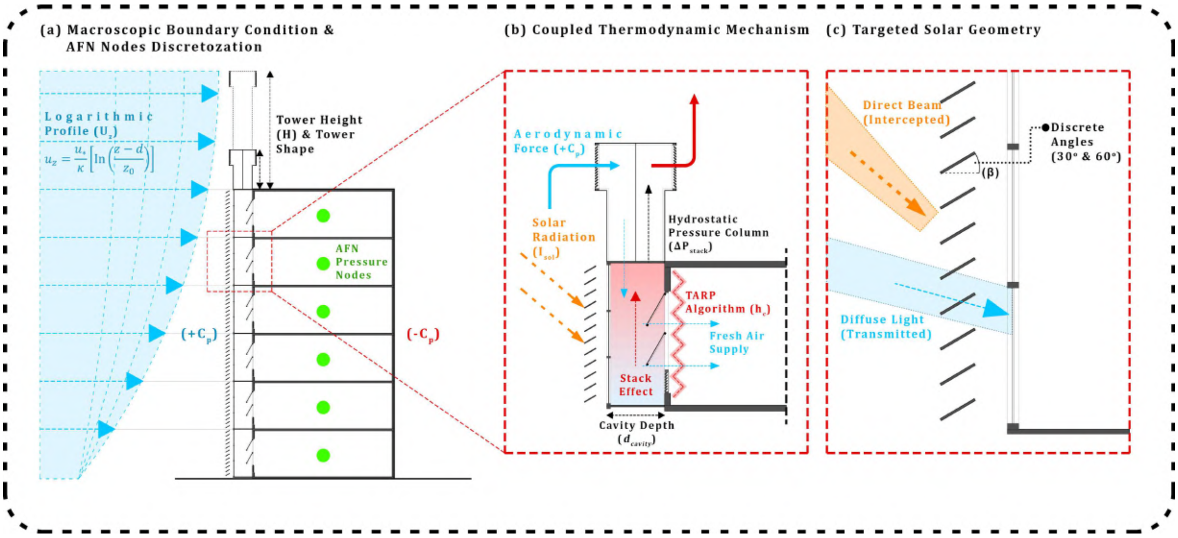


Figure 13. Schematic framework of the integrated passive envelopes system: (a) Macroscopic boundary conditions delineating the logarithmic wind profile and vertical AFN nodal discretization; (b) coupled thermodynamic mechanism illustrating the aerodynamic driving force, hydrostatic pressure column, and TARP-calculated convection coefficients (h_c); (c) Targeted solar geometry optimizing direct beam interception versus diffuse light transmission via discrete louver tilt angles (β)

As illustrated in Figure 14, the anticipated thermophysical responses of each passive element are delineated across the geometric parametric domain defined in Table 7, thereby establishing the conceptual boundary conditions that the phase 1 sensitivity analysis is designated to empirically bound. This framework is predicated on the premise that monotonic performance gains are not a prerequisite; rather, it foregrounds the non-linear thresholds and operational paradoxes that are anticipated to emerge within Jakarta's high-enthalpy equatorial climate. For the DSF (see Figure 14A), it is hypothesized that the interaction between cavity depth and glazing thermal properties will result in diametrically opposed aerodynamic regimes. The low-SHGC glazing will localise solar heat at the outer skin, generating a hotter cavity that will drive buoyancy-driven cavity circulation. The intensity of this circulation will scale proportionally with both the cavity height differential and the thermal gradient between the cavity and the outdoor air, amplifying the stack effect accordingly. By contrast, the high-SHGC glazing will transmit radiation directly into the occupied zone, maintaining a comparatively cooler cavity and inducing a reversed aerodynamic regime, but imposing an internal cooling penalty. For the windcatcher (see Figure 14B), it is predicted that the long-rectangular topology will facilitate access to higher-velocity laminar flow above the urban turbulence boundary, thereby optimising both the air intake and the pressure stack. However, this aerodynamic gain is anticipated to introduce a hygrothermal paradox: the elongated convective shaft will act as a thermal sink, cooling the supply air and consequently increasing its RH to near-saturation levels. This may result in a latent dehumidification penalty that will offset the sensible heat gains.

For the louver system (see Figure 14C), it is anticipated that the concurrent permutation of gap, slat width, inclination angle, and distance from the external glazing will govern three competing objectives simultaneously. It is hypothesized that dense configurations (30 mm gap, 45˚, positioned 300 mm from the glazing) obstruct ambient wind flow, forming a stagnant thermal plenum that converts intercepted solar radiation into conductive heat gain and paradoxically elevates the cooling load. Permeable configurations (270 mm gap, 30˚, positioned 30 mm from the glazing) are expected to eliminate this plenum through continuous convective flushing whilst preserving diffuse daylight transmission and aerodynamic porosity. The geometric entanglement of these four parametric variables precludes single-variable maximisation and establishes the physical rationale for the multi-objective sensitivity analysis presented in Section 3.6.

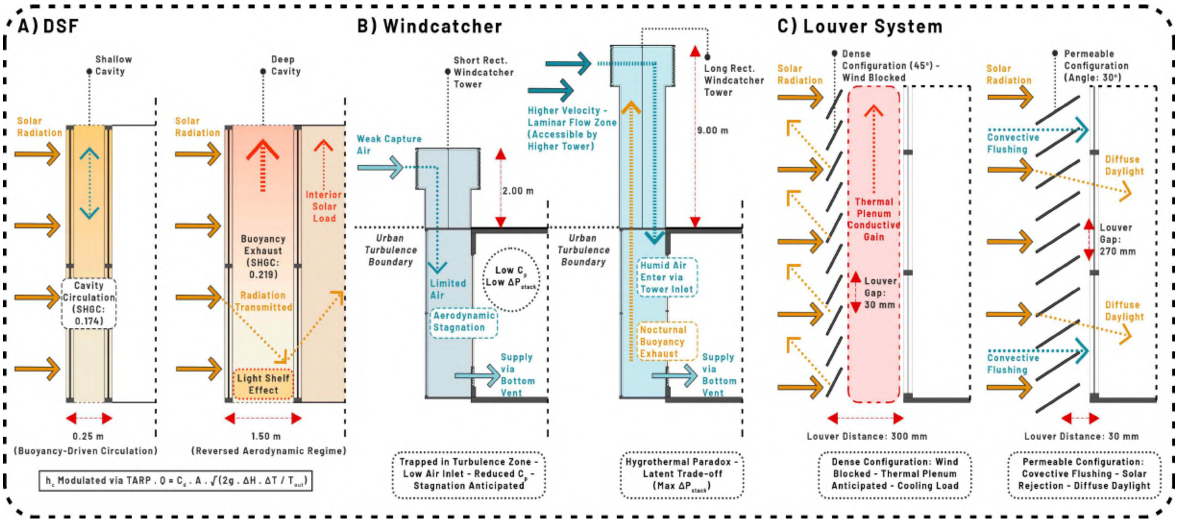
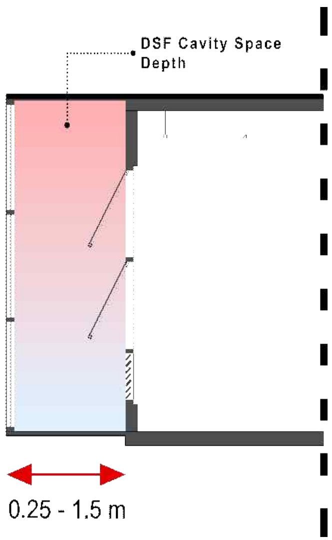
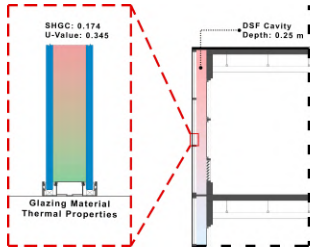
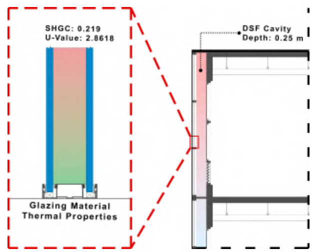
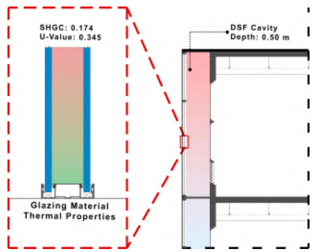


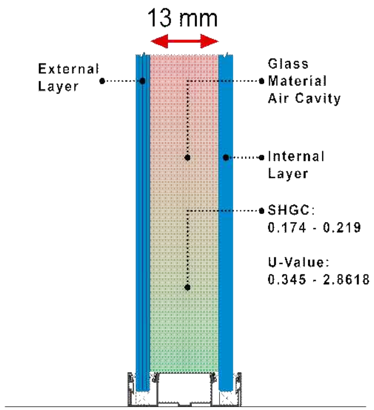
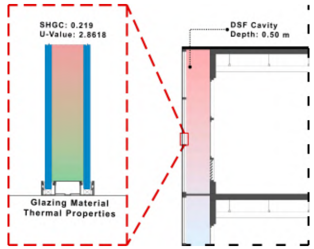
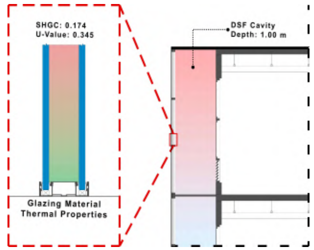
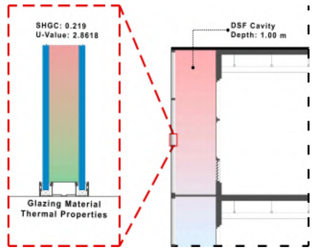
Figure 14. Anticipated aerodynamic and thermal mechanisms of DSF, windcatcher, and louver system under extreme geometric configurations

3.4. Two-Phase Optimization and Assessment Workflow

This research methodology adopted a sequential optimization strategy (Bahdad et al., 2024) structured into two distinct phases to systematically mitigate design conflicts between thermal, visual, and hygienic performance. Phase 1 (component-level optimization) involved the independent simulation of each passive technology to isolate its performance drivers using a parametric approach. To rigorously isolate the marginal thermodynamic impact of each passive strategy against the reference model, specific simulation scenarios were defined. Table 9 delineates the experimental matrix, providing a detailed comparison of the baseline configuration and the targeted intervention cases employed in the initial phase of the parametric sweep.

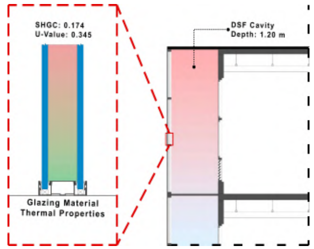
Table 9. Phase 1 building simulation scenarios: Baseline specification and parametric intervention cases

Parameters	Building Scenarios	Element Details	Illustration
Case A (DSF) 	Case A1	• Cavity Space Depth: 0.25 m • SHGC: 0.174 • U-Value: 0.345 W/m^{2}K	
	Case A2	• Cavity Space Depth: 0.25 m • SHGC: 0.219 • U-Value: 2.8618 W/m^{2}K	
	Case A3	• Cavity Space Depth: 0.5 m • SHGC: 0.174 • U-Value: 0.345 W/m^{2}K	

	Case A4	• Cavity Space Depth: 0.5 m • SHGC: 0.219 • U-Value: 2.8618 W/m^{2}K	
	Case A5	• Cavity Space Depth: 1.00 m • SHGC: 0.174 • U-Value: 0.345 W/m^{2}K	
	Case A6	• Cavity Space Depth: 1.00 m • SHGC: 0.219 • U-Value: 2.8618 W/m^{2}K	

<table>
<tr><td></td><td>Case A7</td><td>

- Cavity Space Depth: 1.20 m
- SHGC: 0.174
- U-Value: 0.345 W/m^{2}K

</td><td>
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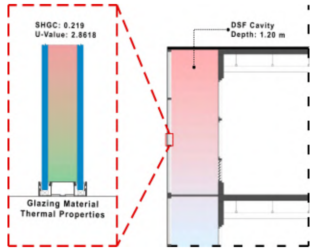
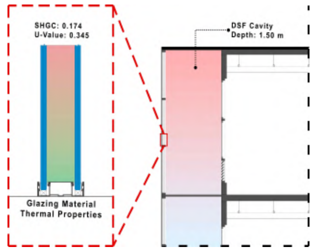
<tr><td></td><td>Case A8</td><td>

- Cavity Space Depth: 1.20 m
- SHGC: 0.219
- U-Value: 2.8618 W/m^{2}K

</td><td>
</td></tr>
<tr><td></td><td>Case A9</td><td>

- Cavity Space Depth: 1.50 m
- SHGC: 0.174
- U-Value: 0.345 W/m^{2}K

</td><td>
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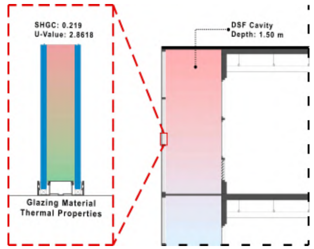
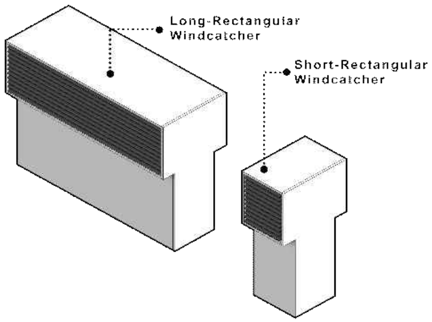
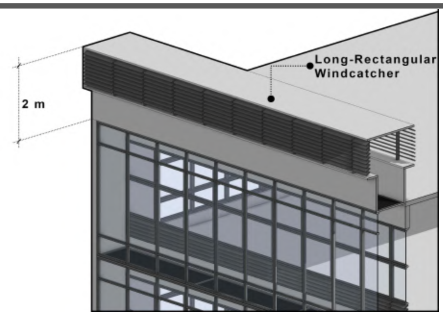
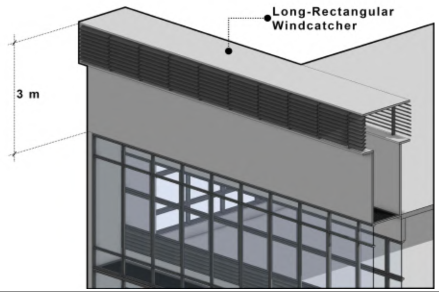
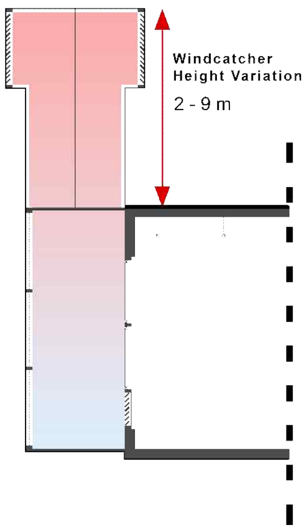
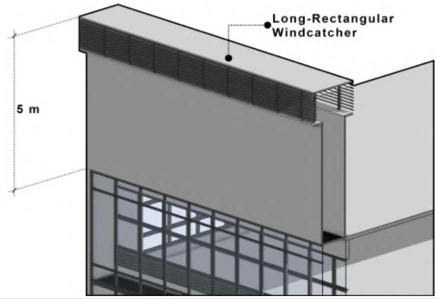
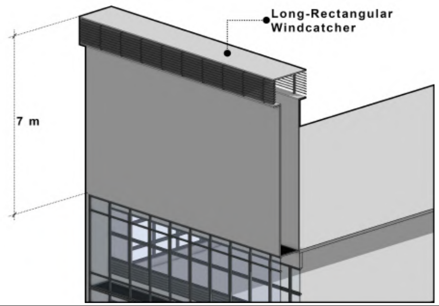
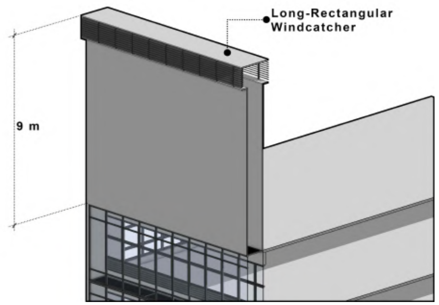
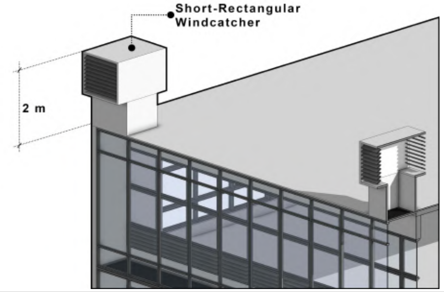
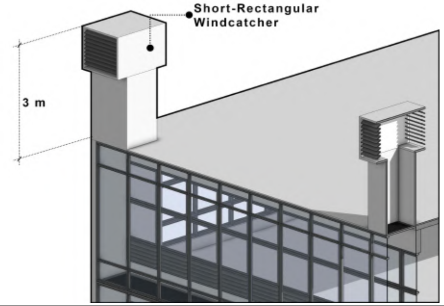
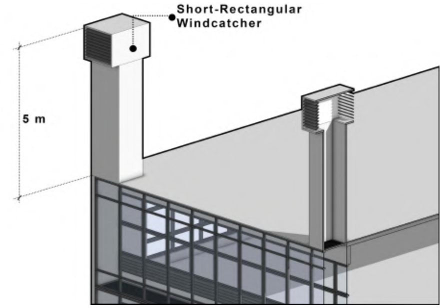
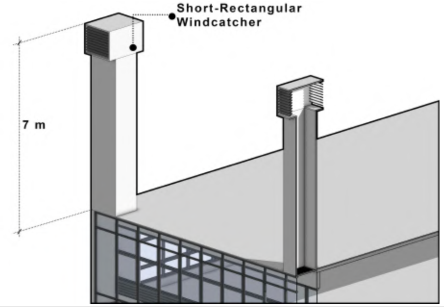
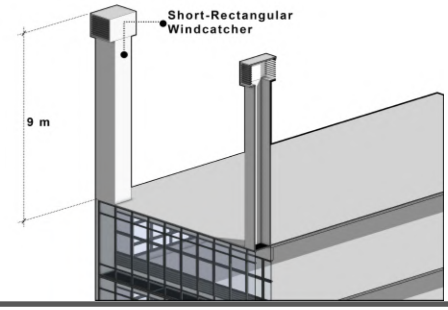
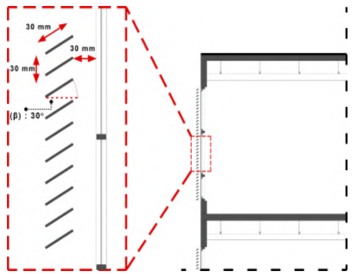
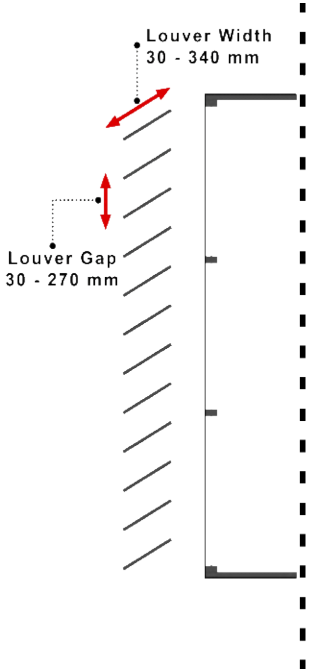
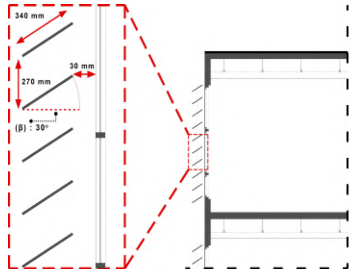
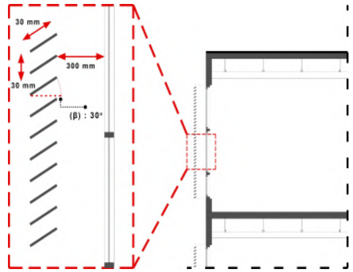
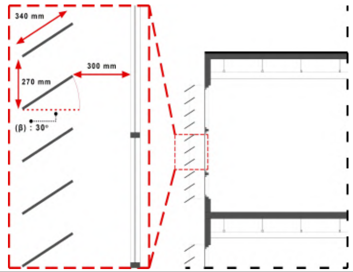
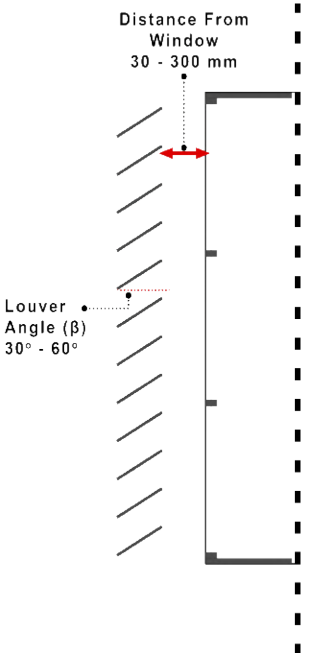
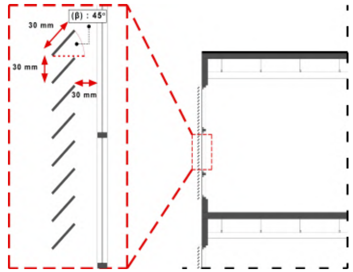
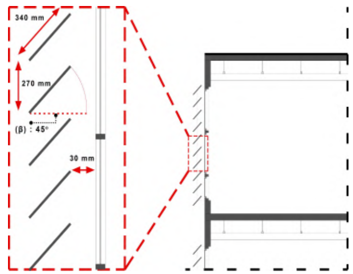
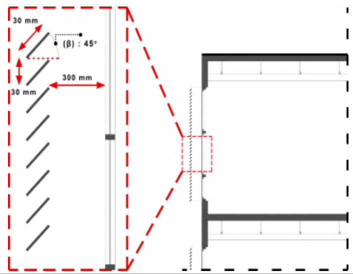
	Case A10	• Cavity Space Depth: 1.50 m • SHGC: 0.219 • U-Value: 2.8618 W/m^{2}K	
Case B (Windcatcher) 	Case B1	• Long-Rectangular Windcatcher • Wind Tower Height: 2 m	
	Case B2	• Long-Rectangular Windcatcher • Wind Tower Height: 3 m	

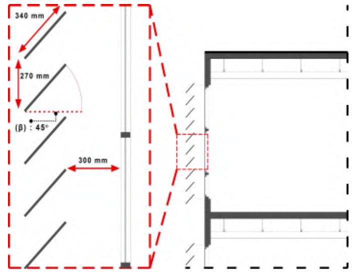
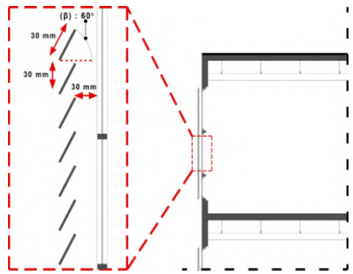
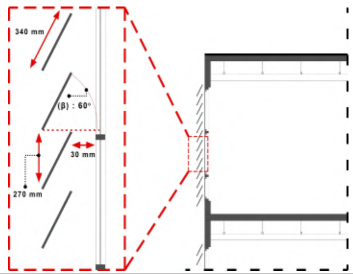
	Case B3	• Long-Rectangular Windcatcher • Wind Tower Height: 5 m	
	Case B4	• Long-Rectangular Windcatcher • Wind Tower Height: 7 m	
	Case B5	• Long-Rectangular Windcatcher • Wind Tower Height: 9 m	

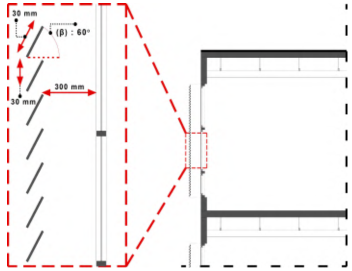
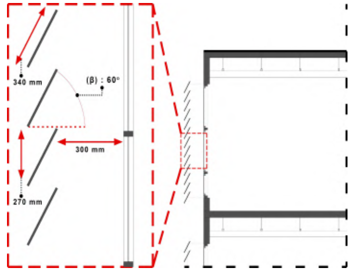
	Case B6	• Short-Rectangular Windcatcher • Wind Tower Height: 2 m	
	Case B7	• Short-Rectangular Windcatcher • Wind Tower Height: 3 m	
	Case B8	• Short-Rectangular Windcatcher • Wind Tower Height: 5 m	

	Case B9	• Short-Rectangular Windcatcher • Wind Tower Height: 7 m	
	Case B10	• Short-Rectangular Windcatcher • Wind Tower Height: 9 m	
Case C (Louvers)	Case C1	• Louver Gap: 30 mm • Width of Louver Slat: 30 mm • Louver Angle: 30˚ • Louver Distance from the Window: 30 mm	

	Case C2	• Louver Gap: 270 mm • Width of Louver Slat: 340 mm • Louver Angle: 30° • Louver Distance from the Window: 30 mm	
	Case C3	• Louver Gap: 30 mm • Width of Louver Slat: 30 mm • Louver Angle: 30° • Louver Distance from the Window: 300 mm	
	Case C4	• Louver Gap: 270 mm • Width of Louver Slat: 340 mm • Louver Angle: 30° • Louver Distance from the Window: 300 mm	

	Case C5	• Louver Gap: 30 mm • Width of Louver Slat: 30 mm • Louver Angle: 45° • Louver Distance from the Window: 30 mm	
	Case C6	• Louver Gap: 270 mm • Width of Louver Slat: 340 mm • Louver Angle: 45° • Louver Distance from the Window: 30 mm	
	Case C7	• Louver Gap: 30 mm • Width of Louver Slat: 30 mm • Louver Angle: 45° • Louver Distance from the Window: 300 mm	

	Case C8	• Louver Gap: 270 mm • Width of Louver Slat: 340 mm • Louver Angle: 45˚ • Louver Distance from the Window: 300 mm	
	Case C9	• Louver Gap: 30 mm • Width of Louver Slat: 30 mm • Louver Angle: 60˚ • Louver Distance from the Window: 30 mm	
	Case C10	• Louver Gap: 270 mm • Width of Louver Slat: 340 mm • Louver Angle: 60˚ • Louver Distance from the Window: 30 mm	

	Case C11	• Louver Gap: 30 mm • Width of Louver Slat: 30 mm • Louver Angle: 60˚ • Louver Distance from the Window: 300 mm	
	Case C12	• Louver Gap: 270 mm • Width of Louver Slat: 340 mm • Louver Angle: 60˚ • Louver Distance from the Window: 300 mm	

Following the definition of these variables, a sensitivity analysis was employed to quantify the hierarchy of influence for each design parameter (j) on total operational energy, using the range (R) method (Tian, 2013). This statistical filtering process effectively identified the high-leverage parameters essential for the subsequent integration phase. Phase 2 (holistic system integration) synthesized these optimized parameters into three distinct scenarios, optimal (scenario 1), moderate (scenario 2), and worst-case (scenario 3), to evaluate complex aerodynamic and thermodynamic synergistic effects. The final configuration was selected using a multi-criteria decision analysis (MCDA) framework, which balanced the minimization of sensible cooling EUI against the minimization of mechanical dehumidification penalties and secondary artificial lighting demands. This rigorous selection process ensures that the final design solution addresses the "sensible vs. latent" dichotomy by prioritizing strategies that reduce sensible loads without inadvertently triggering massive latent moisture intrusion or exacerbating auxiliary fan energy consumption (Mazzetto, 2025).

3.5. Optimization Framework: Objective Function and Design Constraints

The primary objective of this study is the minimization of total cooling EUI (kWh/m^2/year) across a defined parametric domain, subject to a set of multiple-dimensional performance constraints. Formally, the optimization problem is defined as:

The following design constraints govern solution feasibility: (1) spatial daylight autonomy according to sDA300/50%, as prescribed by LEED v4 daylighting credits (USGBC, 2019); (2) mean operative temperature (OT) $\leq$ 27.06°C, corresponding to the upper neutral comfort boundary under SNI 03-6572 and ASHRAE Standard 55 (PMV $\leq$ +0.5); and (3) mean indoor RH $\leq$ 60%, consistent with ASHRAE 62.1 indoor air quality limits. These thresholds represent non-negotiable habitability conditions – any configuration violating them is excluded from the feasible solution set regardless of its sensible EUI performance.

While this study employs a parametric sensitivity framework as opposed to an algorithmic metaheuristic optimization (e.g. genetic algorithm or simulated annealing), the combination of range-based sensitivity index, MCDA weighting, and pareto frontier analysis constitutes a structured multi-objective optimization procedure that is methodologically consistent with established building simulation literature. The legitimacy of applying MCDA as a structured selection mechanism within parametric building simulation is established in the literature - Balali et al. (2023) document its recognized application across passive energy optimization strategies, including natural ventilation envelopes and sun shading devices. Unlike algorithmic MCDA implementations, the weighted sensitivity ranking and pareto frontier analysis employed here constitute a transparent selection procedure – a deliberate methodological choice suited to the 32-case parametric domain and the interpretability requirements of thermodynamic boundary analysis.

To summarize the logic of the two-phase workflow: phase 1 constitutes a component-level parametric exploration where each passive element is assessed independently against the objective function. The MCDA sensitivity ranking then distils the dominant design parameters and informs scenario construction. Phase 2 validates the selected optimal, moderate, and worst-case configurations through full integrated AFN simulation – functioning as a thermodynamic confirmation stage rather than an independent optimization loop. This structure ensures that the phase 2 findings are grounded in empirically ranked parameters, rather than arbitrarily composed scenarios.

3.6. Data Analysis Method: Sensitivity and Thermodynamic Diagnostics

To rigorously quantify the hierarchy of design parameters and diagnose the physical boundaries of the passive system, the data analysis was conducted in two specific stages. First, To identify the dominant design parameters affecting EUI, range (R) sensitivity analysis was applied (Heiselberg et al., 2009; Tian, 2013). The mean performance value $(\overline{K}_{i,j})$ for factor j at level i is determined by the following equation (Eq. 1):

$$\overline{K}_{i,j} = \frac{1}{n}\sum_{k=1}^{n} EUI_{i,j,k} \qquad \textbf{(Eq. 1)}$$

where $EUI_{i,j,k}$ denotes the simulation result (total EUI) for the k - th experiment where factor j is at level i, and n is the number of replicates (Duan et al., 2025). Second, the sensitivity index of the range values (R_j) is subsequently derived by calculating the difference between the maximum and minimum average performance values (Eq. 2):

$$Rj = \max(\overline{K}_j) - \min(\overline{K}_j) \qquad \textbf{(Eq. 2)}$$

Third, To diagnose the thermodynamic limits and validate the decoupling hypothesis, a novel aerodynamic mass-balance assessment was integrated. This analysis evaluates whether the passive volumetric airflow, generated by the synergistic driving forces of the windcatcher and DSF cavity, is sufficient to passively extract internal heat gains without inducing a massive latent penalty. The adequacy of the aerodynamic flow is determined by calculating the required air change rate, derived from the fundamental steady-state sensible heat balance equation established in the ASHRAE Handbook (ASHRAE, 2021) (Eq. 3):

$$ACH_{required} = \frac{H_{internal} \cdot 3600}{\rho \cdot c_p \cdot \Delta T \cdot V} \qquad \textbf{(Eq. 3)}$$

Where V denotes the interior zone air volume (m^3), and $H_{internal}$ represents the accumulated sensible heat load (W), ρ is the air density (kg/m^3), c_p is the specific heat capacity of air (J/kg.K), ΔT is the indoor-outdoor temperature differential, and 3600 is the conversion factor seconds to hours. By comparing the simulated airflow network against this thermodynamic requirement, the "aerodynamic stagnation point" is calculated to determine the exact threshold at which the buoyancy-driven mechanism fails to overcome the system's internal hydrostatic equilibrium.

This failure condition results in inadequate sensible heat removal despite effective solar heat rejection by the external louvers, thereby forcing the auxiliary mechanical fans to engage. Conversely, when ACH_{actual} excessively surpasses $ACH_{required}$, it quantifies the latent penalty threshold – the exact point where the volume of unconditioned tropical air mathematically explodes the mechanical dehumidification load. This methodological innovation addresses a critical gap in the field of tropical building science by shifting the focus from aggregate energy savings to granular diagnosis of aerodynamic stagnation and latent load constraints, thus offering a novel perspective on the energy limits of passive design in hot-humid climates.

3.7. Phase 1: Component-Level Parametric Optimization

This phase involves an independent thermodynamic evaluation of each passive technology, with the objective of isolating the performance drivers and establishing the most effective geometric configuration prior to systematic integration. The assessment emphasizes the decoupling of conductive and radiative heat transfer while balancing the conflicting requirements of secondary artificial lighting demands and total operational cooling energy.

3.7.1. Baseline Building Performance Assessment

To establish a quantitative benchmark for passive optimization, the performance of the baseline model, defined by a standard unshaded envelope and sealed configuration, was first evaluated. The simulation results indicated a critical dependency on mechanical cooling and severe thermodynamic instabilities. The model registered a total EUI of 149.80 kWh/m^2/year, driven almost exclusively by the cooling load (146.51 kWh/m^2/year). This disproportionate energy demand indicates substantial heat ingress through conductive and radiative pathway owing to the lack of external shading.

Furthermore, the indoor environment was found to be highly susceptible to severe tropical heat and moisture accumulation. The reliance on uncontrolled infiltration (mean ACH: 0.0439) failed to provide aerodynamic stability, resulting in a persistent latent heat gain of 3740.18 W and mean RH of 50.68%. This establishes a baseline condition in which the mechanical dehumidification system operates under continuous stress. The energy and mass-balance metrics enumerated in Table 10 function as the thermodynamic comparative standards for the ensuing optimization cases.

Table 10. Baseline building model simulation results

Metrics	Value	Unit
Distric Cooling	351,629.18	kWh/year
Cooling Intensity	146.51	kWh/m^2/year
Interior Lighting	7,890.25	kWh/year
Lighting Intensity	3.29	kWh/m^2/year
Total Building Energy	359,519.69	kWh/year
Total EUI	149.80	kWh/m^2/year
Mean RH	50.68	%
Mean ACH	0.0439	ach
Peak ACH	1.35	ach
MRT / OT	29.64 / 27.06	° (degree) C (Celsius)
Latent Gain	3,740.18	Watt

3.7.2. DSF Parametric Analysis

The parametric optimization of the DSF system was undertaken to thoroughly examine the interplay between cavity depth variation (ranging from 0.25 to 1.50 m) and glazing thermal properties. The analysis identifies case A10 (1.5 m cavity depth; SHGC: 0.219; U-Value: 2.8618 W/m^2K) as the global optimum for total energy performance. This finding calls into question the long-standing assumption that minimizing U-Value is the sole factor determining efficiency. Conversely, the methodology stipulates that this configuration enhances the convective h_c through TARP algorithm, leveraging the stack effect to passively exhaust address heat accumulation within the deep cavity.

- **Aerodynamic Synergy and Sensible Cooling Extraction**

The thermodynamic behaviour of the DSF is characterized by a distinct non-linear trajectory. As demonstrated in Figure 15, the enhancement in cavity depth demonstrates a decreasing return on cooling load reduction. The simulation data indicates that extending the cavity to 1.5 m (cases A9 and A10) effectively minimizes the cooling load to approximately 90.05 – 90.39 kWh/m^2/year. A comparison of the baseline model's cooling intensity of 146.51 kWh/m^2/year to the optimized DSF configuration reveals a substantial reduction in sensible cooling load of approximately 38.3%. This reduction magnitude aligns with the findings of Ayegbusi et al. (2017) in a similar Malaysian tropical context. Specifically, the cooling load trendline (see Figure 15) exhibits a diminishing return characteristic beyond 1.2 m depth. This phenomenon is analogous to Ayegbusi's observation regarding the stabilization of the thermal buffer effect in naturally ventilated cavities, where increasing the air gap yields progressively smaller marginal gains in heat removal. In addition, the predominance of thermal transmittance reduction over solar heat gain control in this deep-cavity configuration corroborates the earlier research of Mulyadi et al. (2011) concerning skin load performance in Indonesia.

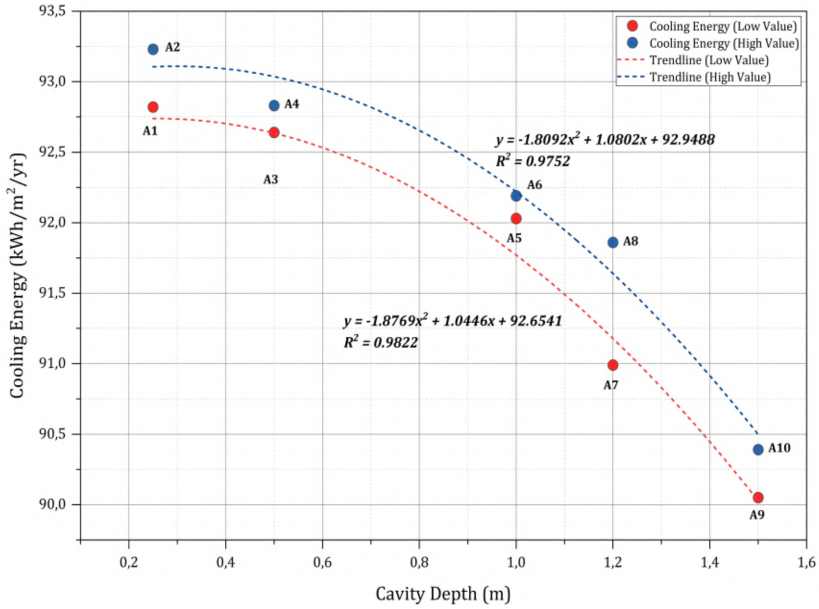


Figure 15. The polynomial trend in cooling energy

To physically substantiate this cooling load reduction, an evaluation of the aerodynamic mechanism is imperative. The thermodynamic synergy between glazing optical properties and cavity geometry is found to be a fundamental factor in determining the buoyancy-driven heat removal capacity in DSF. As demonstrated in Figures 16 and 17, configurations employing low thermal properties (specifically low SHGC) glazing function as an active thermal shield. These external skins intercept shortwave solar radiation, thereby localising thermal accumulation within the cavity space (exceeding 31°C). The localized heating serves to amplify the temperature gradient, thus driving a robust volumetric exhaust mechanism that scales dynamically with cavity expansion. The underlying physical behaviour conforms to the buoyancy-driven natural ventilation model for large openings (Li et al., 2007), where the airflow rate is governed by the following formulation:

$$Q = C_d \cdot A\sqrt{\frac{2g \cdot \Delta H.(T_{cavity} - T_{out})}{T_{out}}} \qquad \textbf{(Eq. 4)}$$

In this formulation, the volumetric airflow rate (Q; m^3/s) is determined through the interaction of three distinct physical domains. The aerodynamic flow resistance is governed by the discharge coefficient (C_d) and the effective cross-sectional area of the openings (A;

m²). Concurrently, the hydrostatic driving force is defined by the vertical separation between the inlet and outlet nodes (ΔH; m) under constant gravitational acceleration (g; 9.8 m/s²). It is crucial to note that the intensity of this exhaust mechanism is contingent on the thermodynamic gradient that is generated between the localized mean air temperature within the cavity (T_{cavity}; K) and the prevailing outdoor environmental conditions (T_{out}; K).

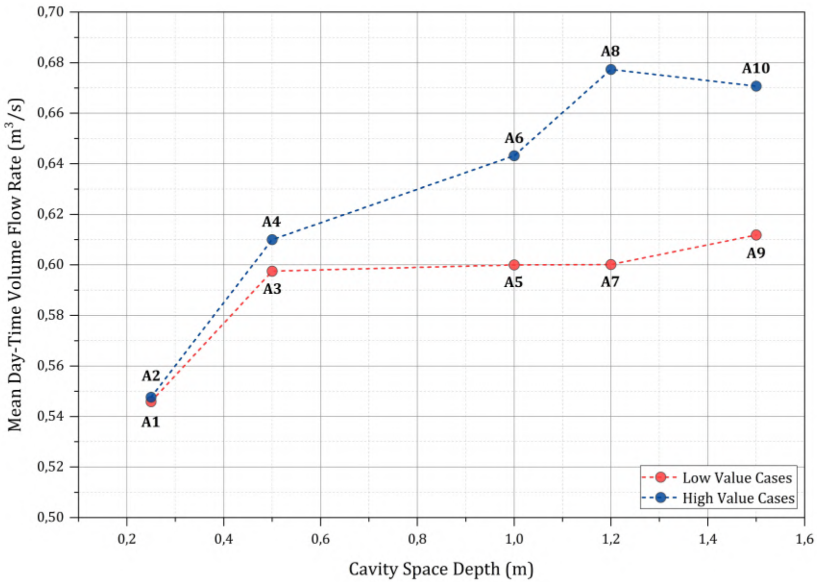


Figure 16. Mean day-time airflow rate

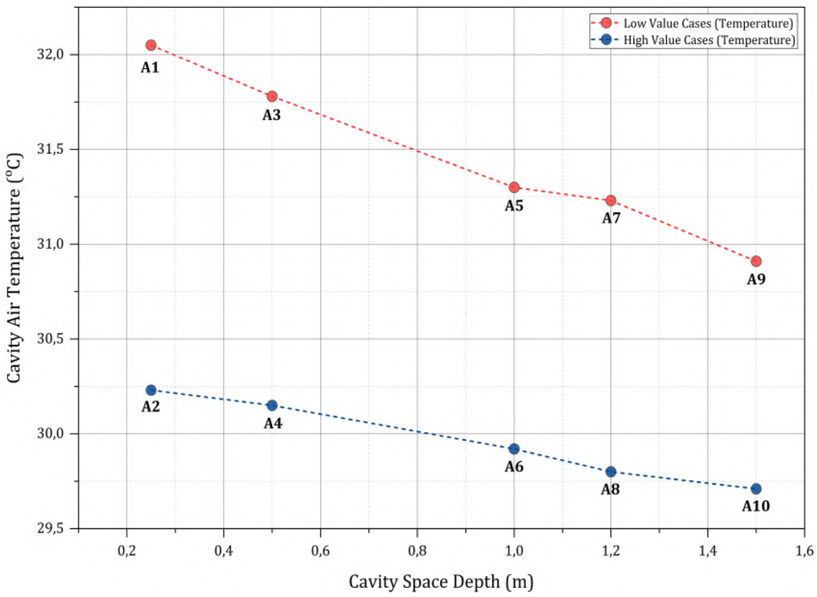


Figure 17. Mean day-time cavity air temperature

- **Transient Aerodynamic Validation: The Buoyancy-Driven Stack Effect**

To validate the mathematical premise of equation 4, a transient analysis of the AFN behaviour during peak solar periods was conducted. The objective of this study is to provide a comprehensive understanding of the internal thermodynamics of the cavity, prior to conducting an evaluation of its aggregate impact on the Total EUI. As demonstrated by the dual-panel correlation in Figure 18, there is a direct physical dependency between incident solar radiation, cavity temperature, and volumetric mass flow rate. During the morning hours, the cavity airflow remains relatively stagnant owing to an absence of a sufficient temperature differential (ΔT) between the inner boundary layer and the outdoor environment. However, as direct solar irradiance aggressively rises and approaches its midday peak, the high-absorptance external skin localises this sensible heat. The localized thermal accumulation triggers a hydrostatic pressure drop. As stipulated by Bernoulli principle within the AFN nodes, this pressure differential exerts an upward force on the lower-density hot air, thereby accelerating the rate of cavity exhaust rate to a maximum of 0.20 m^3/s during the late morning period (approx. 11.00), as clearly illustrated in the transient aerodynamic profile (case A10) (see Figure 18). Subsequently, a distinct thermal lag phenomenon is observed; while exhaust airflow diminishes post-midday as the initial thermal gradient stabilizes, the cavity air temperature continues to elevate due to the envelope's thermal mass retention, reaching its peak of 32.02°C later in the afternoon at 16.00.

This transient profile explicitly proves that the widening of the cavity space up to 1.5 m leads a reduction in aerodynamic boundary layer friction, thereby optimising the effective discharge area (A) and successfully decoupling sensible heat from the occupied zone across the entire geometric spectrum. This macroscopic trend is further validated by Figure 16 and 17, which illustrate the mean day-time airflow rates and cavity air temperatures, respectively. These aggregate metrics confirm that the transient aerodynamic superiority of the 1.5 m cavity consistently translates into optimal daily performance compared to narrower configurations. To quantitatively substantiate this aerodynamic behaviour, Table 11 presents the mean daytime volumetric throughput and flow directionality for all ten DSF configurations. The data reveals a fundamental bifurcation governed by glazing selection rather than cavity depth. Low-SHGC configurations exhibit a consistent dominant upward net flow (supply mode: 62.6 – 73.2%), which confirms that the thermal shield effect generates persistent, buoyancy-driven stack exhaust. Conversely, high-SHGC configurations exhibit a net downward return flow (exhaust mode: 60.1 – 63.3%), suggesting that solar radiation entering the cavity directly reverses the dominant flow direction. Interestingly, mean daytime throughput increases monotonically with cavity depth for both glazing types. It ranges from 0.5402 to 0.6123 m^3/s for low-SHGC cases and from 0.5464 to 0.6730 m^3/s for high-SHGC cases, which corroborates the polynomial trendline in Figure 15.

Table 11. Mean daytime volumetric flow rate and flow directionality across DSF parametric configurations

Case ID	Depth (m)	Glazing	Mean Daytime Q (m^3/s)	Mean Net Flow (m^3/s)	Supply Mode (%)	Exhaust Mode (%)
A1	0.25	Low-value (SHGC: 0.174)	0.5402	+0.4415	73.20	26.80
A2	0.25	High-value (SHGC: 0.219)	0.5464	-0.0639	36.72	63.28
A3	0.50	Low-value (SHGC: 0.174)	0.5940	+0.4624	72.48	27.52
A4	0.50	High-value (SHGC: 0.219)	0.6108	-0.0563	38.31	61.69
A5	1.00	Low-value (SHGC: 0.174)	0.5990	+0.3797	69.58	30.42
A6	1.00	High-value (SHGC: 0.219)	0.6445	-0.0604	39.92	60.08
A7	1.20	Low-value (SHGC: 0.174)	0.6004	+0.3722	69.83	30.17
A8	1.20	High-value (SHGC: 0.219)	0.6706	-0.1001	37.11	62.89
A9	1.50	Low-value (SHGC: 0.174)	0.6123	+0.2402	62.61	37.39

A10	1.50	High-value (SHGC: 0.219)	0.6730	-0.1015	37.66	62.34

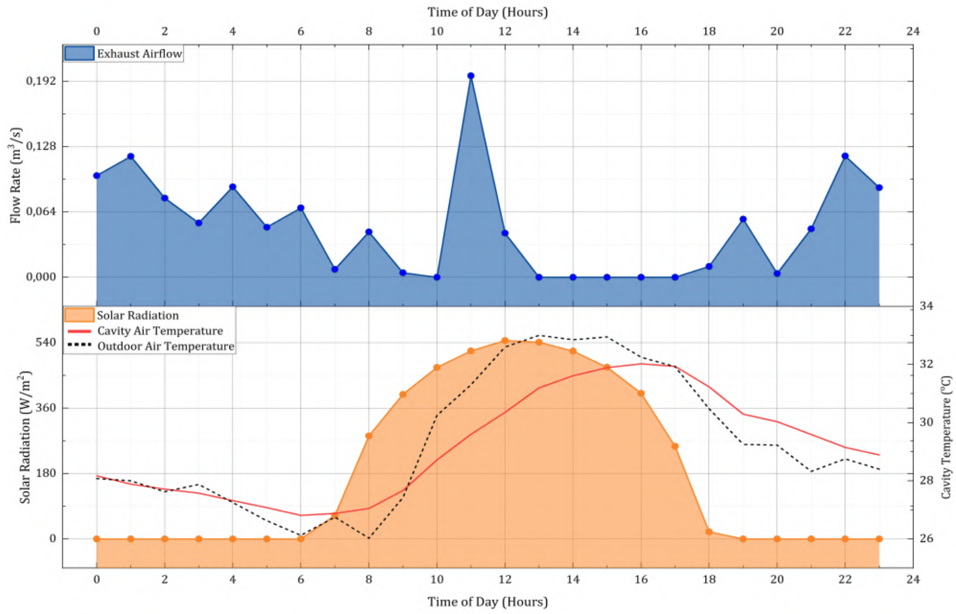


Figure 18. Transient thermodynamic correlation demonstrating the buoyancy-driven stack effect and cavity exhaust acceleration during the peak solar day

Beyond geometric optimization, the correlation between cavity temperature and indoor cooling reveals a distinct thermodynamic paradox driven by glass selection (see Figure 19). In configurations employing low SHGC (the “odd cases”), the external skin functions as a thermal shield. It intentionally absorbs and captures solar radiation, leading to a warmer cavity, while effectively shielding the occupied zone from radiant heat gain. Conversely, high SHGC glazing material (the “even cases”, such as case A10) precipitates the opposite effect. This high-SHGC glazing cnfiguration functions by transmitting radiation directly into the interior, thereby maintaining a relatively cooler aerodynamic cavity (approximately 32.02˚C during peak daytime). However, this process results in an exponential penalty on the building’s internal cooling load. This aerodynamic stabilization across expanding cavities corroborates the empirical trajectories validated by Aziiz et al. (2017), confirming that maximizing natural ventilation in a DSF relies primarily in the absorptive capacity of the external envelope to generate the required thermal driving force.

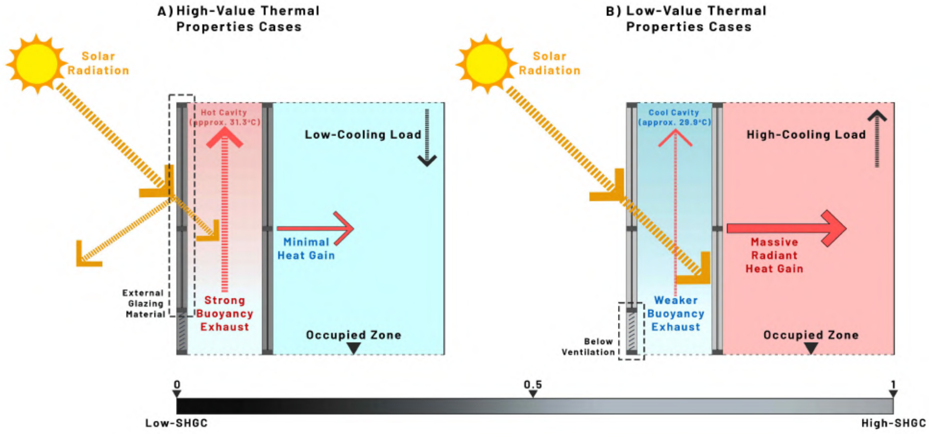


Figure 19. Thermodynamic mechanism of the DSF thermal paradox

- **The Visual-Thermal Trade-off and Optical Translation Paradox**

Concurrent with the process of thermal optimization, the integration of a DSF invariably introduces a quantifiable visual penalty. This can be calculated by measuring the increase in auxiliary lighting loads relative to the baseline building model. As demonstrated by Tong et al. (2021), the pursuit of thermodynamic isolation frequently necessitates the compromise of DA. This critical trade-off is validated by the simulation; even with the optimal architectural configuration, the interior lighting demand in the DSF models represents an increase exceeding 130% compared to the baseline building model's negligible lighting load of 3.29 kWh/m^2/year (see Figure 20). However, in contrast to the prevailing hypothesis that deeper cavities linearly intensify this penalty, the simulation reveals a paradoxical synergic relationship. As demonstrated in the lighting trendline (Figure 20), expanding the cavity geometry to 1.5 m results in a progressive reduction in the lighting load (e.g., decreasing from 8.88 to 7.84 kWh/m^2/year in the low-SHGC configurations). To rigorously isolate and validate this anomaly without optical redundancy, a multi-axis combo chart detailing the sDA trajectory alongside the grouped lighting loads was analysed (see Figure 21). A notable paradox emerged: as the cavity expanded, the sDA declined from 58.20% to 42.81% yet the artificial lighting load concurrently improved. The recorded trajectory deviates entirely from conventional daylighting principles, where lower visual autonomy typically demands higher artificial illumination.

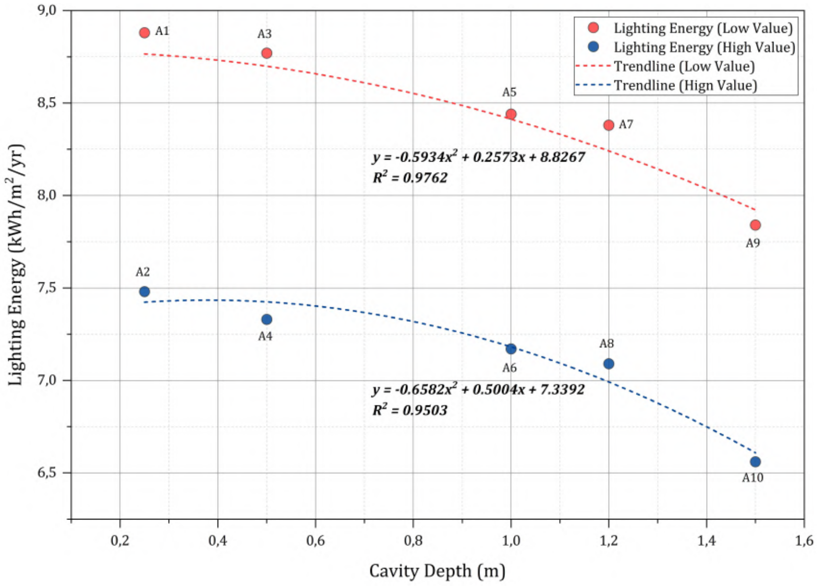


Figure 20. The trend illustrates a significant visual penalty

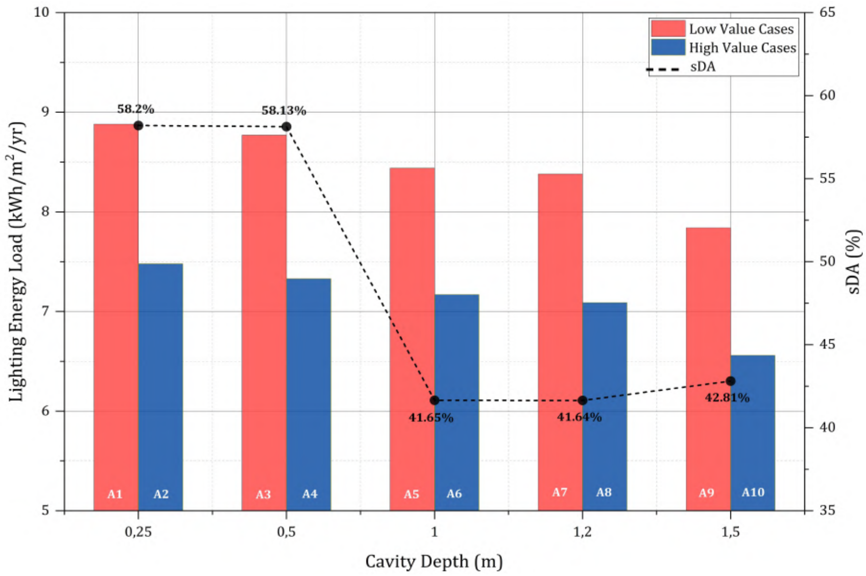


Figure 21. Divergence of sDA and lighting energy across DSF cavity depths

This optical phenomenon can be attributed to the fact that the 1.5 m cavity floor functions mechanistically as an external light shelf (see Figure 22). As asserted by Jung and Lee (2023), augmenting the horizontal depth of a light shelf effectively intercepts direct solar penetration, thereby reducing the over-lit area that exceeds 300 lux, as measured by the sDA. Simultaneously, this augmentation reflects diffuse daylight deeper into the occupied zone, thus compensating for the obstructed sky view factor. This sub-threshold diffuse illuminance (e.g., 150 – 120 lux) is categorized as insufficient by the rigid binary metric of sDA. Nevertheless, it is noteworthy that the device is highly effective in triggering continuous photoelectric dimming sensors, thereby optimizing the total lighting energy load. Consequently, the combination of geometric expansion and selective material transmittance not only buffers thermal transfer but also fundamentally repurposes the cavity structure into a daylight-redirecting device.

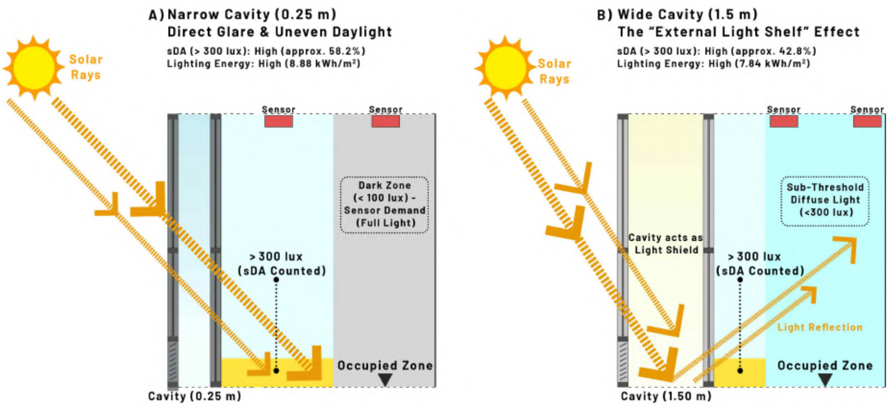


Figure 22. Divergence of sDA and lighting energy across DSF cavity depths

- **Aggregate Energy Performance and Pareto Optimality**

Despite the imposed lighting penalty, the aggregate energy analysis indicates that the cooling savings significantly outweigh the losses in visual performance. As illustrated in Figure 23, case A10 exhibits the most substantial total energy saving potential, with an EUI of 96.94 kWh/m^2/year. This results in a net annual energy reduction of 35.3% compared to the baseline model (149.80 kWh/m^2/year). It is important to note that the analysis emphasises a critical distinction in glass selection. While case A9 (SHGC: 0.174; U-Value: 0.345 W/m^{2}K) exhibited marginally superior cooling performance (90.05 kWh/m^2/year) compared to case A10 (90.39 kWh/m^2/year), it incurred a higher lighting penalty (7.84 kWh/m^2/year) owing to its reduced visual transmittance. Consequently, the trendline confirms that continuing to aggressively block solar radiation beyond the case A10 configuration yields diminishing returns for the total building energy profile.

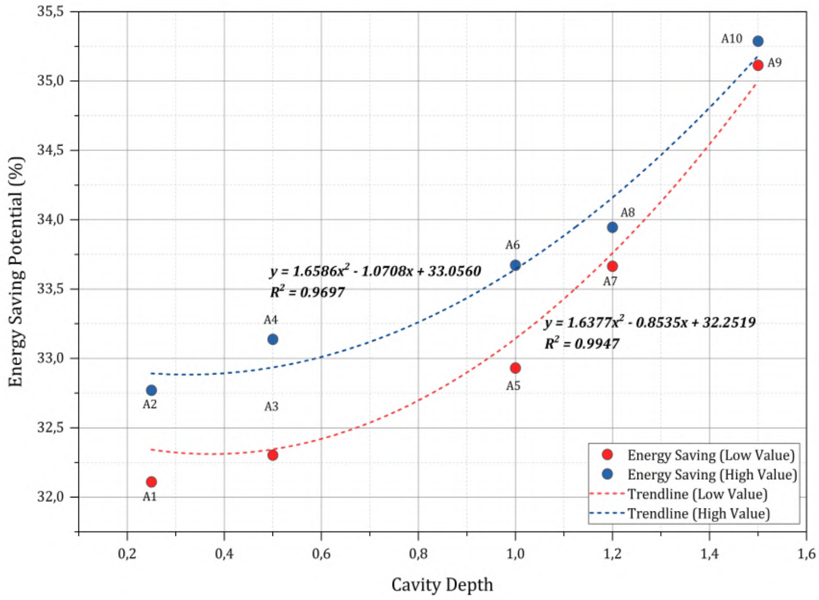


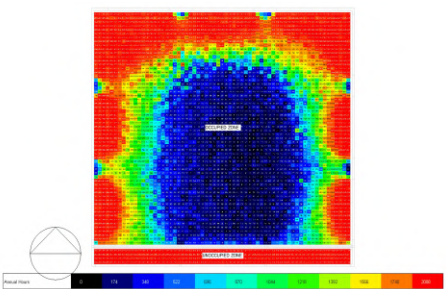
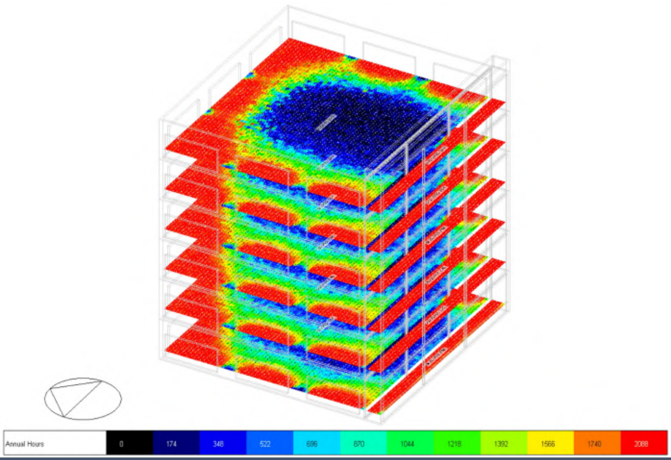
Figure 23. The trendline indicates a positive correlation between cavity depth and net energy savings, peaking at the 1.5 m configuration

- **The Sensible-Optical Equilibrium (Validation of the Tipping Point)**

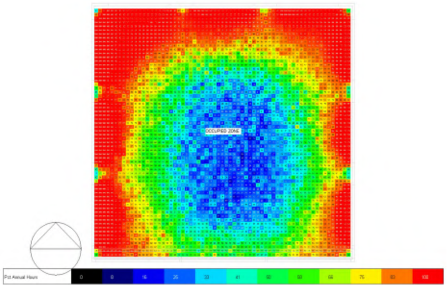
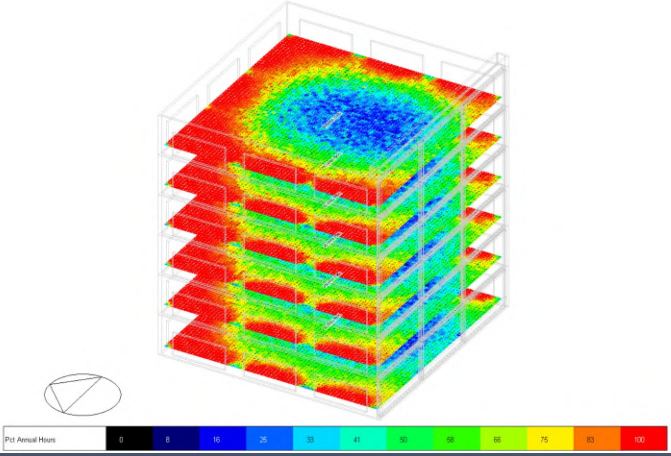
The selection of case A10 as the global optimum necessitates a critical evaluation of the envelope thermodynamic limits. Case A1 (0.25 m cavity space) exhibited a marginally lower thermal transmittance rate into the occupied zone in comparison to the high U-Value (2.8618 W/m^{2}K) glazing utilized in case A10. However, the dual energy breakdown validates that the prioritisation of extreme sensible isolation is mathematically counterproductive to the total EUI goal. This phenomenon aligns with the findings of Barbosa et al. (2015), which substantiate that achieving optimal energy performance in tropical regions frequently requires a judicious trade-off between conductive heat gain and lighting load reduction. The simulation indicates that the negligible rise in sensible heat ingress observed in case A10 is entirely counterbalanced by its exceptional diffuse light transmission properties. This specific material-geometry configuration involves the sacrifice of a fraction of conductive insulation to numerically trigger the photoelectric dimming sensors, thereby reducing the secondary artificial lighting demand. By achieving a net saving of 4.76 kWh/m^2/year in comparison with case A1, case A10 demonstrates that attaining energy efficiency in tropical DSFs is not contingent upon the creation of a perfect thermal vault. Rather, the establishment of a precise equilibrium is paramount, wherein sensible cooling gains perfectly offset the inevitable optical and latent penalties.

Case A10 was thus identified as the optimal holistic scenario, demonstrating clear dominance over the optimization landscape when visual metrics are considered in conjunction with thermal performance. This delicate equilibrium is illustrated in the DSF pareto landscape (see Figure 25), which maps the sDA against the total EUI. The chart provides a clear visual representation of the pareto frontier, with cases A10, A4, and A2 positioned accordingly. Case A10 is a paradigm of the ideal energy compromise (96.94 kWh/m^2/year), accomplished through the acceptance of a suppressed sDA of 42.81%. This is in stark contrast to the other frontier points, cases A4 and A2, which maintain a high sDA300/50% of >55% (USGBC, 2019), yet experience an energy penalty of approximately 3.2 to 3.8 kWh/m^2/year relative to case A10. This distribution provides empirical evidence that optimising energy efficiency in high-temperature climates necessitates a deliberate, calculated trade-off in daylight autonomy (DA) (Ardiani and Koerniawan, 2017; Wong et al., 2008). The spatial distribution of this suppressed daylight autonomy across the DSF parametric configurations is visualized in Figure 24, confirming that the sDA gradient follows the cavity depth trajectory – deeper cavities yield lower sDA as the cavity floor increasingly functions as a light shelf that intercepts direct solar penetration.

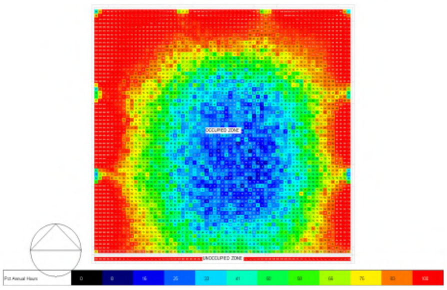
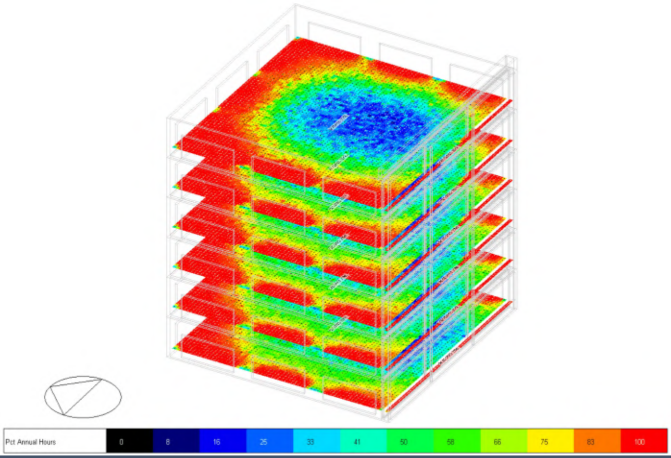
Energy Optimal Case A10 (sDA: 42.81%)

Floorplan	Isometry
	

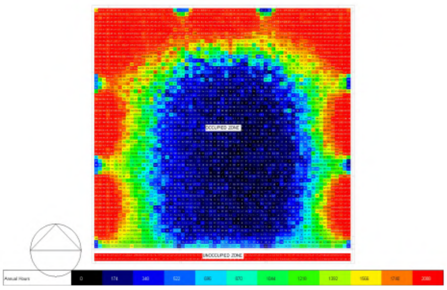
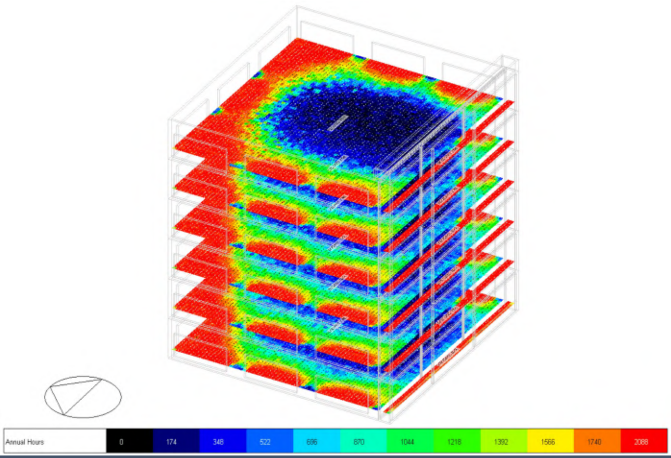
sDA Optimal Case A2 (sDA: 58.20%)

Floorplan	Isometry
	

Case A4 (Pareto Frontier) (sDA: 58.13%)

Floorplan	Isometry
	

Moderate Performance Case A5 (sDA: 41.65%)

Floorplan	Isometry
	

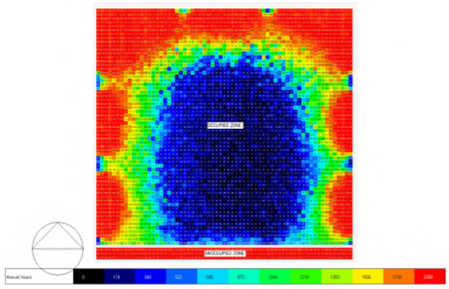
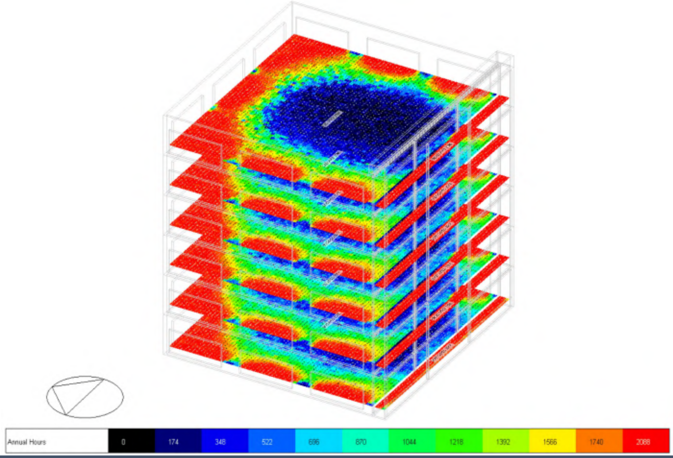
Worst Case A8 (sDA: 41.64%)	
Floorplan	Isometry
	

Figure 24. sDA300/50% false-colour distribution across phase 1 DSF parametric configurations

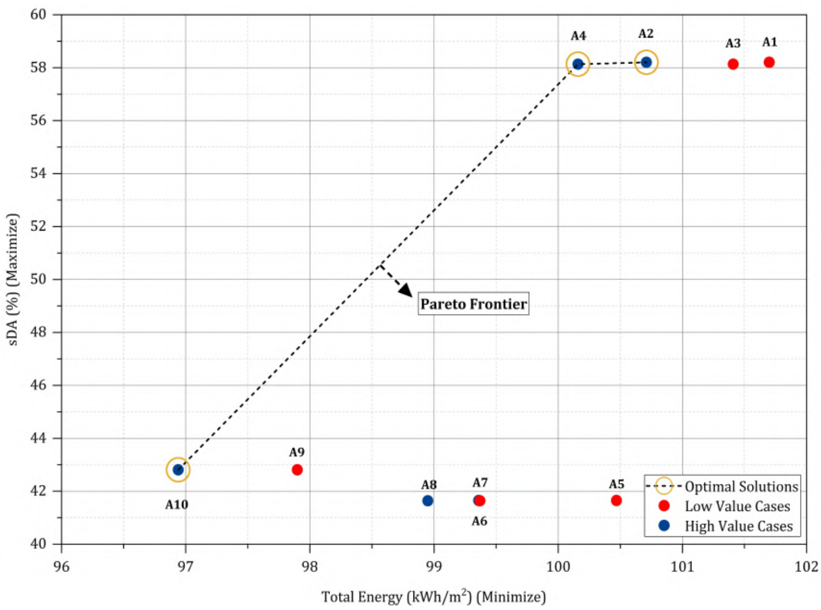


Figure 25. The scatter plot identifies case A2, A4, and A10 on the pareto frontier

To contextualize magnitude of the above findings, Tabel 12 benchmarks the performance of case A10 against comparable DSF optimization studies conducted in tropical or humid climates, as well as a number of other climatic conditions. This comparison confirms that while DSF applications have been predominantly evaluated in temperate contexts, the sensible cooling reductions documented here are empirically consistent with –

and in several instances exceed – outcomes reported in similar Southeast Asian and equatorial studies.

Table 12. Benchmarking of DSF A10 cooling reduction against comparable climate studies

Study	Climate / Location	Optimal Cavity	Other Variables	Cooling Reduction / Other Results	Key Findings
Bahdad et al. (2024)	Tropical (Penang, Malaysia)	0.5 m depth	Glazing materials (U-Factor: 1.66 – 1.80 W/m^2K); SHGC (0.29 – 0.63); Tvis: 40.9 – 75.4	Cooling load reduced by 26.20% (-86.88 kWh). Total EUI reduced by 25.33%	Combination of DSF-IGUs and dynamic louvers significantly enhances thermal comfort and reduces cooling needs
Lim and Ismail (2023)	Tropical (Kuala Lumpur, Jakarta, Phnom Penh, Nha Trang)	1.0 m depth	Orientations (N, E, S, W)	Cooling energy reduced by > 40% (e.g., Kuala Lumpur) – (SSF: 1665 MWh to DSF: 865 MWh)	Even without shading devices, DSF outperforms SSF in all tropical classifications
Ayegbusi et al. (2017)	Hot-Humid (Malaysia)	0.6 m depth	Orientations (N, E, S, W)	~ 48% reduction (East: dropped from 418.16 to 215.65 kWh/m^2)	A 0.6 m ventilated cavity provides the most efficient reduction in cooling load and OTTV
Dewi et al. (2013)	Hot-Humid (Taiwan)	1.2 m depth	Glass: clear, reflective, laminated, double glazed Low-E	Up to 34.69% cooling energy reduction	Wider air gaps combined with Low-E glass minimizes solar heat transmission and cooling energy
Ardiani and Koerniawan (2017)	Tropical (Tangerang, Indonesia)	Not mentioned specific depth	Perforated metal 50%, single, double, triple glazing – (triple glazing best); SHGC (0.00 – 0.86); U-Value	100,374 kWh (triple glazing) vs. perforated metal (126,674)	Perforated metal without natural ventilation increases cooling load compared to tripe

			(1.5330 – 6.7069 W/m^{2}K)		glazed-DSF
Mulyadi et al. (2011)	Equator / Hot-Humid (Indonesia)	2.00 m depth	Glazing material thickness (6,8 10, 12 mm); Properties (U-Value: 2.19 – 3.6 W/m^{2}K; SC: 0.20 – 0.52)	12% reduction in annual total energy consumption; 10 -11% reduction in total heat load	Thicker glass reduces solar heat gain; wider distance lowers thermal transmittance without causing condensation
Qahtan (2019)	Tropical (Kuala Lumpur)	1.2 m depth	-	N/A (Field measurement) ; Temp: Indoor temp. maintained at 25.3-26.4 ˚C; West solar radiation up to 236.4 W/m^2	Existing DSF requires advanced solar control on West façades to protect against direct tropical solar radiation
Aziiz et al. (2018)	Tropical Humid (Bandung, Indonesia)	0.5 m depth	One-way silvery glass	N/A (Focus on temperature difference)	0.5 m gap showed the lowest deviation (8.92˚C) to prevent overheating
Aziiz et al. (2017)	Tropical Humid (Bandung, Indonesia)	0.5, 1.0, 1.5 m air gap width	One-way silvery glass	N/A (Focus on fluid dynamics)	Natural ventilation in high-rise building is strongly related to the air gap distance to generate buoyancy
Barbosa et al. (2015)	Tropical Humid (Rio de Janeiro)	Cavity (25 – 100 cm); Tapered cavity	Shading (none, concrete, aluminum)	N/A (Focus on thermal comfort) – thermal comfort acceptance improved to 70.6%	Internal shading device and tapered cavity are the most influential parameters to maximize annual comfort
Pomponi et al. (2017)	Tropical Humid (Rio de Janeiro)	1.5 m depth	Orientations (N, NE, NW, S, SW, SE)	N/A (Focus on adaptive comfort); Comfortable	DSF provides significant comfortable indoor

				hours achieved for 61%	conditions passively for large portions of the year
Yasa (2015)	Mediterranean (Turkey)	1.60 m depth	Aperture settings	N/A (Focus on PMV/PPD); Wind speed increases from 5 m/s (0.8 m depth) to 7 – 8 m/s (1.60 m)	The distance between envelopes directly influences atrium natural ventilation & PMV-PPD comfort levels
Radhi et al. (2013)	Hot Arid (UAE)	0.5 – 1.5 m	Glazing (single, double, triple); Orientations	17 – 20% cooling energy savings on a typical summer	Outer screens heavily impact convective heat transfer
Kang (Kang et al., 2019)	Temperate (Yongin, South Korea) - Summer	20 mm depth	Clear glass 6 mm; Low-E glass 6 mm; Argon 12 mm; U-Value: 1.38 W/m^2K; SHGC: 0.50	Summer cooling load decreased by 57.1%; Peak temperature reached 36˚C (cavity)	Slim DSF effectively exhaust cavity heat and significantly reduces summer cooling loads
Khoshbakht et al. (2017)	Temperate (Shefield, UK) & Subtropical (Queensland, Australia)	500 mm depth	Glazing properties: Y-Value = 1.5 W/m^2K; Shaft box window façade vs. Triple-glazed windows	51% energy savings (Temperate); 16% energy savings (Subtropical)	DSFs yield considerable 16% energy savings in subtropical climates; though more suited for temperate regions
This Study - Case A10	**Tropical Equatorial (Jakarta, Indonesia)**	**1.5 m depth**	**SHGC: 0.219; U-Value: 2.8618**	**38.3% cooling reduction**	**Conductive pathway dominance confirmed via three-pathway decomposition**

3.7.3. Windcatcher Parametric Analysis (Thermal Stack Tower Characterisation)

Subsequent to the thermal decoupling achieved by the DSF, the second stage of optimization concentrated on the aerodynamic performance of the roof-mounted windcatcher. The parametric sweep investigated the interplay between tower topology (long-rectangular vs. short-rectangular) and vertical elevation (2 to 9 m). To rigorously quantify this aerodynamic contribution, the airflow performance was evaluated using component-level monitoring at the AFN Link node, which corresponds to the transition from the tower to the occupied zone. It is imperative to differentiate this metric from a whole-building balance: while a whole-building measurement would yield higher ACH values owing to envelope infiltration and fenestration, such a metric would obscure the specific contribution of the windcatcher. Therefore, the performance metrics reported in this study are indicative of the net pure volumetric airflow generated exclusively by the device. This effectively isolates the aerodynamic efficiency of the windcatcher from external noise, such as that resulting from building leakage or fenestration infiltration. This sealed baseline configuration – a building model devoid of windcatcher – registered a mean daytime operational infiltration ACH of 0.0439, a condition of near-aerodynamic stagnation driven exclusively by uncontrolled envelope leakage. This value establishes the comparative thermodynamic benchmark against which all ten windcatcher parametric cases are evaluated in the ensuing analysis. The parametric evidence presented in this section – specifically the dominant of buoyancy-driven nocturnal exhaust over wind-induced supply as the primary energy contributor – substantiates a recharacterization of this device. Hereafter, the element is designated the TST, a term that accurately reflects its empirically confirmed operative mechanism.

- **Fundamental Aerodynamic and Physical Mechanism**

The computational airflow domain comprises two interactive pressure zone: the wind-chimney (TST shaft) and the occupied zone (Figure 26). Mass transfer between them is not a simple linear conduit; it is governed by the pressure differential (ΔP) between wind-driven dynamic pressure at the inlet and buoyancy-driven stack effect within the shaft (Hunt & Linden, 1999). At the roof-level inlet, the effective capture area (A_{inlet}) varies parametrically between 1 x 20 m (long-rectangular) and 1 x 1.2 m x 3 units (short-rectangular), with the outdoor wind velocity governed by the logarithmic boundary layer profile (V_{out} = 0.8 – 2.0 m/s). Following standard wind-driven ventilation principles (ASHRAE, 2021), the volumetric air input is physically expressed in Eq. 5:

$$Q_{in} = C_d \cdot A_{inlet} \cdot V_{out} \qquad \textbf{(Eq. 5)}$$

Where C_d represents the aerodynamic discharge coefficient of inlet openings. Once captured, the downward kinematic flow is subsequently modified by internal shaft friction and the thermal gradient across the 2 – 9 m vertical drop. The airflow transfer into the occupied zone strictly follows the fundamental principle of mass continuity incompressible flow (Çengel & Cimbala, 2006), formulated in Eq. 6:

$$A_1 \cdot V_1 = A_2 \cdot V_2 \qquad \textbf{(Eq. 6)}$$

In the simulation environment, the terminal openings transferring air from the shaft to the occupied zone are modelled as ventilation grilles. To account for the local acceleration and aerodynamic resistance induced by the grille components, the continuity principle is modified using a discharge coefficient, following standard space air diffusion practices (ASHRAE, 2021). The resultant ventilation velocity passing through the terminals (V_{vent}) is determined by Eq. 7:

$$V_{vent} = \frac{Q_{shaft}}{C_{d(vent)} \cdot A_{gross}} \qquad \textbf{(Eq. 7)}$$

Where A_{gross} is the total geometric area of the terminal opening and $C_{d(vent)}$ represents the aerodynamic discharge coefficient of the ventilation grille. However, the magnitude and direction of V_{vent} are not static; they are dynamically governed by the net pressure differential within the 1-m wide vertical wind-chimney. Fundamentally, the downward kinetic energy supplied by the wind must continuously overcome the opposing upward buoyancy force ($\Delta P_{thermal}$) generated by indoor heat accumulation. This continuous pressure balance mathematically dictates the real-time mass transfer rate into the occupied zone, establishing the baseline aerodynamic boundary of the model.

Instead of treating the boundary conditions as an isolated linear conduit, the simulation employs the AFN nodal calculation to resolve the global aerodynamic behaviour. The AFN model abstract the complex 3D building geometry into a macroscopic network of pressure nodes connected by airflow linkages. To calculate the definitive mas flow rate ($\dot{m}$) across all global linkages – ranging from the roof-level wind tower inlets to the perimeter windows – the simulation strictly employs the empirical power-law relationship, as governed by the standard EnergyPlus Airflow Network engine (U.S. Department of Energy, 2014) (Eq. 8):

$$\dot{m} = C_Q \cdot (\Delta P)^n \qquad \textbf{(Eq. 8)}$$

Where C_Q is the air mass flow coefficient of the specific linkage, n is the flow exponent, and ΔP is the net pressure differential. The algorithm distinguishes between the varied architectural elements inherently through the assignment of the flow exponent (n); large structural openings such as windcatcher grilles are governed by a fully turbulent exponent of 0.50, whereas narrow envelope gaps operate at an exponent of 0.65. By evaluating the dynamic equilibrium between wind-derived from the EPW data and the opposing internal thermal buoyancy (ΔP), this iterative solver simultaneously calculates the steady-state mass balance. Ultimately, this mathematical execution yields airflow rates (m^3/s) and air changes per hour (ACH), which are directly coupled into the thermal energy engine to determine the cooling energy savings of the passing system.

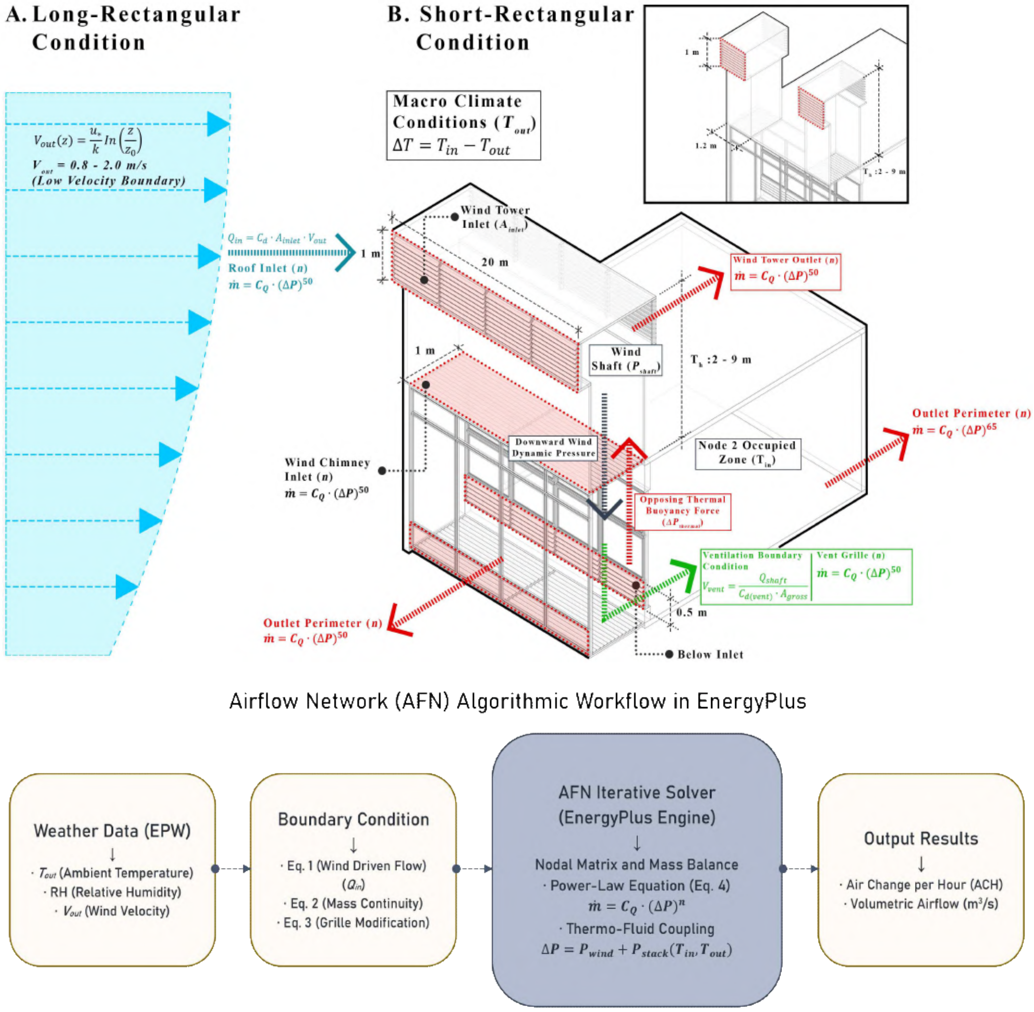


Figure 26. Boundary conditions and control volumes, and algorithmic workflow and the fundamental power-law calculation engine of the AFN model within the EnergyPlus simulation environment

- **Airflow Mechanics: Flow Inversion and Functional Shift**

A thorough aerodynamic investigation of the optimized configuration (case B5: long-rectangular; 9 m height) reveals a complex flow inversion phenomenon that acts as the primary driver for the system's performance (see Figure 27). The simulation data indicates a distinct functional shift between day and night operations. During the daytime, the system operates under wind-driven pressure to deliver a robust true net supply of +0.080 m^3/s. However, rather than acting as a conventional passive supply device around the clock, the 9 m vertically extended wind tower undergoes a massive functional shift during nocturnal hours, operating predominantly in a buoyancy-driven exhaust mode. This phenomenon is quantitatively substantiated by the observed decline in true net airflow, which exhibits a definitive negative vector of -0.017 m^3/s. This phenomenon suggests that as wind pressure diminishes, thermal buoyancy forces become predominant, resulting in the ascent of warmer indoor air through the tower, thereby effectively expelling accumulated heat. Crucially, by acting as a powerful thermal chimney, case B5 actively induces an upward draft that pulls fresh air from the lower envelope layers, functioning as the primary engine for whole-building aerodynamic exchange.

The functional shift comparison (see Figure 28) provides a quantitative assessment of this phenomenon. Case B5 demonstrates a highly stable regime, resisting the severe aerodynamic stagnation rate across all parametric configurations at a mere 1.4%, significantly outperforming the short-rectangular counterparts (e.g. case B6) which exhibit a higher susceptibility to aerodynamic stagnation (3.2%). The long-rectangular profile's dynamic adaptability ensures uninterrupted air exchange, regardless of external wind direction. This finding serves to substantiate the geometric superiority of the longitudinal typology as established by (Satwiko and Tuhari, 2016), who observed that elongated openings exhibited reduced sensitivity to wind incidence angles in comparison to conventional square ports.

Furthermore, the minimal stagnation rate (1.4%) substantiates that the 9 m height engenders sufficient hydrostatic pressure to overcome the neutral pressure plane. This phenomenon is quantitatively evidenced by the drastic increase in exhaust mode duration, from 22.8% in the case B1 (wind tower height: 2 m) to 42.4% in case B5 (wind tower height: 9 m). Crucially, this mode expansion translates into a progressive aerodynamic reversal that emerges once tower height exceeds the 5 m threshold (cases B3-B5 and B8-B10); while the 2 m tower of case B1 continues to push a positive supply flow during the night (+0.010 m^3/s), the 9 m tower of case B5 exhibits the strongest nocturnal inversion across all parametric configurations, extracting air at -0.017 m^3/s. This proof serves to confirm that the vertical extension is indeed successful in shifting the outlet into the negative pressure zone required for effective buoyancy-driven discharge. While (Ghadiri et al., 2011) proposed lower optimal heights (6 m) for arid climates, this study substantiates that in humid tropical conditions, a vertically extended stack is imperative to optimize the buoyancy differential during transitional periods.

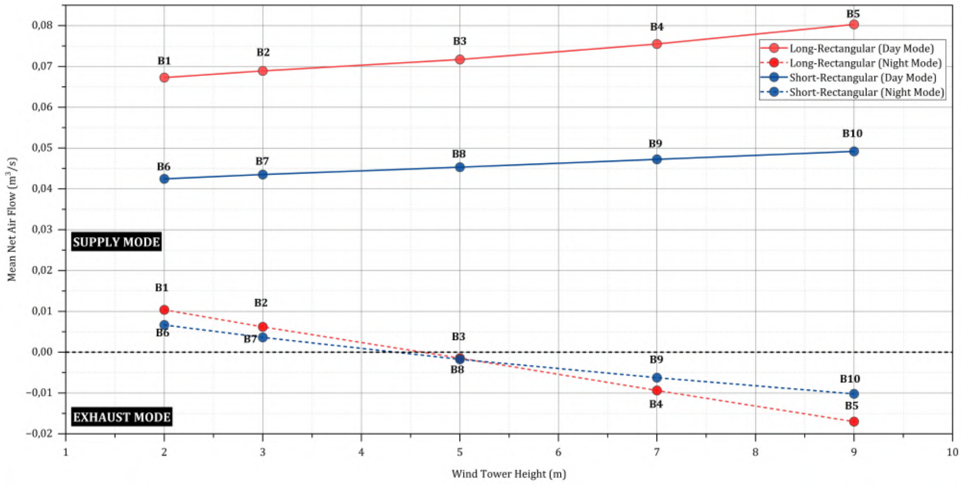


Figure 27. The mean net airflow (m^3/s) across different heights, highlighting the distinct aerodynamic behaviours during diurnal and nocturnal cycles

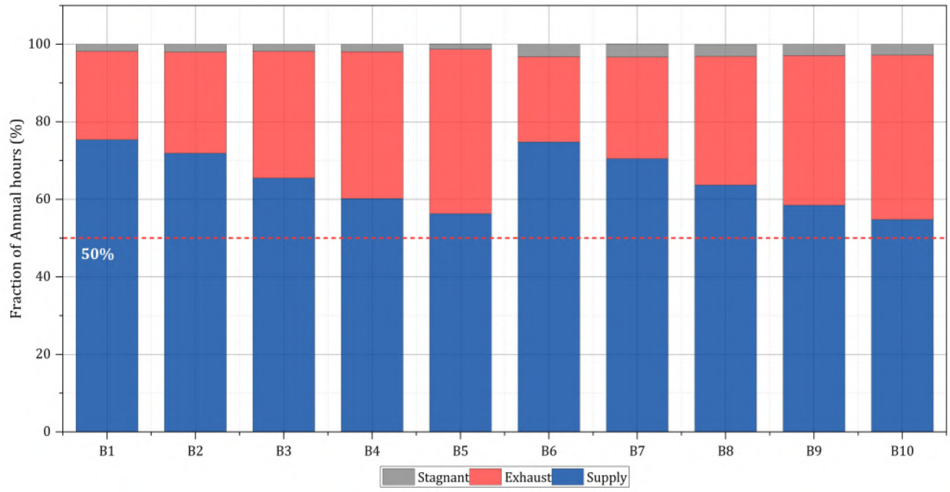


Figure 28. The stacked bar chart reveals the operational mode distribution

- **Volumetric Aerodynamics and the Hygrothermal Paradox**

The primary aerodynamic objective of the TST is to maximize the volumetric air change rate (ACH) to effectively flush accumulated structural heat. However, the rigorous net airflow diagnostic reveals a profound aerodynamic limitation of passive ventilation in the equatorial region. When evaluated holistically, the integration of the B5 TST nearly doubled the natural ventilation capacity of the building. The whole-building ACH escalated from a baseline of 0.0439 ach to 0.0840 ach, representing a remarkable +91% enhancement in aggregate fresh air induction. Paradoxically, despite this nearly doubled aerodynamic performance, the resultant ventilation rate (0.0840 ach) fulfils only 12.3% of the required ASHRAE 62.1 mandate (0.684 ach). Benchmarked against the ASHRAE Standard 62.1 minimum outdoor air requirement of 0.684 ach – derived from Q = (2.5 L/s/person x 351 persons) + (0.3 L/s/m^2 x 2400 m^2) = 1,598 L/s, normalized to the total building volume 8,400 m^3 (occupancy density: 0.188 person/m^2 at 1st floor and 0.138/m^2 at upper floors). This reveals a profound empirical truth: the choked flow phenomenon. This reveals a profound empirical truth: the choked flow phenomenon. It serves as strong evidence that even a geometrically optimized passive TST operating at maximum potential is fundamentally incapable of overcoming the aerodynamic limits required to satisfy minimum volumetric ventilation mandates within a dense, high-temperature tropical climate. The full parametric spectrum – 0.0113 ach (case B6) to 0.0189 ach (case B5) for standalone 24 hours TST performance – is quantified in Figure 29. This phenomenon can be attributed to the augmented cross-sectional area, which effectively mitigates internal flow resistance and aerodynamic flow separation. This physical principle is corroborated by the research conducted by Alsailani et al. (2021), who demonstrated that the optimization of the internal geometrical profile of passive ventilation towers significantly minimises pressure loss and amplifies the overall ventilation rate.

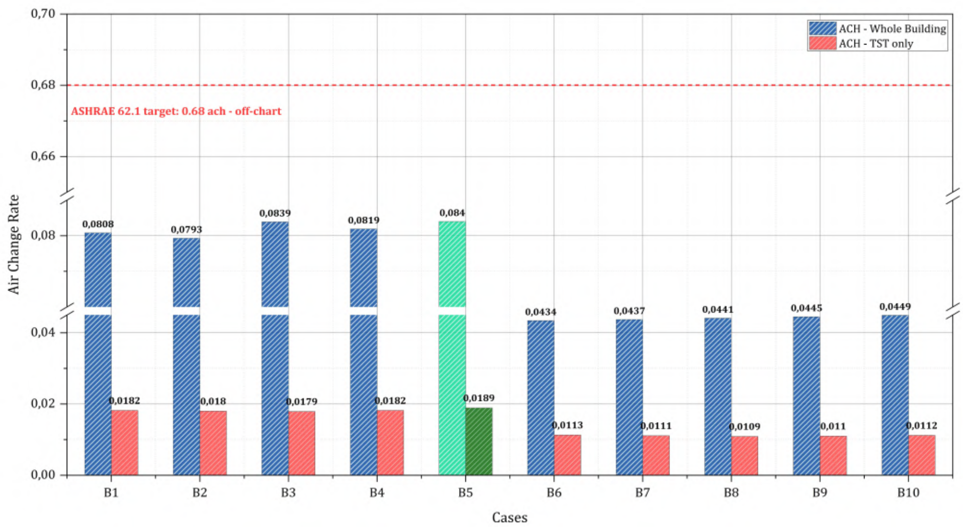


Figure 29. TST parametric ACH performance relative to sealed baseline and ASHRAE 62.1 mandate

The volumetric output of this aerodynamic hierarchy is presented in Figure 30, which shows the daytime mean flow rate and daily ventilation volume across all ten parametric cases. B5 daytime supply surpasses its shorter counterpart (B1) despite greater internal shaft length, demonstrating that the vertical extension's hydrostatic pressure gain outweighs the added viscous friction – as depicted in Figure 30 – it nevertheless signifies the superior aerodynamic capacity of a solitary tower over the multi-shaft configurations. The comprehensive parametric array unveils a systematic divergence driven by topology. The long-rectangular cases (B1 – B5) sustain daytime net flows of 0.067 – 0.080 m^3/s, generating daily volume exchanges of 6,262 – 7,481 m^3/day. In contrast, the short-rectangular cases (B6 – B10) exhibit a net flow rate of 0.0425 – 0.0492 m^3/s, resulting in a daily volume of 3,895 – 4,502 m^3/day – a decline of 37.8 – 39.8% at equivalent tower height. The B5-to-B6 differential is the most diagnostic: B5 delivers 92% greater daily ventilation volume than B6, despite the latter's superior energy efficiency, constituting the primary aerodynamic justification for B5's selection as the holistic optimum. This empirical finding confirms that dividing a single large shaft into three smaller shafts (B6 – B10) severely increases internal viscous friction, destroying aerodynamic momentum and doubling the stagnation.

The comparative AFN performance summary in Table 13 further contextualizes this trade-off through metrics not previously reported. The long-rectangular topology's aerodynamic superiority is most evident in its operational mode distribution: B5 allocates 42.4% of annual hours to buoyancy-driven exhaust mode, compared to 22% in B6, thus confirming that the extended vertical shaft maintains a persistent thermal draft independent of wind availability. The stagnation period of 1.4% in B5 versus 3.2% in B6 further demonstrates the 9 m tower's capacity to sustain continuous air exchange under low-wind conditions. The nocturnal flow inversion – defined as a functional shift where B5 dedicates 42.4% of its operational time to exhaust mode compared to only 22% in B6 – reveals a fundamental mechanistic divergence: B5 actively purges accumulated structural heat during nocturnal hours through buoyancy-driven exhaust, whilst B6's lower exhaust capacity indicates thermal stagnation outside operational hours.

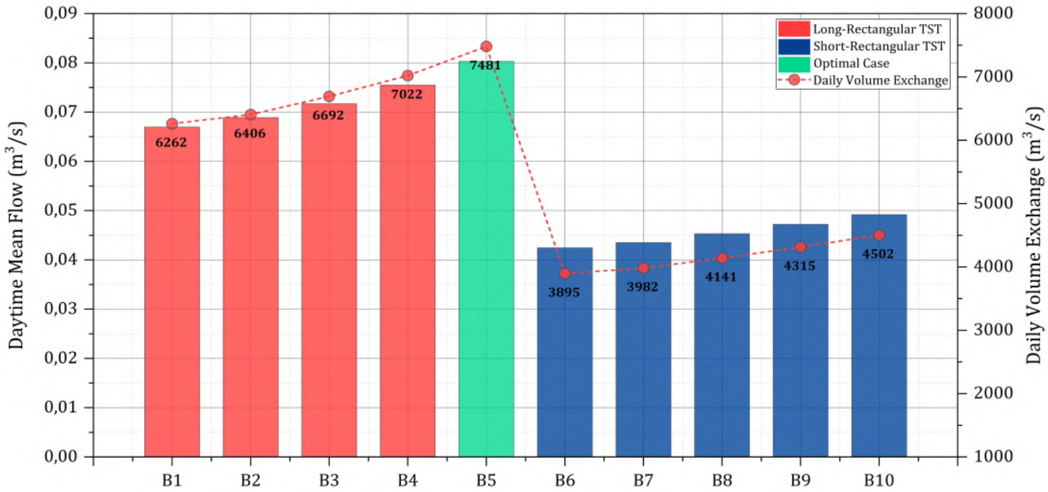


Figure 30. Daytime mean volumetric flow and daily volume exchange capacity across all ten parametric case

Table 13. B5 vs. B6 complete AFN performance comparison: aerodynamic topology trade-off across key operational metrics

Metric	B5 (Long-rectangular; Height 9 m)	B6 (Short-rectangular; Height 2 m)	Unit	Δ / Ratio
TST-specific ACH (annual)	0.0189	0.0113	ach	× 1.67
Whole-Building ACH	0.0840	0.0434	ach	× 1.94
Daytime Mean Flow	0.0803	0.0425	m^3/s	× 1.89
Night-time Mean Flow	-0.0170	0.0067	m^3/s	Inversion vs. Supply
Daily Volumetric Exchange	7,481	3,895	m^3/day	× 1.92
Peak Supply Flow	0.2898	0.1489	m^3/s	× 1.95
Peak Exhaust Flow	-0.3797	-0.1241	m^3/s	× 3.06
Supply Mode	56.30	74.80	%	-18.50 pp
Exhaust Mode	42.40	22	%	+20.40 pp
Stagnant Flow	1.40	3.20	%	-1.80 pp
Mean Net Airflow	0.0803	0.04246	m^3/s	× 1.89
Cooling EUI	123.27	121.84	$kWh/m^2/year$	+1.43
TST only ACH vs. ASHRAE 0.68	2.8	1.7	%	B5 = 1.65× more

However, the empirical temperature data reveals a counterintuitive thermal paradox that is directly tied to the tower's vertical geometry. As the tower height increases from 2 to 9 m, the elongated vertical shaft significantly augments the internal convective surface area, thereby acting as a substantial thermal sink that successfully cools the transiting supply air. The simulation records indicate a progressive decrease in the mean daytime supply temperature, from 29.74˚C (in the 2 m tower) to 28.89˚C (in the 9 m tower), which is well below the mean outdoor temperature of 30.74˚C. Notwithstanding this rational cooling benefit, optimizing passive ventilation (ACH) paradoxically results in an escalation in the aggregate cooling demand. As demonstrated in Figure 31, as the long rectangular tower height increases, the cooling intensity climbs from 122.83 to 123.27 kWh/m^2/year. This phenomenon unveils the "hygrothermal paradox", a distinctive occurrence within the tropical environment.

Crucially, while the tall tower successfully cools the air, the fundamental laws of psychrometric principles dictate that lowering the dry-bulb temperature of a constant-moisture air mass invariably increases its RH. Empirical evidence from the internal TST nodes corroborates this thermodynamic shift; the mean daytime RH escalates from 70.22% in the 2 m configuration (case B1) to a near-saturation level of 73.73% on the 9 m configuration (case B5). The intricate mechanistic transition driving this hygrothermal penalty – from the aerodynamic boundary layer capture down to the psychrometric shift and the subsequent HVAC dehumidification overload – is comprehensively illustrated in the diagrammatic evaluation presented in Figure 32. Consequently, the mechanical system is subjected to a state of intense dehumidification to neutralize this elevated moisture concentration. This phenomenon, defined here as the latent trap, highlights a critical boundary: the success of sensible temperature reduction in tall towers becomes the direct catalyst for humidity control failure owing to psychrometric shifts. This indicates that the energy penalty incurred by extracting airborne moisture significantly exceeds the sensible cooling benefits provided by the thermal mass of the tower. This finding suggests that unconditioned volumetric flushing in tropical climates is constrained by a rigid thermodynamic limit that is primarily determined by humidity, as opposed to dry-bulb temperature.

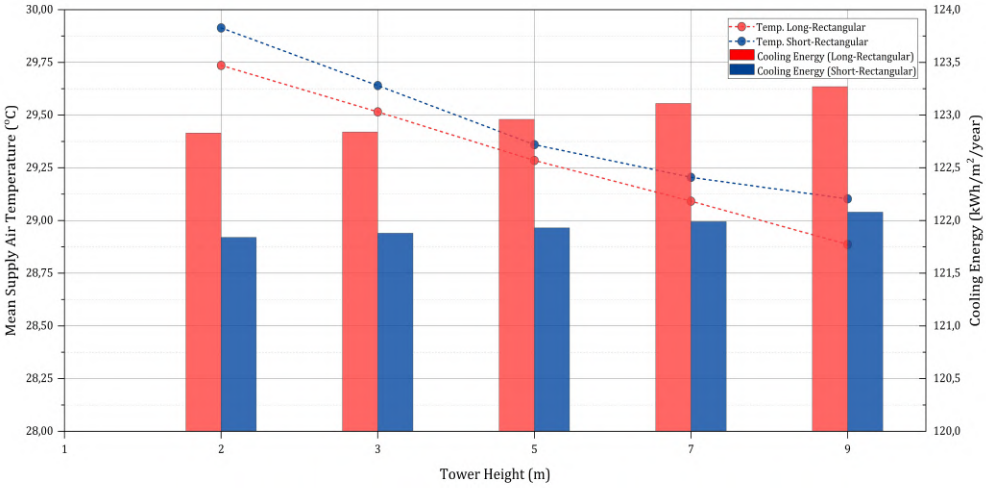


Figure 31. The hygrothermal paradox: Declining supply temperatures overwhelmed by latent cooling penalties

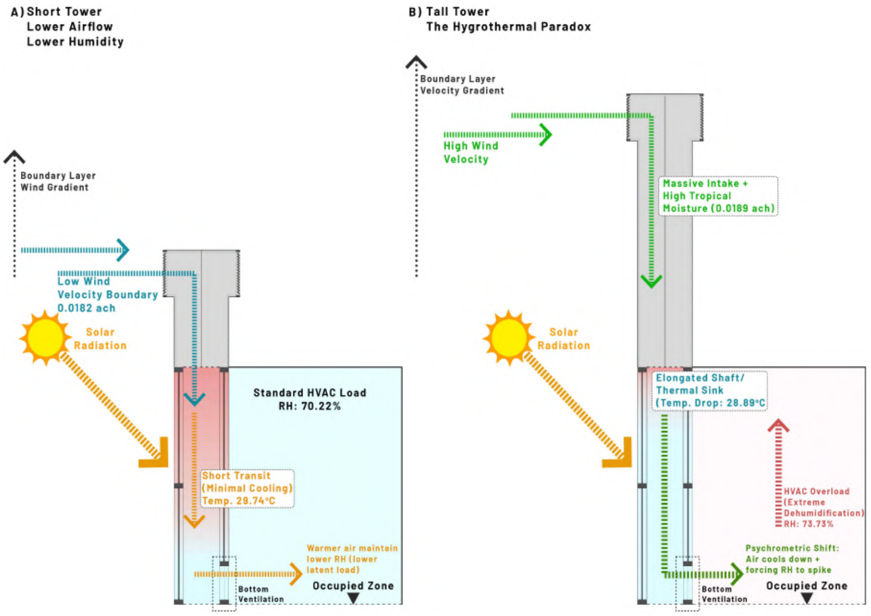


Figure 32. The hygrothermal paradox mechanism: Sensible cooling induces severe latent load penalties

- **The Cooling Penalty: The Stack Effect Limit**

Achieving the aforementioned aerodynamic improvements introduces a counterintuitive thermodynamic paradox: maximizing passive ventilation does not directly correspond with a reduction in cooling load. As demonstrated by the polynomial regression in Figure 33, increasing the tower height led to a marginal but consistent escalation in the cooling energy trajectory. To rigorously isolate and diagnose this anomaly across different architectural footprints, a multi-axis combo chart detailing the volumetric air change rate (ACH) alongside the grouped cooling loads for all ten configurations was analysed (see Figure 34). Regardless of the geometry, a universal thermodynamic conflict emerged. The successful amplification of volumetric airflow (ACH) – which spans from 0.0193 ach in the compact short-rectangular variant (case B6) to an optimal 0.0371 ach in the elevated long-rectangular configuration (case B5) during daytime operational hours – acts as the direct catalyst for the increased cooling energy. The long-rectangular geometry actively amplifies this effect; its superior capture area maximizes the volumetric flow rate (Q), which inadvertently maximizes the latent energy penalty.

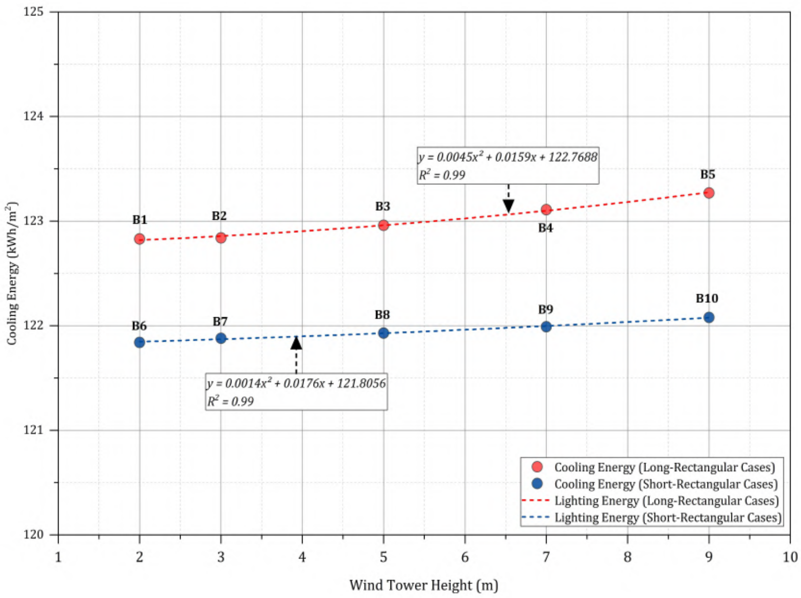


Figure 33. The trendline for cooling energy in TST cases

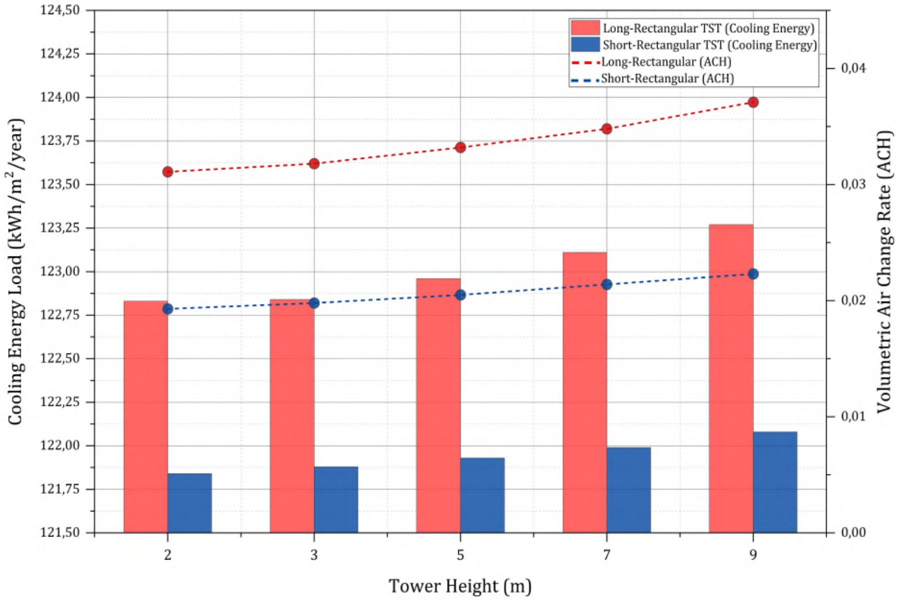


Figure 34. The ventilation-cooling paradox: Increasing volumetric airflow inadvertently escalates cooling loads

This counterintuitive escalation is governed by a "dual-penalty" mechanism specific to the integrated façade architecture (as previously conceptualized in Figure 32). Firstly, in accordance with the tropical environmental constraints examined by Bernal et al. (2024), the taller 9 m tower effectively penetrates higher boundary layers, capturing greater wind velocities and amplifying the hydrostatic pressure column (ΔP_{stack}). As established, this introduces a significant latent load owing to the high moisture content of the equatorial climate. Secondly, and most critically, because this supply air is routed downward through the wind tower chimney before entering the occupied zone via bottom vents, it undergoes complex psychrometric interactions. Consequently, the integrated system injects a substantial hygrothermal intrusion – combining outdoor latent moisture with fluctuating sensible heat – forcing the mechanical system into a state of intense dehumidification and sensible cooling. Notwithstanding, the aggregate cooling energy savings remained substantial, ranging from 15.86% to 16.84% (refer to Figure 35), thereby substantiating the efficacy of the TST in eradicating internal heat accumulation, albeit with diminishing returns at elevated altitudes.

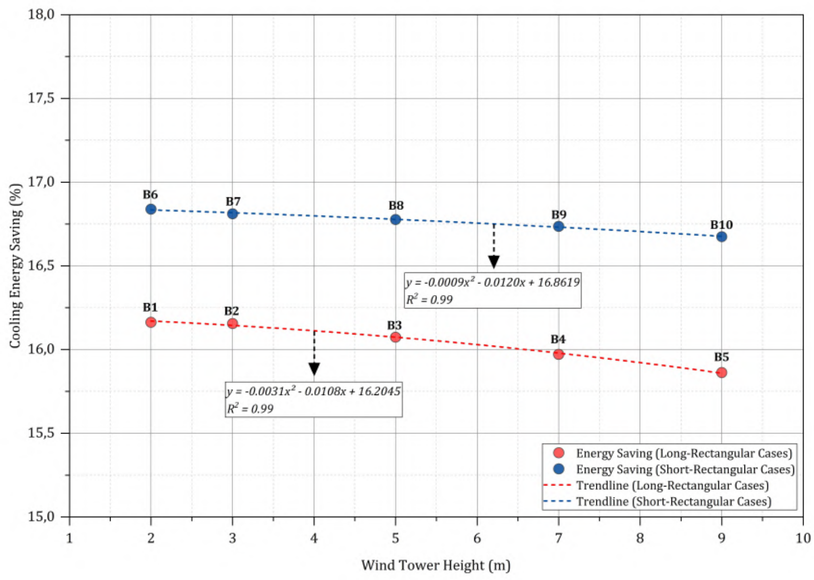


Figure 35. The system maintains a consistent energy savings of >15% relative to baseline, validating the efficacy of buoyancy-driven heat removal

- **Thermodynamic Boundary Diagnosis: $ACH_{Required}$ and Sensible Heat Removal**

To resolve the apparent contradiction between the TST's measured air change rate and its demonstrated energy savings efficacy, a thermodynamic boundary analysis is conducted by applying Eq. 3 (Section 3.5) – wherein the sign of $\Delta T = T_{indoor} - T_{outdoor}$ governs whether convective cooling is thermodynamically achievable – directly applicable to the TST operational context. This diagnostic quantifies the ACH that would theoretically be required for the TST to achieve passive convective thermal cooling – the cooling mechanism presumed by ventilation standards developed for temperate climates. For any temperate climate where T_{outoor} falls below the conditioned indoor setpoint, Eq. 3 yields a positive $ACH_{required}$, confirming the thermodynamic validity of ventilative cooling benchmarks in such contexts and identifying the precise thermodynamic origin of the 2-6 ach standard cited for passive cooling applications. However, when the actual Jakarta equatorial conditions are applied – as established in the simulation framework ($T_{outdoor}$ = 30.74˚C; T_{indoor} = 26.56˚C; ΔT = -4.18˚C) – the denominator of Eq. 3 becomes negative, yielding $ACH_{required}$ = -3.41 ach. A negative ACH requirement is physically undefined: it constitutes mathematical proof that convective thermal cooling via ventilation is thermodynamically impossible when outdoor air temperature exceeds the indoor setpoint, regardless of the volumetric flow rate achieved. As illustrated in Figure 36, this diagnostic compares the temperate reference condition against the Jakarta equatorial condition ($ACH_{required}$ = undefined/negative), confirming that the 0.020 ach achieved by case B5 does not represent a shortfall relative to any temperate-derived ACH target – rather, it reflects the aerodynamic ceiling of the passive device operating within a thermodynamic regime in which such targets are physically inapplicable.

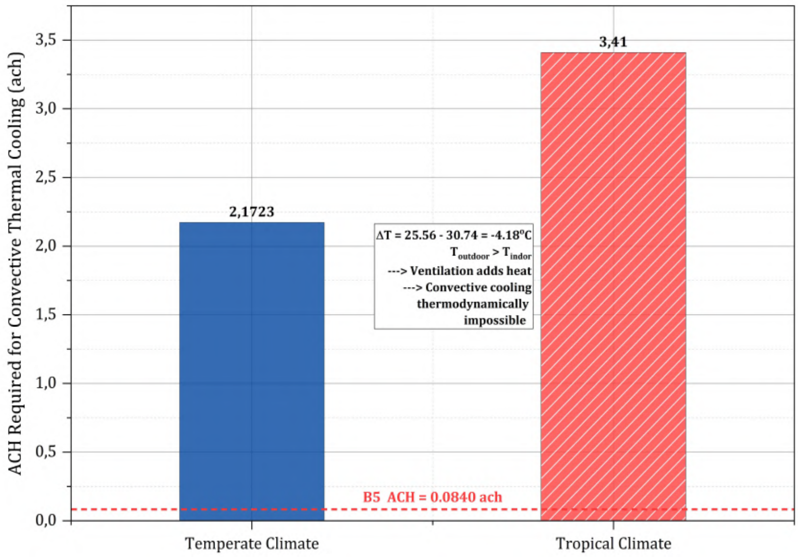


Figure 36. Equation 3 thermodynamic boundary diagnostic: $ACH_{required}$ under temperate vs. Jakarta equatorial conditions

Consequently, the thermodynamically correct performance metric for the TST in Jakarta's equatorial climate is not ACH magnitude, but direct sensible heat removal from the occupied zones, as quantified by the AFN Zone Ventilation Sensible Heat Loss Rate output variable. As demonstrated in Figure 37 and substantiated by the ventilation sensible heat removal analysis data, the air intakes (long-rectangular topology) extract an average of 143.1 W of sensible heat from the six occupied floors in case B5 during the daytime. Despite the aerodynamic advantages of B5 in volumetric airflow (whole building ACH: 0.0840 vs. 0.0434), AFN thermal diagnostics reveal that the short-rectangular topology achieves higher sensible heat extraction (161 – 170 W) whilst exhibiting a comparatively lower ACH (0.0113 ach). In case B6, this results in 60.5 W of heat being extracted on the sixth floor, compared to 41.8 W in case B5. This compensates for the lower volumetric throughput through superior spatial reach. Consistent heat removal across all ten parametric configurations is ultimately sustained by the buoyancy-driven nocturnal exhaust mechanism shown in Figure 28 – an independent thermodynamic mode – which accounts for the 15.86 – 16.84 % aggregate cooling energy reduction in Figure 34. A more precise comparison of the performance of these two metrics (ventilation sensible heat removals vs. ACH) can be observed in Table 14 and Figure 38, which presents the scatter plot cluster for each TST topology.

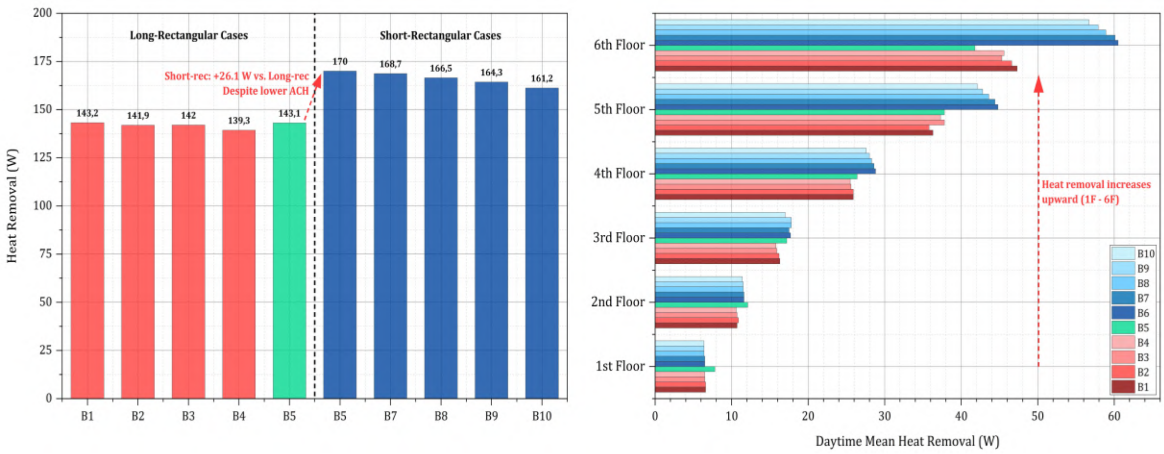


Figure 37. AFN Zone Ventilation Sensible Heat Loss rate - daytime mean, all ten TST parametric cases

Table 14. Ventilation sensible heat removal vs. ACH daytime mean data

Case	TST-Specific ACH	Daytime Sensible Heta Removal (W)	Daily Airflow Volume (m^3/day)	Cooling EUI Reduction (%)
B1	0.0182	143.2	6,262	16.16
B2	0.0180	141.9	6,406	16.15
B3	0.0179	142.0	6,692	16.07
B4	0.0182	139.3	7,022	15.97
B5	0.0189	143.1	7,481	15.86
B6	0.0113	170.0	3,895	16.84
B7	0.0111	168.7	3,982	16.81
B8	0.0109	166.5	4,141	16.78
B9	0.0110	164.3	4,315	16.74
B10	0.0112	161.2	4,502	16.67

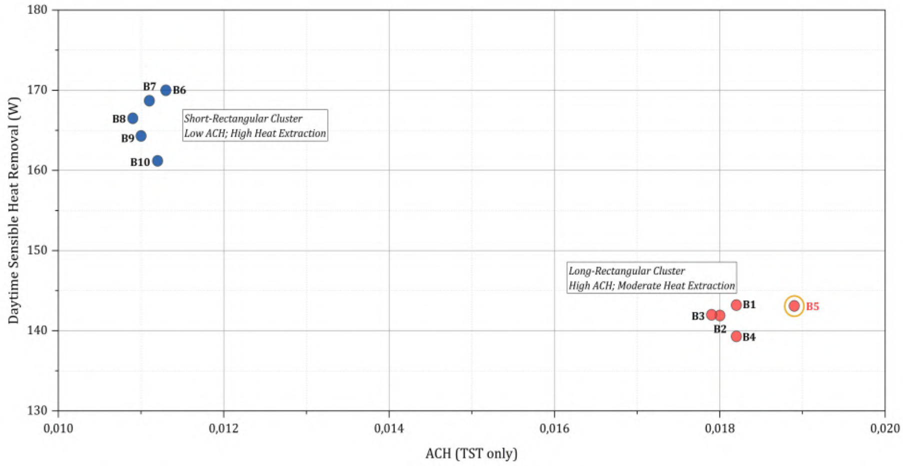


Figure 38. Decoupling: ACH magnitude vs. ventilation sensible heat removal

- **Performance Trade-off: The Strategic Shift Toward Aerodynamic Resilience**

Ultimately, the selection of case B5 as the holistic optimum is underpinned by a rigorous trade-off analysis between these conflicting thermal and aerodynamic metrics, which is visualized in the performance optimization landscape (Figure 39). The scatter plot reveals two distinct clusters: the energy focus group (short-rectangular) and the aerodynamic focus group (long-rectangular). Case B5 is situated as the strategic compromise. By accepting a negligible energy penalty of 1.173% (compared to the energy-optimum case B6), the system ensures superior ventilation effectiveness. Rather than prioritizing maximal volumetric supply, case B5 unlocks unparalleled aerodynamic stability, registering the lowest stagnation rate across all parametric configurations at a mere 1.4%. Furthermore, it exploits a massive functional shift towards buoyancy-driven exhaust (42.4% operational time) to act as a critical nocturnal thermal relief valve.

This resilient aerodynamic advantage prioritizes maximum convective flushing over marginal energy savings. This approach advances the foundational tropical ventilation principles established by Satwiko and Tuhari (2016) who demonstrated that in Indonesia's warm-humid climate, maintaining continual airflow is a non-negotiable prerequisite for preventing indoor heat and pollutant accumulation. Building upon their premise, the data indicate that maximizing aerodynamic breathability and purging structural heat through scaled geometries should be prioritized over maximal energy conservation to effectively eradicate structural heat stagnation. By effectively doubling the baseline natural infiltration rate (+91%), B5 maximizes aerodynamic performance within passive physical limits.

Crucially, the empirical reality that this maximized output still only achieves 12.3% of the ASHRAE target robustly justifies the necessity of the proposed hybrid decoupled framework for subsequent optimization stages.

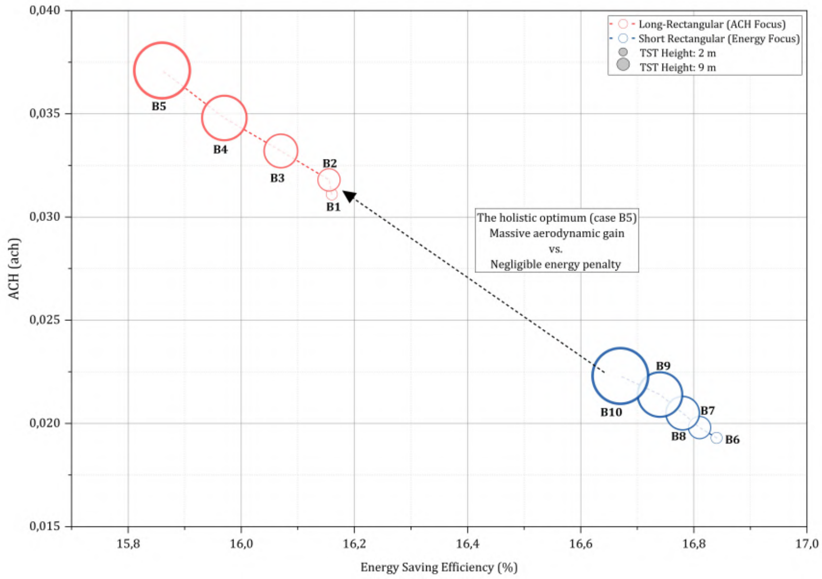


Figure 39. The strategic decision to select case B5, trading a minimal energy penalty for a significant gain in ventilation effectiveness (TST tower ACH performance during daytime operational hours)

The ACH values recorded for case B5 in phase 1 standalone assessment (whole building ACH: 0.084 ach) are substantially lower than those reported in windcatcher studies conducted in arid or temperate climates, where ACH commonly exceeds 5. However, this divergence is not an anomaly – it is the thermodynamic consequences of Jakarta's high-enthalpy equatorial condition, where $T_{outdoor}$ persistently exceeds T_{indoor}. Table 15 places these findings in comparative context, demonstrating that the B5 performance is consistent with the limited body of windcatcher research in hot-humid tropical Southeast Asia, and that the recharacterization from windcatcher to TST is empirically justified.

Table 15. ACH validation of TST case B5 against windcatcher studies in tropical and other climate conditions

Study	Climate / Location	Tower Topology / Geometry	ACH Achieved	Other Results	Key Finding
Bernal et al. (2024)	Tropical (Panama City)	Short rectangular, long rectangular, short semicircular, long semicircular, short/long extension; Best: Short semicircular	5 ach (75% improvement)	OT: 24 – 35 ˚C; RH: 43 – 69%	Short semicircular: significantly increase air exchange rates while maintaining temperature and humidity within tropical comfort limits
Nejat et al. (2021)	Tropical (Malaysia)	Roof mounted two sided windcatcher with upper/side wing walls; Height: 1.5 m (based on 1:10 scale of 150 mm); Upper wing wall length (10 – 50 cm); Best: 50 cm upper wing wall	ACH: Up to 35 h^{-1} (maximum at 50 cm)	Cooling capacity: 1.6 – 9.6 kWh (at 0.5 – 4.0 m/s wind speed); Indoor airflow rate: 0.7 m/s; Temp.: < 29˚C; RH: < 80%	50 cm upper wing wall à offering sufficient passive cooling power to offset internal heat gains
Satwiko and Tuhari (2016)	Warm Humid (Tropical) (Yogyakarta, Indonesia)	Longitudinal hybrid windcatcher with exhaust fans; Inlet height (1 m vs. 3 m above roof); Best: Inlet height 1 m above roof	-	Indoor air velocity: 0.19 – 1.01 m/s (with 1 – 5 m/s outdoor wind); Air velocity via exhaust fan (calm days): 0.05 m/s	The hybrid configurations with exhaust fan guarantees continuous ventilation and physiological cooling in tropical basements even when outdoor wind absent
Kubota et al. (2022)	Hot-Humid (Tropical) (Tangerang, Indonesia)	Application Topology: Windcatcher for High-rise apartment	-	Ventilation rate: 1.2 to 1.4 times increase	Implementing windcatchers in tropical high-rise residential buildings

					successfully multiples the natural ventilation rate
Khalid et al. (2025)	Subtropical (Lahore, Pakistan)	Ventilation modes: Fan + WC, AC + WC, AC, Heater + WC, No Heater + No WC; Best for Summer: Fan + WC / AC + WC	-	PM2.5: > 250.5 μg/m^3 (hazardous); CO_2: Higher in summer AC + WC mode; PMV/PPD: Neutrality achieved only in AC + WC mode	While wind chimneys (WC) combined with fans/AC improve thermal satisfaction, heavy dust/pollution in the region necessitates the integration of air filtration
Ghadiri et al. (2014)	Hot & Arid (Yazd, Iran)	Tower height on a square four-sided H-form plan (tested heights: 3.5 to 10.5 m); Best: 6 m height	-	Significant improvement in indoor air temperature reduction up to 6 m; Improvements stall beyond 9 m	For a conventional windcatcher without evaporative cooling ponds, 6 m is the optimum; Taller towers decrease mass flow rate or provide insignificant gains
Saif et al. (2021)	Hot & Arid (Kuwait)	System Integration: AC only; WC only: WC + evaporative cooling; Best: WC + evaporative cooling	ACH: Median 0.56 (range 0.41 – 0.65)	PMV: 100% comfortable; PPD < 20%; Energy use: 38 kWh/m^2 (Summer); RH: median 60%	Integrating WC into an active air loop with evaporative cooling saves energy but severely limits ACH (0.56) compared to direct natural systems
Heidari et	Arid	One sided	ACH: 3 to	Cooling load:	Using

al. (2024)	(Tehran)	horizontal windcatcher integrated with a cross-flow evaporative channel; Height/length: 8 m; Window aperture size (0 – 100%); Best: > 20%	20 h^{-1} (scales linearly with wind up to 6.4 m/s)	up to 12,000 W; Energy: ~50% savings vs. vertical	horizontal windcatcher topology requires the building's outlet to be opened at least 20% to prevent back-pressure and maintain ACH above 3
Nejat et al. (2025)	Temperate – Cold (15 European Cities)	Two sided windcatcher; Height: standard modern commercial design	ACH: 20 h^{-1} to 35.7 h^{-1}	Airflow: 600 L/s: CO_2: Effective dilute	Even in lower wind speed, two sided topologies provide ACH values (20-35) that vastly exceed ASHRAE minimum standard
Shayegani et al. (2024)	Temperate (Viena, Austria)	Roof mounted one sided windcatcher; Tower height: 1.5, 2.0, 2.5, 3.0 m); Best: Height 2.5 m; Inlet: 90 x 140 cm	-	Energy: lowest annual energy demand (1686 kWh); Thermal load: optimized for warm seasons	Taller is not always better; restricted to 2.5 m height with a specifically widened inlet (90 x 140 cm) creates optimal pressure capture
Katona et al. (2022)	Oceanic climate (temperate)	Two sided windcatcher; Inlet deflector curve shape (empty, circle, parabola, hyperbola) + deflector extensions; Best: Parabola + 1 m modified deflector	ACH: average 7.50 h^{-1} (up to 7.70 h^{-1} with deflector)	Airflow: reduced vortices	Parabolic inlet shape smoothly funnels air downwards, maximizing ACH while minimizing counter-productive turbulent vortices
Alsailani et	General	One sided	ACH proxy:	Airflow	Single sided

al. (2021)	Aerodynamic	windcatcher; Height: 1.5 m; Inlet extensions, corner radius, guide vanes; Best: straight inlet extension (S/D = 0.375) + 7 guide vanes	multiples ventilation deeply into multi-storey	velocity: increased by 130 – 199%	façade topology requires the exhaust outlet to be physically enlarged to prevent back-pressure at lower floors
Noroozi and Veneris (2018)	Cold Desert (Qaen, Iran)	Traditional four-sided windcatcher with wetted pad; Height: 2.5, 3.5, and 5 m; Best: 5 m height + closed pad	-	Airflow: mass flow rate increased at low winds; Cooling load: doubled (1.2 kW at 5 m); RH: increased 39 – 64%	Using closed evaporative pads at an optimal 5 m height drastically improves the cooling load when outdoor wind velocities drop below 3 m/s
This Study - Case B5	**Tropical Equatorial (Jakarta, Indonesia)**	**Long Rectangular; Height: 9 m**	**0.0840 ach (whole building); 0.0371 (TST tower - day time operational hours)**	**Energy saving: 15.86% relative to baseline**	$\boldsymbol{ACH_{required}}$ **= -3.41; Wind-driven cooling thermodynamically impossible; recharacterized as TST**

3.7.4. Louver System Parametric Analysis

With the thermal decoupling function of the DSF and the aerodynamic contribution of the TST established, the focus of optimization shifts to the external louver system as the final parametric layer governing direct solar interception and envelope porosity. The optimization of the external louver system in this study is governed by a dual-physics objective: intercepting the high-incidence direct beam radiation inherent to the tropical solar geometry while preserving aerodynamic permeability and diffuse daylight transmission. External placement of the louver system – positioned in front of the glazing rather than as an inter-pane device – was adopted to enable pre-glazing solar interception: direct beam radiation is attenuated before transmission into the occupied zone.

The application of external louvers as a sunshade on a building is illustrated in Figure 40 to elucidate the implementation of louver elements in phase 1 in the simulation system. At the 30 mm offset (case C2), the louver sits flush against the glazing, maximizing solar blockage whilst maintaining the 270 mm aerodynamic gap for convective flushing – as confirmed by the bifurcation data in Figure 43. The 340 mm slath width, although unusual in conventional practice, is consistent with the range chord depth documented in tropical high-performance façade literature, including Dinapradipta et al. (2024). According to the aggregate performance metrics, the simulation identified case C2 (Slats gap: 270 mm; Slats width: 340 mm; Slats angle: 30˚; and distance from external window: 30 mm) as the holistic optimum. This configuration effectively reconciles the antagonistic requirements of solar heat rejection and luminous sufficiency. According to the aggregate performance metrics, the simulation identified case C2 (Slats gap: 270 mm; Slats width: 340 mm; Slats angle: 30˚; and distance from external window: 30 mm) as the holistic optimum. This configuration effectively reconciles the antagonistic requirements of solar heat rejection and luminous sufficiency.

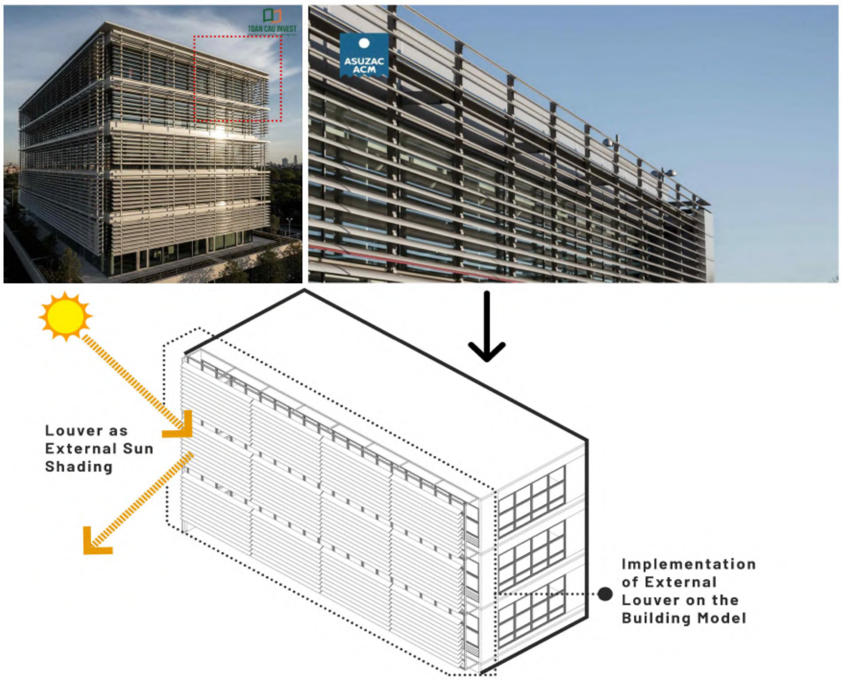


Figure 40. The implementation of external louver in the mid-rise building (https://toancauinvest.vn/en/operable-sun-louver-system/; Accessed: June 10th, 2026)

- **Energy Performance and Trade-off Analysis**

The simulation results, as illustrated in the cooling load trendline (Figure 41), indicate that the implementation of case C2 achieved a superior cooling efficiency, reducing the annual cooling load to 94.51 kWh/m^2/year. This represents a substantial 35.49% reduction relative to the unshaded baseline model (146.51 kWh/m^2/year). This magnitude of reduction is empirically consistent with the tropical case study in Jakarta by Suryandono et al. (2021), which reported that optimized external louvers could yield cooling energy savings of approximately 30-40%, depending on the WWR. A salient finding of the analysis is the revelation of a counterintuitive geometric phenomenon. While steeper slat angles (e.g., 60° in case C10) are generally hypothesized to provide maximum shading, the data indicates that the 30° angle combined with a wider slat dimension (340 mm) in case C2 resulted in a more balanced thermodynamic outcome. This finding is consistent with the conclusions of Shaeri et al. (2022) and Bramiana et al. (2025), whose simulations demonstrated that precise geometric tuning of slat angles is essential for minimizing solar heat gain without inducing a thermal trap. Furthermore, Saraswati et al. (2025) observed that moderate angles (15-30 °) frequently surpass steeper inclinations in preserving adequate daylight illuminance (UDI).

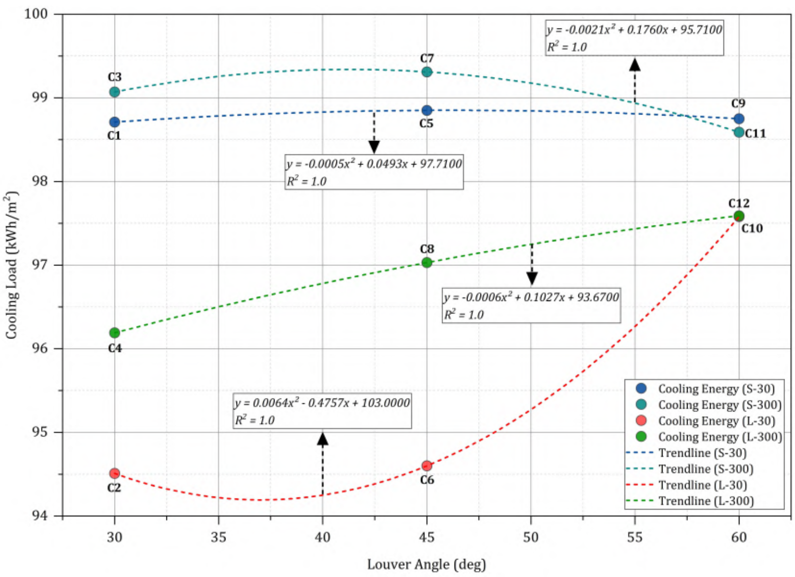


Figure 41. Polynomial regression confirms that precise geometric tuning minimizes solar gain while preserving convective heat dissipation

However, this thermal gain introduces a quantifiable visual penalty. As illustrated in Figure 42, the shading obstruction led to an escalation in artificial lighting consumption. Specifically, case C2 exhibited a lighting load of 5.98 kWh/m^2/year, which is indicative of a substantial increase compared to the negligible baseline load of 3.29 kWh/m^2/year. Nevertheless, this penalty is significantly lower than the penalties associated with “aggressive shading” scenarios. For instance, case C7 (45° angle; 30 mm gap) exhibited a surge in lighting demand to 8.27 kWh/m^2/year, attributable to excessive daylight obstruction. This inverse correlation, in which thermal shielding compromises visual autonomy, lends credence to the necessity of MOO approach. This approach, as advocated by Alafchi et al. (2025) and Bahdad et al. (2024), aims to reconcile the conflict between conductive heat gain reduction and luminous permeability.

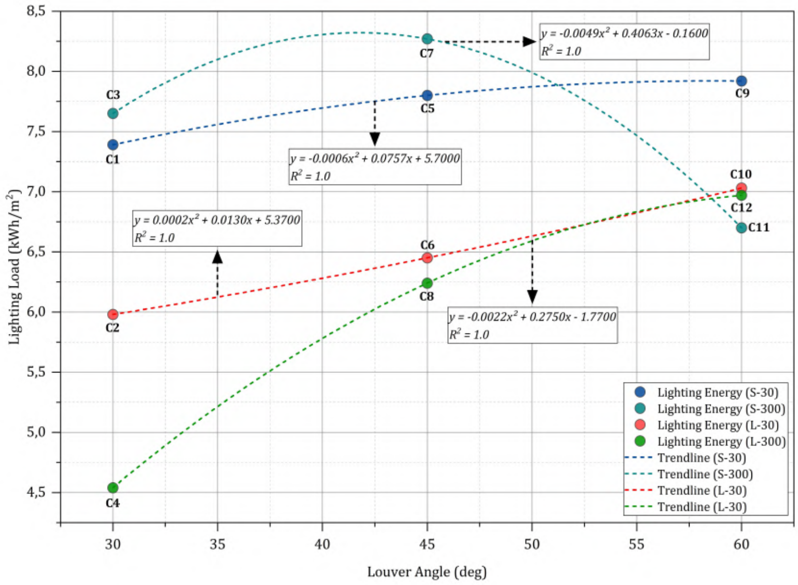


Figure 42. The trajectory illustrates the visual penalty, where aggressive shading obstruction causes an exponential surge in artificial lighting demand

To rigorously resolve this conflict and uncover the underlying physical mechanism, a quadrant clustering analysis was conducted, mapping the visual penalty (lighting load) against the thermal penalty (cooling loads) for all configurations (see Figure 43). The scatter plot unveils a distinct aerodynamic bifurcation, indicating a clear separation of the variables (model cases). The data unequivocally separates into two different performance regimes, dictated entirely by the louver gap (porosity). The dense configurations (gap 30 mm – red cluster) universally fail, creating a heat trap effect characterized by severe cooling loads (>98 kWh/m^2/year) and diminished luminous permeability. This geometric failure occurs because micro-louvers restrict the aerodynamic flushing of the façade. As demonstrated in the following mechanism (see Figure 44), the positioning of a dense louver system 300 mm from external glazing material (e.g. case C7) results in the formation of a stagnant thermal plenum. The ambient wind is obstructed, leading to the accumulation of convective heat against the glass, while the solid obstruction significantly restricts the sky view factor.

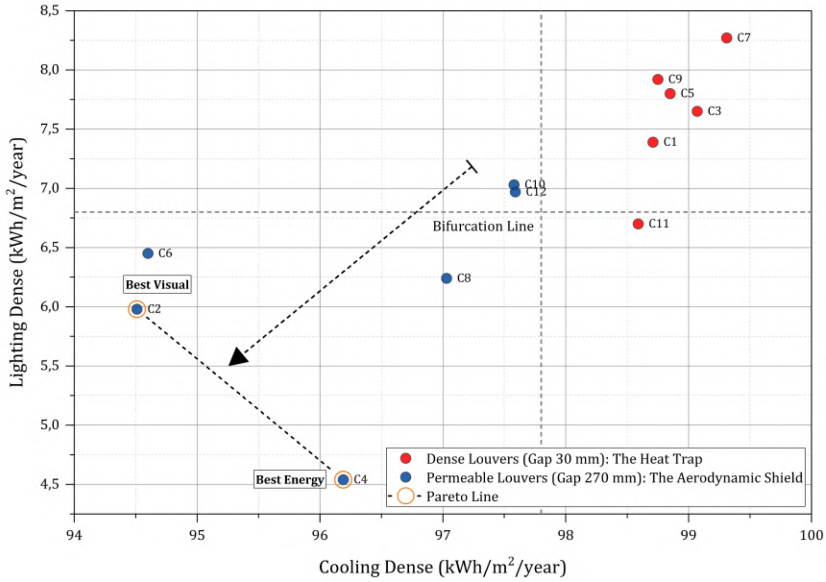


Figure 43. Louver performance landscape: Aerodynamic bifurcation and porosity clustering

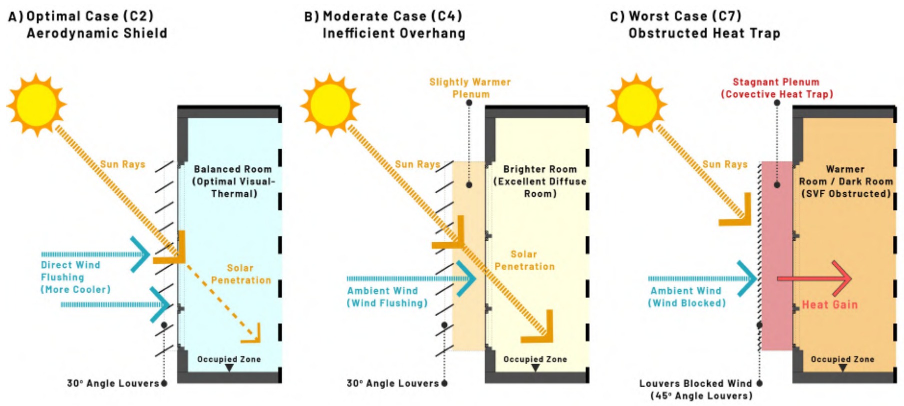


Figure 44. Thermodynamic impact of louver porosity on convective flushing

Conversely, the permeable configurations (gap 270 mm - blue cluster) successfully balance the variables, establishing the pareto-efficient frontier (Figure 43). As illustrated in Figure 44 A, the optimal case C2 positions the wide slats mere 30 mm from the glass, thereby ensuring maximum solar interception, while the 270 mm gap facilitates robust ambient wind flushing, thus dissipating sensible heat. A distinct trade-off is observed in the moderate case C4 (Figure 44 B): extending the louver distance to 300 mm improves diffuse daylight penetration (lowering the lighting load to 4.54 kWh/m^2/year), but the excessive distance allows partial solar penetration, penalising the cooling efficiency. This bifurcation phenomenon is in alignment with the findings of Mazzetto (2025) who demonstrated that the integration of shading elements and natural ventilation is imperative for optimal thermal comfort. The isolation of solar control without ensuring sufficient convective airflow can result in a substantial thermodynamic penalty, particularly in hot-humid climates. It can be posited that this aggregate assessment confirms the superiority of case C2 as the non-dominated pareto solution, which achieved the lowest total EUI of 100.48 kWh/m^2/year. This results in a net energy reduction of 32.9% from the baseline model, as illustrated in the total energy saving potential (see Figure 45).

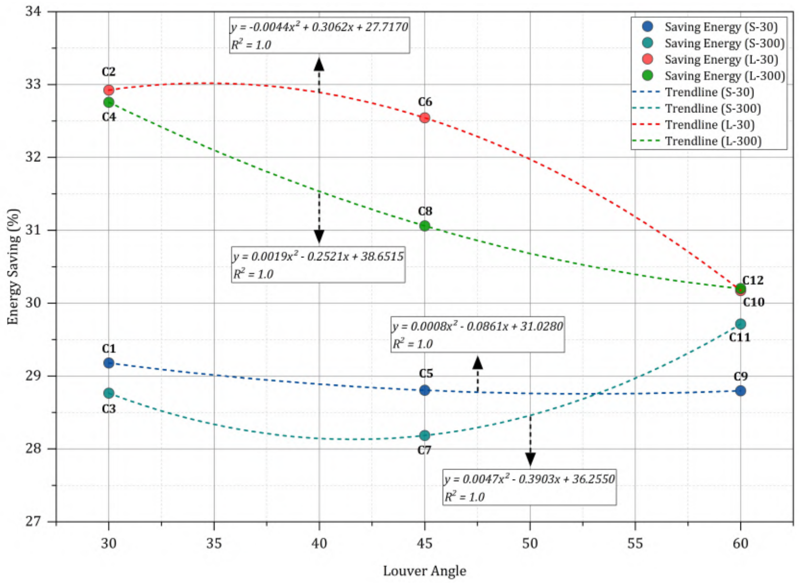


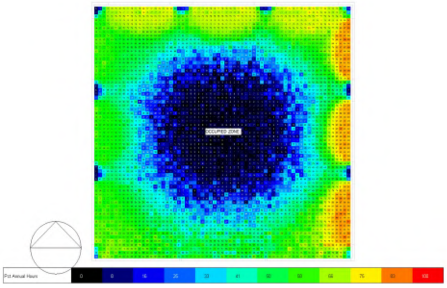
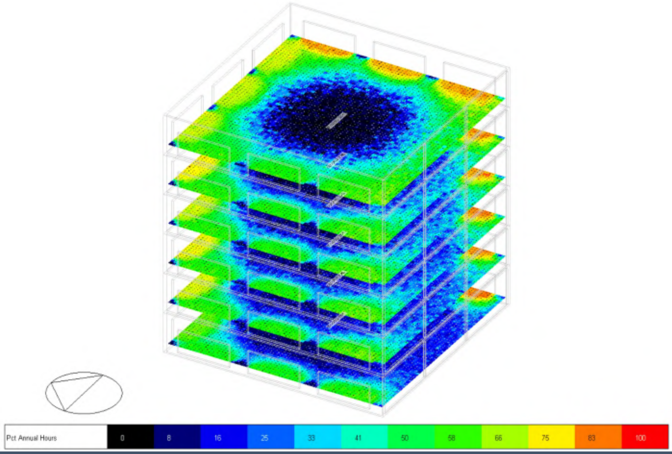
Figure 45. Polynomial trends confirm larger slat dimensions maximize energy saving potential

- **Pareto Optimization Landscape**

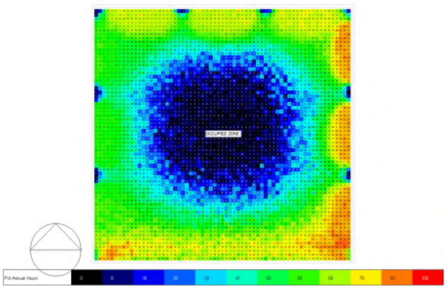
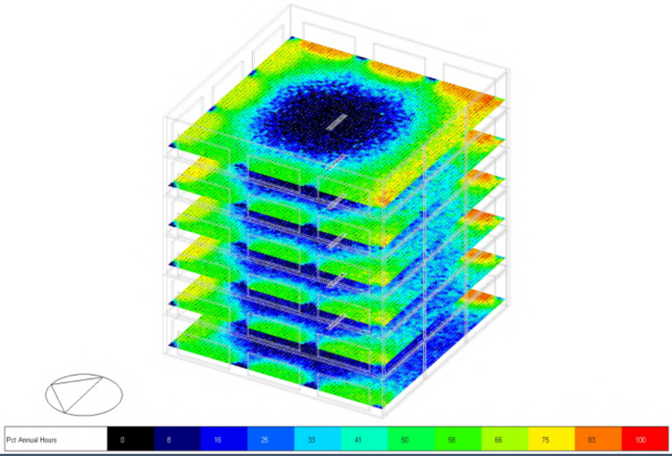
To rigorously validate the selection of case C2, a pareto frontier analysis was conducted by mapping sDA against total EUI. As delineated in louver optimization landscape (Figure 47), the scatter plot unveils a distinct efficiency frontier defined by the interplay between aerodynamic porosity and optical obstruction. Case C2 establishes the energy optimum vertex (lowest EUI: 100.48 kWh/m^2/year), prioritizing solar heat rejection and convective flushing through wide slats (340 mm) at a compact 30 mm window offset (distance from the external glazing material), which inevitably suppresses visual admission (sDA: 30.64%). Conversely, case C11 represents the visual optimum, achieving the highest sDA of 44.21% through a combination of micro-slats (30 mm width), a 60˚ slat angle, and an extended 300 mm offset. This specific geometry minimizes optical obstruction and maximizes the sky view factor, allowing profound diffuse light penetration. However, this luminous maximization induces a severe thermodynamic penalty; the dense 30 mm gap fundamentally restricts aerodynamic flushing, creating a thermal trap that drives the total EUI up to 105.29 kWh/m^2/year. Meanwhile, case C4 functions as strategic intermediate solution on the non-dominated frontier, offering a moderate visual improvement (sDA: 33.33%) for a negligible energy compromise (EUI: 100.73 kWh/m^2/year).

The placement of case C2 on this frontier substantiates the conclusion that achieving complete energy reduction in hot-humid climates is thermodynamically unfeasible without a calculated compromise in sDA. This empirical finding corroborates the observation of Khidmat et al. (2022), demonstrating that effectively managing trade-offs requires precise geometric tuning rather than single-variable maximization. The spatial implication of this trade-off is made directly legible through the sDA false-colour distribution across all configurations (Figure 46), confirming that case C2 – despite its energy optimum position on the pareto frontier – suppresses sDA to 30.64%, a quantifiable visual cost that the aerodynamic bifurcation analysis in the following section contextualises within the broader thermodynamic performance landscape.

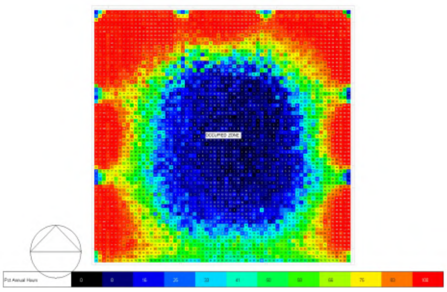
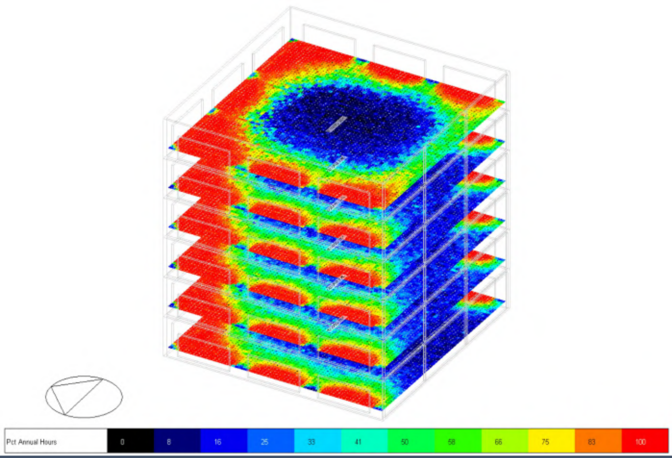
Energy Optimal Case C2 (sDA: 30.64%)

Floorplan	Isometry
	

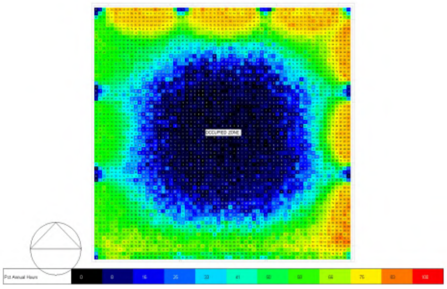
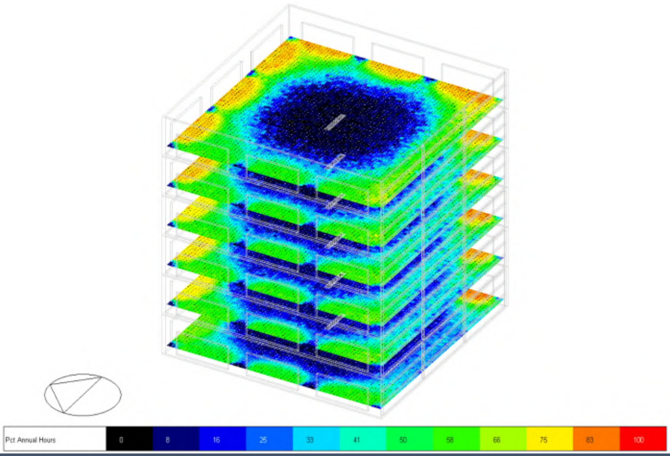
Case C4 (Pareto Frontier) (sDA: 33.33%)

Floorplan	Isometry
	

Case C12 (Pareto Frontier) (sDA: 41.37%)

Floorplan	Isometry
	

Moderate Performance Case C7 (sDA: 30.91%)

Floorplan	Isometry
	

sDA Optimal Case C11 (Pareto Frontier) (sDA: 44.21%)

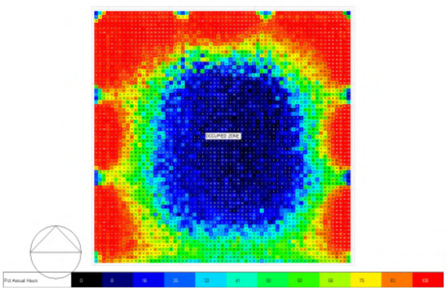
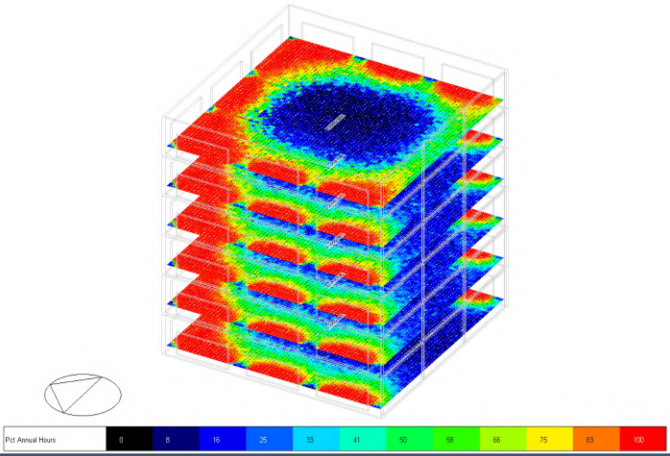
Floorplan	Isometry
	

Figure 46. sDA300/50% false-colour distribution across phase 1 louver parametric configurations

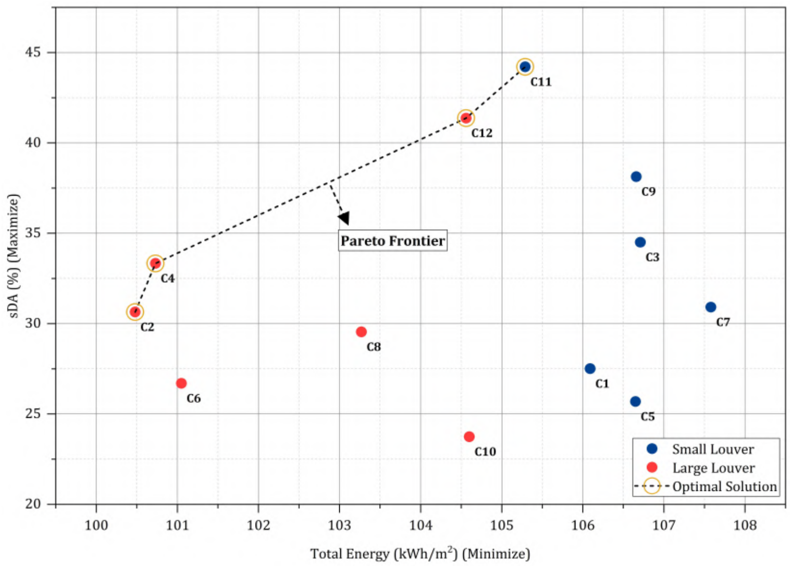


Figure 47. Efficiency frontier delineates optimal energy-daylight trade-offs

- **Aerodynamic Porosity and Convective Heat Dissipation**

Beyond optimizing solar interception and daylight admission, the configuration of the louver system exerts a fundamental influence on the aerodynamic flow resistance of the envelope. The simulation data reveals a critical thermodynamic division based strictly on the louver gap (porosity). As illustrated in Figure 48, configurations characterized by a restricted 30 mm gap consistently demonstrate elevated cooling loads across all slat angles. This phenomenon, known as the thermal penalty, arises from the dense configuration of louvers that impede the ambient wind flow, thereby creating a boundary layer of stagnant air against the external glazing. In the absence of adequate convective airflow, the solar radiation that is intercepted by the structure is converted into significant conductive heat gain, thereby transforming the façade into a thermal trap. In contrast, expanding the gap to 270 mm (permeable configurations) alleviates this aerodynamic bottleneck. This engineered porosity facilitates continuous convective heat dissipation, actively stripping the accumulated sensible heat from the building skin before it can conduct into the occupied zone. This phenomenon underscores the notion that in high-temperature tropical climates, the efficacy of external shading is not solely governed by optical cutoff angles. Rather, it is strictly constrained by the aerodynamic necessity to prevent structural heat stagnation. Consequently, it is imperative to prioritize the maximization of physical porosity as a prerequisite to circumvent the “choked flow” boundary that impedes natural ventilation.

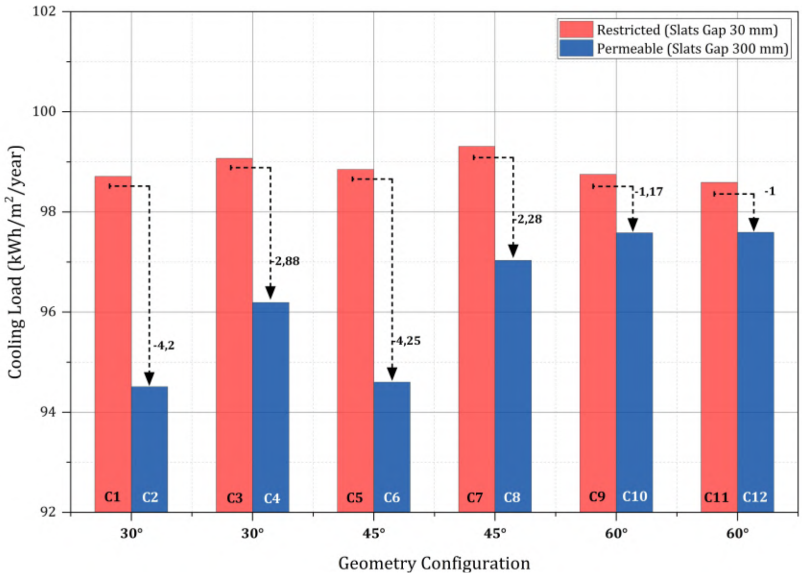


Figure 48. Permeable louvers gaps enhance convective flushing, consistently reducing annual cooling loads

The cooling optimization achieved by case C2 (35.49%) is empirically consistent with findings from comparable louver optimization studies in tropical and several climatic conditions. Table 16 benchmarks C2 against prior work, confirming the magnitude of the reduction while highlighting that the aerodynamic bifurcation phenomenon documented in this study – where restricted gaps universally increase cooling load regardless of slat angle – has not explicitly reported in the reviewed literature.

Table 16. Cooling reduction benchmarking of louver case C2 against prior external shading studies in tropical and other climatic conditions

Study	Climate / Location	Geometry	Result	Method	Key Finding
Saraswati et al. (2025)	Tropical (Surabaya, Indonesia)	90˚ louver angle	Met daylight standard (UDI) 100% (300 – 3000 lux); DF 3.2%	Simulation (Revit, Insight)	Shading device with a 90˚ louver angle perform optimally in meeting daylight illuminance standards without sacrificing natural light distribution
Dinapradipta	Tropical	Two-Bent slat	DF	Simulation	Louver with

et al. (2024)	(Surabaya, Indonesia)	(LB Type) with 90˚ opening angle; Slath width: 34 cm; Distance from window: 30 cm; Slat distance: 27 cm	discrepancies minimized to 2.6% for even daylight distribution; Reached 6.4%	(Radiance)	multiple bent slat geometry effectively reflect high altitude tropical incident light and scatter it evenly into the innermost room areas
Bramiana et al. (2025)	Tropical (Semarang, Indonesia)	Moderate angles (30˚, 45˚) for midday	DA: achieved 70% during work hours; Average illuminance at 1.00 p.m. was 593 lux (0˚) and 578 lux (15˚)	Simulation (Revit)	Flatter angles maximize early morning daylight but increase glare, whereas steeper angles effectively reduce glare at midday in tropical studios
Suryandono et al. (2021)	Tropical (Jakarta, Indonesia)	Type D (30 mm; 1:1 gap ratio) & Type E (1:2 ratio)	Cooling energy reduced by 17% (which peaked at 934.79 kWh/m^2); UDI improved by 16.78 points (98.25 – Type E average)	Simulation (Rhino + Honeybee)	L-shaped mini louvers a highly economical solution with a 0.86-year payback period, outperforming traditional overhangs
Khidmat et al. (2022)	Tropical / Temperate (Jakarta, Birmingham, Sidney)	Overhang 30 cm, spacing 30 cm, blade size 40 cm, rotation 5˚	UDI improved to 76.6% (Jakarta), 65.6% (Birmingham 80% improvement), and 65.8% (Sydney, 79.4% improvement); cooling energy reduced by 3.26% (Jakarta	MOO simulation (Octopus + Ladybug)	Spacing and blade size are the most influential parameters driving objective performance for balancing daylight maximization and cooling energy minimization
Olaniyan et al. (2023)	Tropical (Ogbomoso, Nigeria)	Exterior louver system	Indoor solar gains reduced significantly by 56.55%;	Simulation (DesignBuilder)	The external louver system demonstrated superior

			Mitigate high diffuse solar radiation (peaking at 114.43 kWh/m^2 in March) and direct solar radiation (84.48 kWh/m^2)		effectiveness in minimizing solar gains compared to side-fins and standard overhangs in tropical envelopes
Riviere and Damour (2023)	Hot and Humid (Reunion & Mayotte)	Balanced 50% shading density (coverage); Slat thickness: 3 cm; Slat width: 5 – 25 cm; Slat angle: 0˚ - 50˚ (> 20˚: optimal)	100% shading gives excellent solar protection (Cm 0.28) but eliminates daylight; DA: increased by 8%; AC: energy reduced by 40% when blade spacing equals twice the blade width	Simulation (EnergyPlus; jEPlus)	It is impossible to find a single optimal angle for all parameters; a trade-off is required between solar shading effectiveness and ensuring adequate blade spacing for natural ventilation
Dutta et al. (2017)	Tropical (Kolkata, India)	Automated exterior venetian blinds (dynamically adjusted to 90% shading factor during 10.00 a.m. – 2.00 p.m.) (South-facing window shading)	9.8% average annual savings; Thermal heat gain reduced	Simulation (TRNSYS)	South-oriented shading most effectively reduce cooling loads and discomfort
Bahdad et al. (2024)	Tropical (Malaysia)	Dynamic shading louvers integrated within DSF insulated glazed units	Thermal dissatisfaction reduced by 28.72%; Cooling load reduction: 26.2%	Simulation (Rhino + GA)	Dynamic louvers drastically reduce cooling loads and discomfort
Alafchi et al. (2025)	Semi-Arid (Iran)	Denser light-guiding shading	PMV shifted to comfortable range (+0.13	Simulation (Honeybee)	Design enhancing daylight generally raises cooling

		devices (LSD) optimized from 0.4-3 m spacing and 0-2 m depth ranges	to +0.23); UDI: 59.08 – 60.92%; ASE: decreased 25.66%; ~30 $kWh/m^2/yr$ EUI reduction (max)		loads
Abedini et al. (2025)	Hot & Dry (Qom / Tehran, Iran)	Horizontal wooden H-louvers: 1 m depth, 4 fins, 0.3 m spacing, 50% WWR	TCP improved to 74.18%; sDA: 100%; ASE: reduced to 9.38%; 14.95% EUI reduction	MOO simulation	H-louvers provide best thermal comfort and glare reduction – effectively blocking sunlight from multiple angles compared to vertical louvers
Shaeri et al. (2022)	Hot-Humid, Hot-Dry, and Cold (Iran)	Horizontal slats (2 mm thick, 30 cm width, 30 cm spacing); Angled at +60° or +70° (South / West)	21 – 28.29% energy consumption reduction; 51% increase in adjusted UDI via dynamically adjusted louvers	Simulation (DesignBuilder)	The location of the window area and the louver angle are the most critical factors affecting thermal comfort and energy reduction across diverse climates
Ma et al. (2024)	Moderate (Qingdao, China)	Centre-mounted louver ventilation	Improved UDI; 34.4% total energy reduction (83.81 $kWh/m^2/year$ to 55 $kWh/m^2/year$)	Simulation (Rhino + Ladybug)	Shading must be considered early in the design process; louver ventilation strategies must be flexibly adjusted according to seasonal changes
Nicoletti et al. (2023)	Mediterranean (Rome, Italy)	Venetian blind (1 mm thick, 2.5 cm width, 1.875 cm spacing, bi-directional; angle: 0 - 180°)	Up to 38.7% cooling savings	Simulation (EnergyPlus)	ANN effectively tracks optimum angle for shading devices
This Study – Case C2	**Tropical Equatorial (Jakarta, Indonesia)**	**Gap: 270 mm / slat width: 340 mm / angle: 30°/**	**35.49% cooling reduction (cooling EUI:**	**Parametric + Pareto (EnergyPlus)**	**Aerodynamic bifurcation identified; Permeable**

		offset from external window: 30 mm	94.51 kWh/m^2/yr)		configuration is non-dominated pareto optimum

3.7.5. Comparative Sensitivity and Dominant Cooling Mechanism

As demonstrated in Sections 3.6.2 through 3.6.4, the component-level parametric analyses established the individual performance optima of the DSF, TST, and louver system. However, a critical question regarding the thermodynamics of the system remains unresolved by isolated parametric sweeps. This is despite the fact that the psychrometric boundary analysis in Section 3.2 clearly identifies solar shading of windows as the sole passive strategy with demonstrable efficacy in Jakarta's equatorial climatic conditions, attaining 33.2% and 31.1% coverage of dry and rainy season operational hours, respectively. The question therefore arises as to why the DSF (case A10: cooling load 90.39 kWh/m^2/year) consistently outperforms the louver system (case C2: 94.51 kWh/m^2/year), which is architecturally defined as a direct solar interception device. This section resolves the aforementioned paradox through the implementation of a dual analytical framework: Firstly, a cross-system sensitivity ranking via R index (see Eq. 2), and secondly, a decomposition of the heat gain pathway derived from EnergyPlus zone-level outputs.

- **Parametric Sensitivity Hierarchy**

The application of the index across all 32 parametric configurations results in the generation of a cross-system performance hierarchy (see Figure 49). The DSF cavity depth has been found to demonstrate the highest sensitivity index (R = 2.81 kWh/m^2/year), followed by louver gap porosity (R = 2.63 kWh/m^2/year), TST aerodynamic topology (R = 1.06 kWh/m^2/year), louver slat angle (R = 1.01 kWh/m^2/year), louver distance from the external glazing (R = 0.80 kWh/m^2/year), and TST height (R = 0.44 kWh/m^2/year). DSF glazing SHGC registers the lowest overall sensitivity index across all parameters (R = 0.39 kWh/m^2/year). This verifies that, within the DSF parametric domain, cavity geometry, as opposed to glazing optical properties, is the primary performance driver. It is also demonstrated that, once aerodynamic porosity (louver gap) has been optimized, the physical offset distance of the slat assembly exerts negligible incremental influence on the cooling load. The predominance of cavity depth as the paramount parameter across all systems is indicative of a thermodynamic mechanism that is inherently divergent from the solar interception function exhibited by the louver system. As the cavity depth scales from 0.25 m (mean level K = 93.025 kWh/m^2/year) to 1.50 m (K = 90.22 kWh/m^2/year), the system transitions from a static optical barrier to an active thermal exhaust mechanism – a transition that no single louver geometric parameter can replicate. This distinction constitutes the fundamental rationale for the superior aggregate cooling performance exhibited by DSF, a claim that is quantitatively substantiated through the heat gain decomposition analysis that is presented below.

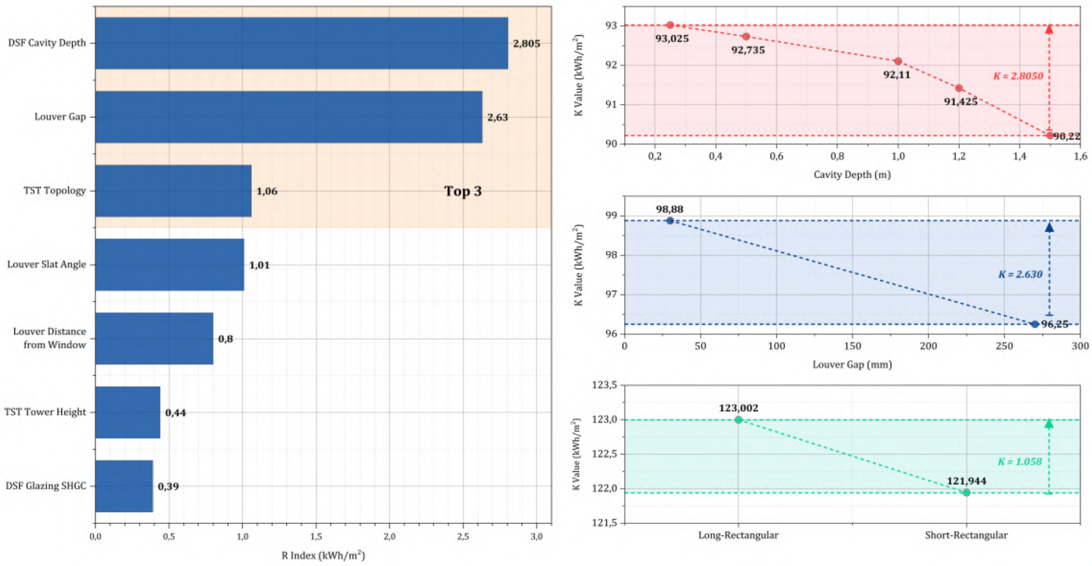


Figure 49. Parametric sensitivity hierarchy and mean cooling load response across design parameter levels

- **Heat Gain Pathway Decomposition: Resolving the Psychrometric Paradox**

The R sensitivity analysis (Section 3.5) was extended to determine the annual sensible cooling load of each optimal configuration. This was decomposed into three constituent pathways – solar radiative gain, internal sensible gains, and conductive heat flux (residual) – derived from EnergyPlus zone-level hourly outputs. These were integrated across 8,760 timesteps and normalized by 2,400 m^2, with arithmetic closure errors confirmed below 0.01 kWh/m^2/year for all configurations (see Figure 50; Table 17). The decomposition data reveals a counterintuitive finding that directly resolves the psychrometric paradox. It has been demonstrated that louver C2 achieves a marginally lower solar radiative load (18.07 kWh/m^2/year) than DSF A10 (19.23 kWh/m^2/year). This finding serves to empirically confirm the hypothesis that louver is the superior direct solar shading device, thereby aligning with the psychrometric mandate outlined in Section 3.2. This demonstrates that louver C2 exhibits a reduced internal gain load (43.69 vs. 49.91 kWh/m^2/year for DSF), attributable to its enhanced diffuse daylight transmission (sDA: 30.64% vs. 42.81% - Section 3.6.2), which effectively suppresses the contribution of artificial lighting heat to the occupied zone.

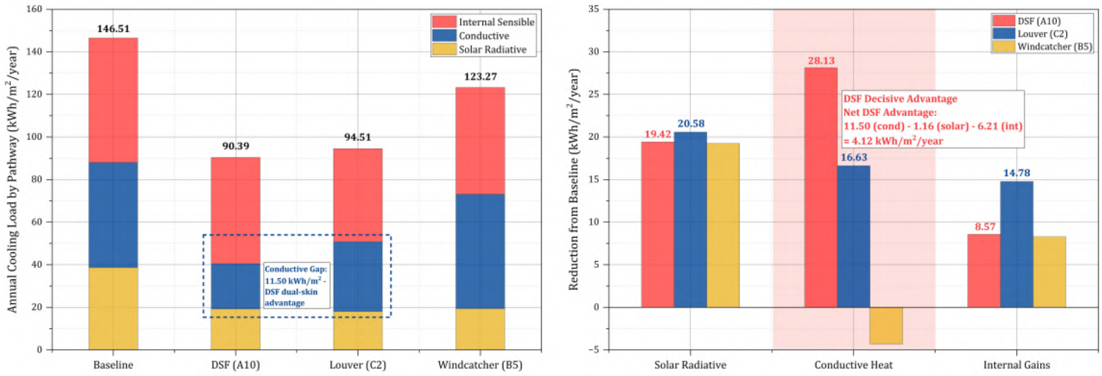


Figure 50. Solar, conductive, and internal heat gain pathway decomposition across optimal passive configuration

Table 17. Heat gain pathway decomposition of optimal configurations

Configuration	Cooling Load (kWh/m^2/year)	Solar Radiative (kWh/m^2/year)	Conductive heat Flux (kWh/m^2/year)	Internal Sensible (kWh/m^2/year)
Baseline	146.51	38.65	49.38	58.48
DSF(case A10)	90.39	19.23	21.25	49.91
Louver (case C2)	94.51	18.07	32.75	43.69
TST (case B5)	123.27	19.39	53.70	50.19

The margin of decision is confined exclusively to the conductive heat flux pathway. The DSF A10 registered 21.25 kWh/m^2/year, whilst the louver C2 recorded 32.75 kWh/m^2/year – a differential of 11.50 kWh/m^2/year, constituting the DSF’s singular thermodynamic advantage (see Table 17). This phenomenon can be attributed to the dual-skin construction, which comprises an outer skin that localizes solar radiation as cavity heat. This heat is then actively removed by the buoyancy-driven exhaust mechanism (Eq. 4) before conduction across the inner glazing boundary. In contrast, the single-skin louver assembly does not possess this heat removal pathway. The net pathway balance is quantitatively precise: the conductive advantage of DSF (11.50 kWh/m^2/year) is partially offset by the solar and internal gain advantages of louver (1.16 kWh/m^2/year and 6.22 kWh/m^2/year, respectively). The net DSF superiority is predicted to be 4.12 kWh/m^2/year, which corresponds exactly to the observed cooling load differential (94.51 – 90.39 = 4.12 kWh/m^2/year). This closure confirms that the three-pathway framework accounts for the totality of the observed performance divergence without residual unexplained variance. The hypothesis is further corroborated by TST B5, which, despite exhibiting comparable solar reduction to DSF (19.39 vs. 19.23 kWh/m^2/year), has a conductive load (53.70 kWh/m^2/year) that exceeds the baseline (49.38 kWh/m^2/year) by 4.31 kWh/m^2/year. This quantitative evidence confirms the hygrothermal convective penalty identified in Section 3.6.3.

- **Synthesis: Implication for MCDA Scenario Definition**

The cross-system sensitivity ranking and heat gain decomposition collectively establish three evidence-based principles that govern the MCDA scenario boundaries defined in section 3.6.6 (see Figure 51). Firstly, the data confirm that the psychrometric mandate for solar shading is both necessary and insufficient in terms of achieving the desired outcomes. Solar interception, as demonstrated by the louver system, is incapable of closing the 11.50 kWh/m^2/year conductive differential that is eliminated by the DSF's dual-skin geometry. Secondly, it is evident that DSF cavity depth, the parameter with the highest sensitivity index, is the most leverage-efficient design variable across all 32 parametric configurations. This renders its maximization (1.5 m, case A10) the non-negotiable anchor of the optimal scenario. Thirdly, the TST's conductive penalty structurally precludes its inclusion as a primary cooling mechanism; its role within the integrated system is aerodynamic rather than thermal, a distinction that is justified by the separate diagnostic framework applied in Section 4.3. The evidence shows that the three aforementioned principles establish that the thermodynamic supremacy of DSF in Jakarta's equatorial climate is not a function of superior optical shading – a role in which the louver demonstrably excels – but rather of its capacity to eliminate the conductive heat pathway that governs 50.1% of the total cooling load reduction achieved by the optimal DSF configuration.

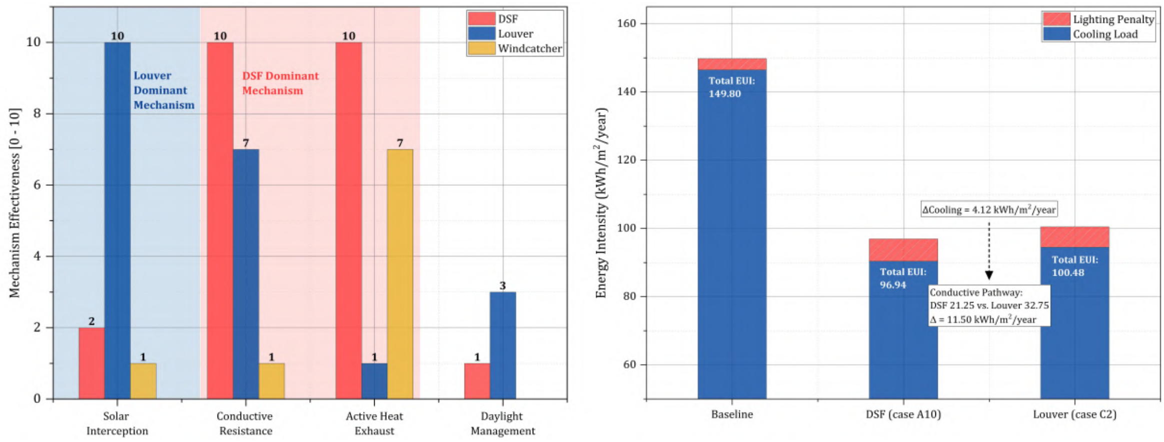


Figure 51. Comparative thermophysical mechanism profile and total EUI breakdown (DSF vs. louver) by passive system

3.7.6. Summary of 1st Phase Simulation: MCDA and Scenario Definition

To transition from isolated component sensitivity to holistic integration, MCDA was executed for the purpose of synthesizing the parametric hierarchy that had been established in phase 1. As demonstrated in the preceding component-level analyses, the evaluation of the disparate performance metrics of the DSF, TST, and louver system was conducted through the utilisation of a pareto-based optimisation framework. This approach enabled the objective assessment of the conflicting trade-offs between total EUI, sensible cooling constraints, and visual energy penalties. In contrast to the conventional single-variable maximization approach, this MCDA methodology employs three quantitatively isolated architectural extremities to define the boundary conditions for the subsequent phase 2 evaluation. The boundary conditions established here are defined relative to the baseline total EUI of 149.80 kWh/m^2/year and cooling intensity of 146.51 kWh/m^2/year, which serves as the thermodynamic reference for the integrated scenario evaluation presented in phase 2.

Optimal integration (**scenario 1**) is the amalgamation of the thermodynamic optima identified in the preceding analysis, specifically the coupling of the deep-cavity DSF (case A10: 1.5 m depth with high-value glazing thermal properties) for maximum buffer capacity, the high-capacity long-rectangular TST (case B5: 9 m) to maximize buoyancy-driven ventilation potential, and the permeable aerodynamic louver (case C2: 270 mm slat gap, 30° slat angle, 340 mm slats width, and 30 mm distance from the external glazing material) to facilitate convective heat dissipation. This configuration is hypothesized to theoretically maximize the sensible load decoupling capabilities of the building envelope. Critically, scenario 1 is not expected to be a complete solution; it is the starting point for identifying where passive optimization structurally fails. **Scenario 2** (moderate) functions as an intermediate validation benchmark, integrating the standard performance tiers (cases A5, B1, and C4) to evaluate the point of diminishing returns in geometric sizing. If scenario 2 consistently occupies the intermediate position across all metrics, it validates that the parametric sensitivity rankings from phase 1 translate predictably into integrated system behaviour – a necessary condition for the thermodynamic boundary analysis to be considered representative rather than anecdotal. In contrast, **scenario 3** (worst-case) combines the least effective geometric parameters, namely the restrictive shallow-cavity DSF (case A1), the stagnant short-rectangular TST (case B6), and the obstructed louver configuration (case C7), to establish a lower baseline for performance failure. The three-scenario boundary architecture is deliberately structured as an extremal bracketing framework – not a densely sampled pareto sweep – wherein in the optimal and worst-case configurations define the thermodynamic poles of the parametric space, with the moderate scenario confirming the monotonicity of the performance trajectory between them. This extremal approach is methodologically sufficient for the objective of thermodynamic limit definition, as the boundaries – not the interior – govern the claim.

The definition of these boundary conditions stems from a rigorous comparative analysis of all 32 simulated parametric configurations. The comprehensive performance dataset (see Appendix) and the corresponding parametric efficiency heatmap for every iteration are detailed in Figure 52, and the synthesis of these distinct tiers is prioritized here. This strategic classification is illustrated in the performance radar (see Figures 53, 54, and 55), which superimposes the efficiency footprints of the selected cases, emphasizing the trade-offs between sensible energy conservation (cooling load, total EUI, energy saving), aerodynamic permeability (ACH, daytime volumetric airflow), and where applicable, daylight performance (sDA). The comprehensive quantitative specifications for these scenarios are detailed in the consolidated simulation matrix (Table 18). Moreover, to elucidate the physical integration of these complex systems, Figures 56, 57, and 58 present the schematic configurations comparison, which illustrates the system overview, detailed mechanism, and component assembly for each scenario. These consolidated configurations function not merely as design options, but as the critical boundary conditions required to diagnose the “choked flow” phenomenon and validate the sensible-latent decoupling hypothesis in the forthcoming Phase 2 integrated system analysis.

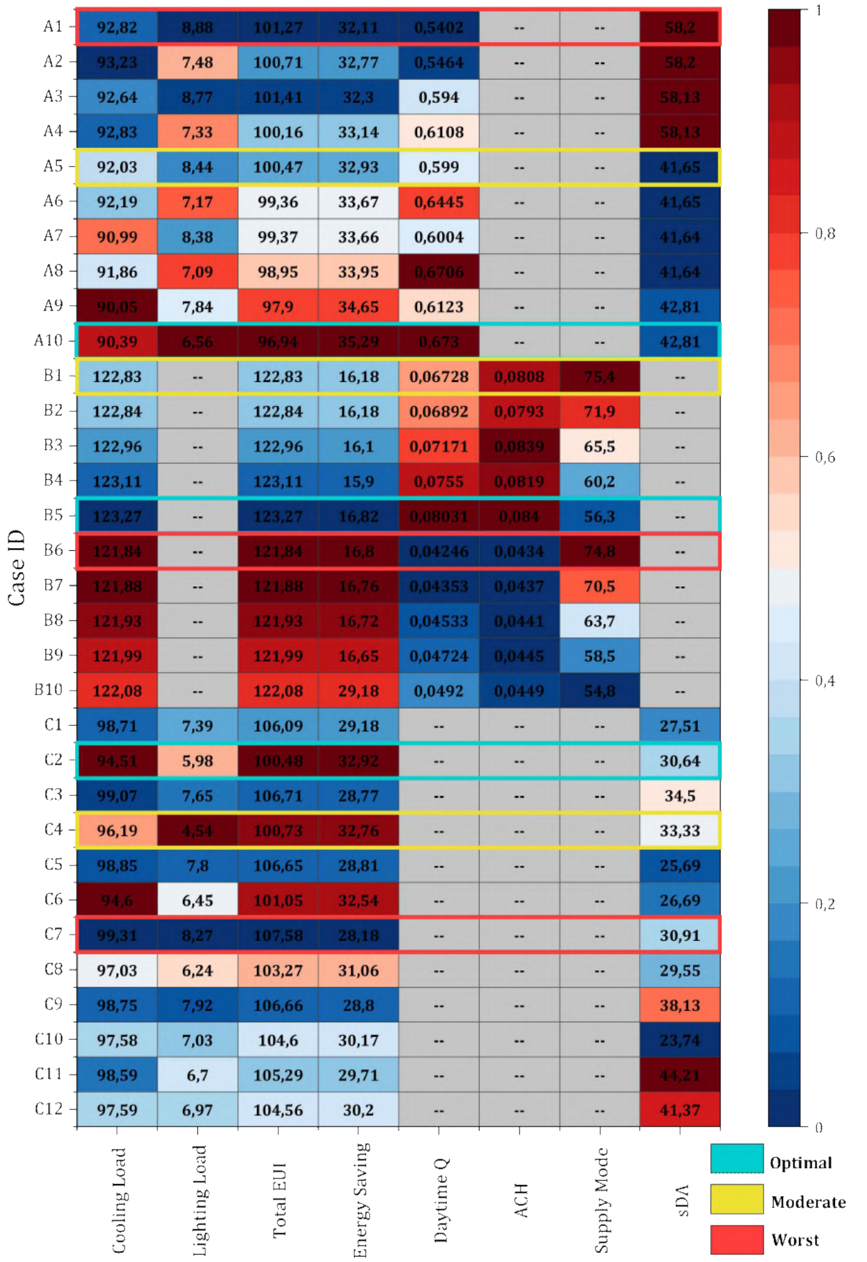


Figure 52. Comprehensive parametric efficiency and case selection matrix

The radar plot for the DSF element (see Figure 53) illustrates a discernible thermodynamic expansion from case A1 to the optimal configuration (case A10). As depicted in the case A1 scenario, the performance footprint is constricted, indicating a limited capacity to mitigate solar gain. Conversely, case A10 exhibits a substantial expansion of the efficiency envelope, particularly along the cooling load and total EUI axes. This shift indicates that deepening the air cavity serves as a critical thermal buffer, effectively decoupling the indoor environment from intense ambient thermal loads. While this deepened cavity allows a calculated regression in thermal transmittance owing to its specific glazing properties, the substantial net gain in total energy efficiency (driven by the reduction in lighting energy demand) justifies case A10 as the superior holistic boundary. It is imperative to note that the daytime *Q* axis, which signifies mean daytime volumetric throughput from Table 18, substantiates that case A10's enhanced thermal performance is not merely attributable to passive insulation, but rather is predominantly driven by its stack effect capacity (0.6730 m^3/s). The concurrent expansion of the energy saving axis (35.29% relative to the baseline) validates that aerodynamic enhancement and aggregate energy reduction are co-dependent outcomes of the 1.5 m cavity geometry.

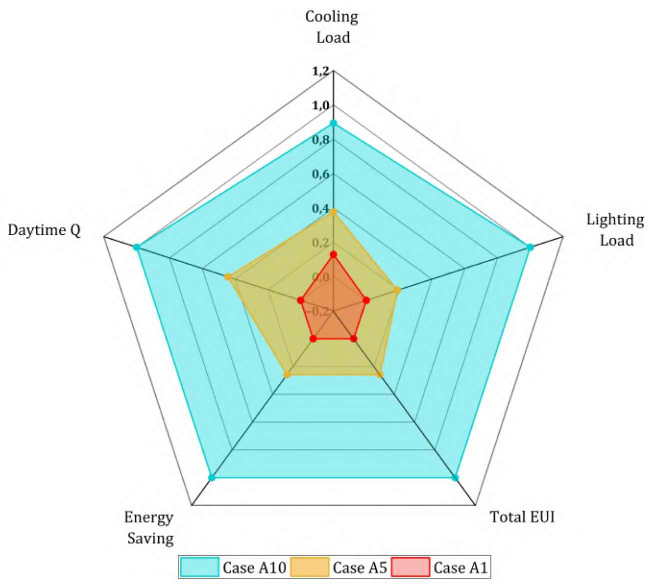


Figure 53. DSF element performance radar

The TST comparative analysis reveals a pronounced dichotomy between energy conservation and ventilation effectiveness. The radar profile for case B6 appears to be efficient on the energy axis; however, this is a symptom of "choked flow," where restricted air intake prevents heat ingress, but at the cost of paralyzing the natural ventilation rate. Conversely, the optimal case B5 demonstrates a deliberately asymmetric performance polygon – achieving maximum scores on the aerodynamic axes (ACH, daytime Q, exhaust mode) while accepting minimum scores on the energy axes (cooling, energy saving). Whilst this configuration necessitates a calculated reduction in cooling efficiency, it attains the highest volumetric airflow capacity required to dissipate accumulated structural heat. This observed asymmetry, therefore, cannot be considered a deficiency; rather, it serves as tangible evidence that challenges the conventional wisdom surrounding the ventilation-cooling paradox. This paradox posits that maximization of buoyancy-driven heat flushing in tropical conditions is thermodynamically incompatible with the minimization of cooling energy. This finding serves to provide substantiation for the hypothesis that case B5 represents the functional optimum, as it prioritises continuous aerodynamic flushing over the deceptive energy savings that would be achieved in a worst-case scenario characterized by stagnation – the central trade-off that justifies the hybrid decoupled framework proposed.

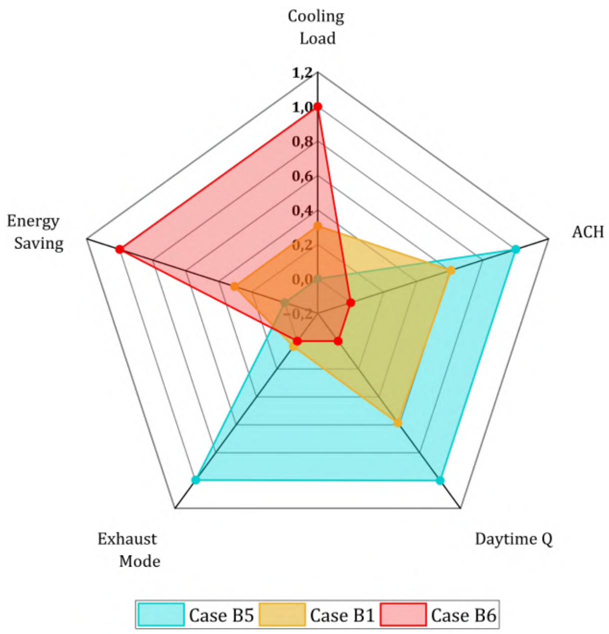


Figure 54. TST element performance radar

The parametric evolution of the louver system illustrates a transition from the obstructed footprint of the worst-case scenario (case C7) to a finely tuned equilibrium in the optimal configuration. The radar plot for case C7 demonstrates negligible performance across all metrics, thereby confirming that aggressive slat angles have a detrimental effect on both daylight admission and convective heat dissipation. The analysis further underscores the critical trade-off between the visual-centric case C4 and the balanced case C2. While case C4 attains the zenith of lighting performance, it exhibits a compromise in thermal defence. Consequently, case C2 is identified as the pareto-optimal solution, exhibiting energy-dominant performance with maximum scores on the three primary energy axes at the cost of calculated daylight autonomy compromise (sDA: 0.3371 – lower than case C4). This trade-off is not a limitation but a deliberate thermodynamic decision: case C2 prioritizes the elimination of the 4.80 kWh/m^2/year cooling penalty over daylight maximization.

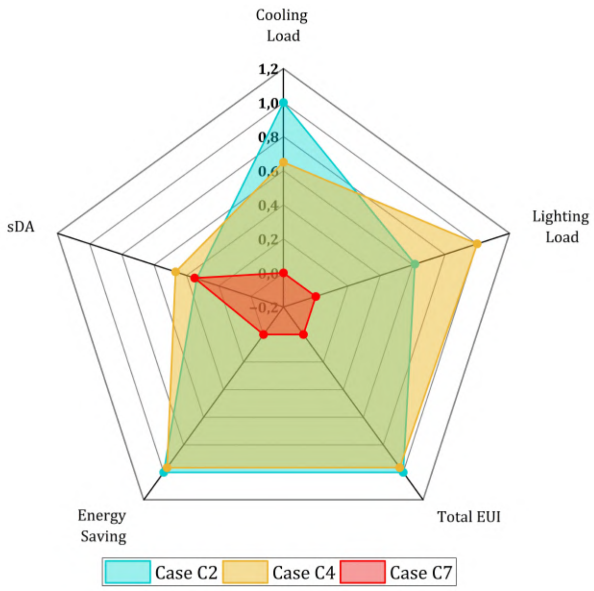


Figure 55. Louver element performance radar

Table 18. Comparative specification of geometric variables and thermodynamic properties of the three benchmark integrated scenarios

Passive Sub-System	Design Parameters	Scenario 1 (Optimal)	Scenario 2 (Moderate)	Scenario 3 (Worst-Case)
DSF	**Case ID**	**Case A10**	**Case A5**	**Case A1**
	Cavity Depth (m)	1.5	1	0.25
	SHGC (W/m^2K)	0.219	0.174	0.174
	U-Value	2.8618	0.345	0.345
TST	**Case ID**	**Case B5**	**Case B1**	**Case B6**
	Tower Topology	Long-Rectangular	Long-Rectangular	Short-Rectangular
	Tower Height (m)	9	2	2
	Inlet Cross Section Area (m^2)	20	20	3.6
Louvers	**Case ID**	**Case C2**	**Case C4**	**Case C7**
	Slat Inclination Angle (˚)	30	30	45
	Slat Width (mm)	340	340	30
	Slat Gap (mm)	270	270	30
	Distance from External Glazing (mm)	30	340	340
Methodological Function	Boundary Definition	Theoretical Upper Bound	Intermediate Benchmark	Failure Baseline (Lower Bound)

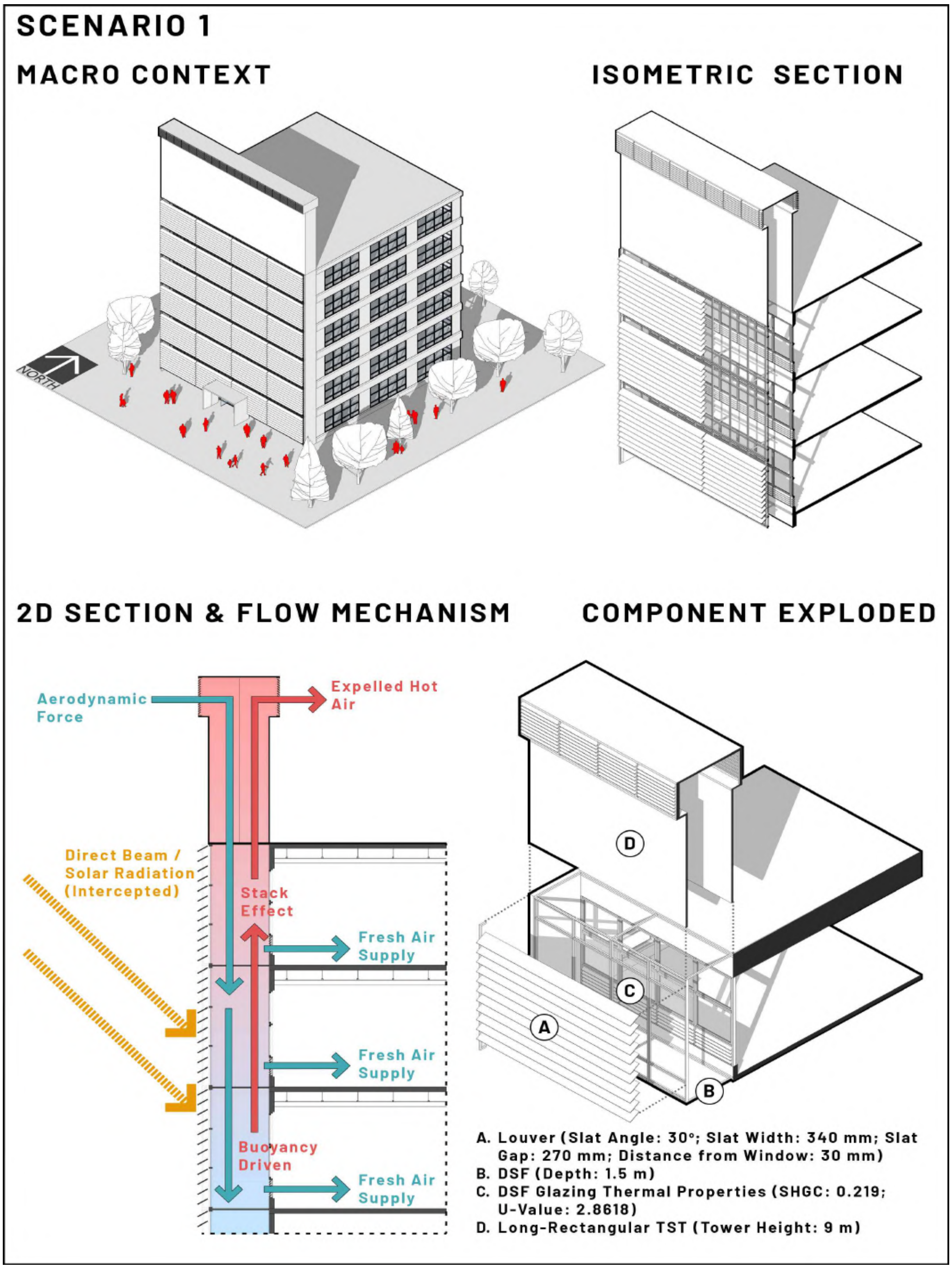


Figure 56. Scenario 1 system overview

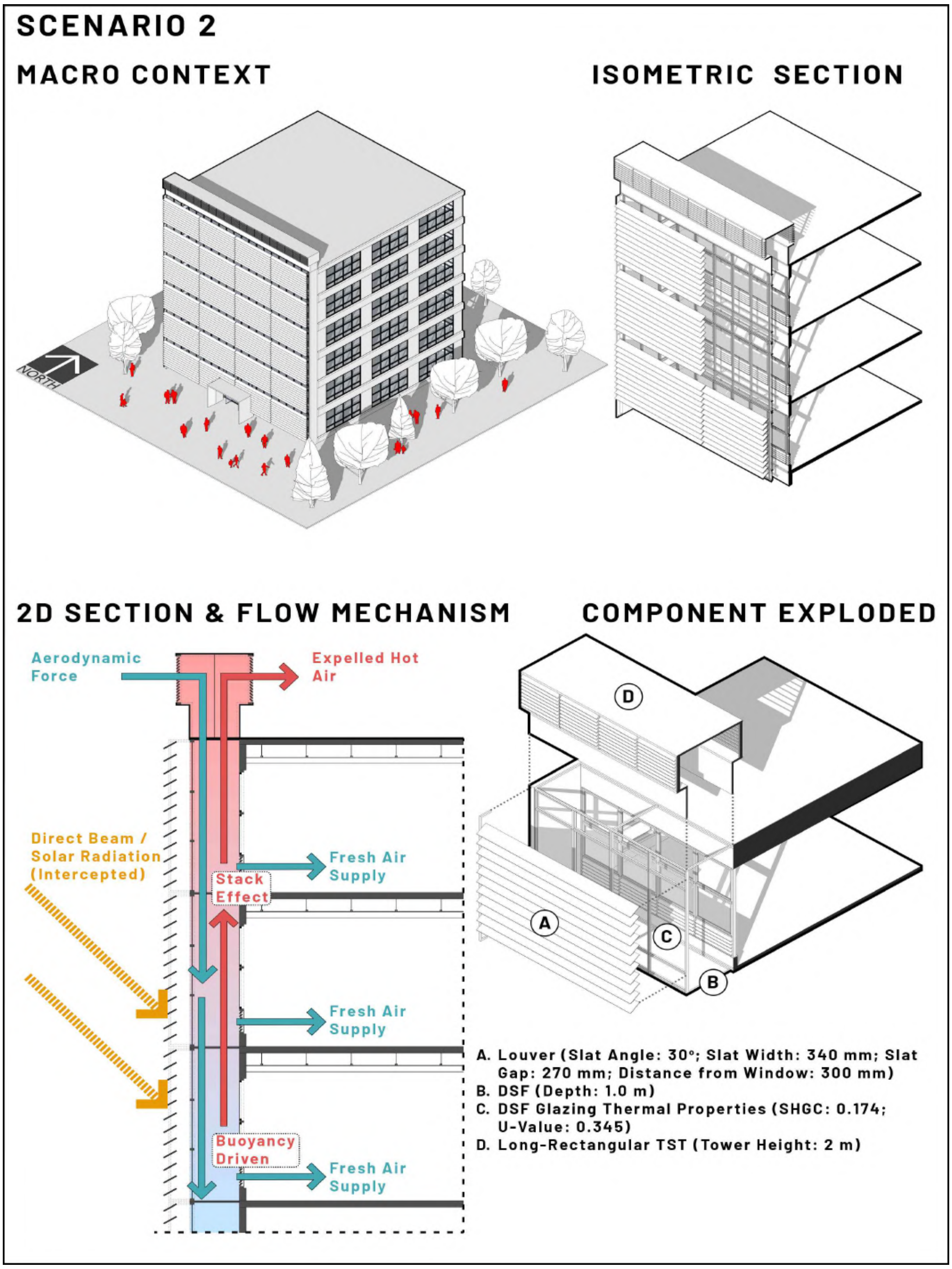


Figure 57. Scenario 2 system overview

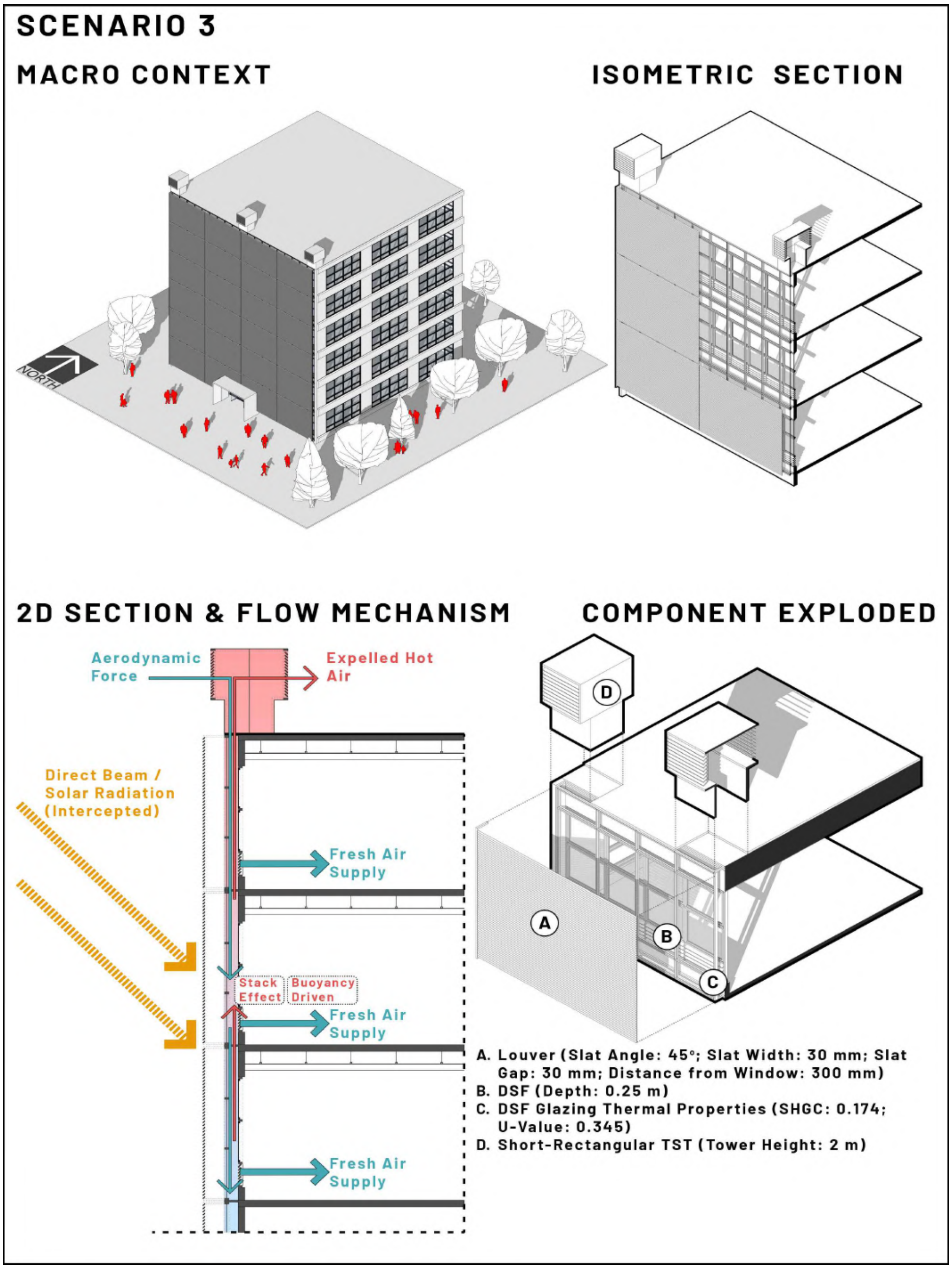


Figure 58. Scenario 3 system overview

The component-level optimal established in Chapter 3 – namely the 1.5 m deep-cavity DSF (case A10), the 9 m long-rectangular TST (case B5), and the permeable louver array (case C2) – are operationalized in this chapter through their systematic integration into three distinct performance scenarios. In integrated system, the louver's external position serves a function that extends beyond standalone solar control: by intercepting direct beam radiation before it reaches the DSF cavity, it reduces the thermal load that the cavity must subsequently buffer and exhaust. Diffuse light transmission is preserved simultaneously, maintaining daylighting sufficiency in the occupied zone, this establishes a deliberate sequential thermal cascade – louver as radiation filter → DSF as conductive buffer → TST as buoyancy-driven exhaust – where each component manages a distinct thermodynamic layer rather than operating in isolation. As a departure from the isolated parametric sensitivity of phase 1, the coupled thermodynamic assessment presented herein evaluates the aerodynamic and hygric synergies that emerge when these subsystems function as a unified envelope. The analysis is structured across two sequential diagnostic tiers. Firstly, the integrated performance assessment (Sections 4.1 – 4.3) is employed to quantify the energy efficiency profile, hygric loading and aerodynamic stagnation of the integrated system. Secondly, the thermodynamic boundary definition (Sections 4.4 – 4.6) is used to identify the aerodynamic and latent limits at which passive envelope performance is exhausted. This establishes the empirical justification for the hybrid decoupled framework advanced in Chapter 5.

4. Integrated Passive Envelope Simulation and Thermodynamic Limit Definition

4.1. Simulation Results of Phase 2: Integrated Performance Assessment

In this phase, the holistic performance of the integrated passive envelope system is evaluated by juxtaposing the three scenarios – scenario 1 (designated optimal), scenario 2 (designated moderate), and scenario 3 (designated worst-case) – against the baseline building model. Departing from the single-variable sensitivity analysis conducted in Phase 1, this section delineates the coupled thermodynamic interactions among the DSF, TST, and louvers. The deployment of comparative delta evaluation and multi-objective pareto front analyses enables the visualization of the performance trajectory across varying configuration cases, with a focus on decoding the thermodynamic conflict between sensible heat rejection and daylight admission.

4.1.1. Energy Efficiency Profile and Thermodynamic Trade-offs

The integration of passive strategies resulted in a fundamental shift in the building's energy profile, with the dominant load drivers altering from external solar gain to internal operational dynamics. Empirical simulation data reveals a distinct "visual-thermal trade-offs," whereby aggressive sensible cooling optimization invariably results in severe lighting energy penalties, necessitating a decoupled design approach.

A. Cooling Load Trendline: The Paradox of Airtighness

As demonstrated in Figure 59, the baseline model cooling demand is 146.51 kWh/m^2/year, symptomatic of a thermally porous envelope with an elevated SHGC. The integrated passive strategies were found to be highly effective in suppressing this load. However, the comparative trajectory reveals a non-linear threshold of diminishing returns. In a counterintuitive outcome, scenario 3 (worst-case) achieved the lowest sensible cooling load (95.25 kWh/m^2/year), outperforming the optimally designated scenario 1 (102.14 kWh/m^2/year). This margin does not denote superior architecture; rather, it manifests the "choked flow" phenomenon (see Figure 60). The aerodynamic bottlenecks and restrictive louver geometries of scenario 3 effectively transformed it into a sealed bunker with severe ventilation stagnation (ACH: 0.19), thereby minimising both solar heat and natural ventilation intake. In contrast, scenario 1 resulted in active envelope breathability. While this architectural permeability successfully promoted deep ventilation, it simultaneously vacuumed significant tropical moisture into the occupied zone via buoyancy-driven stack effect, causing a latent gain spike to 3,553.99 W (see Figure 61). Consequently, the marginally elevated cooling energy in scenario 1 can be attributed to the inherent penalty of mechanical dehumidification, as opposed to a thermal resistance failure.

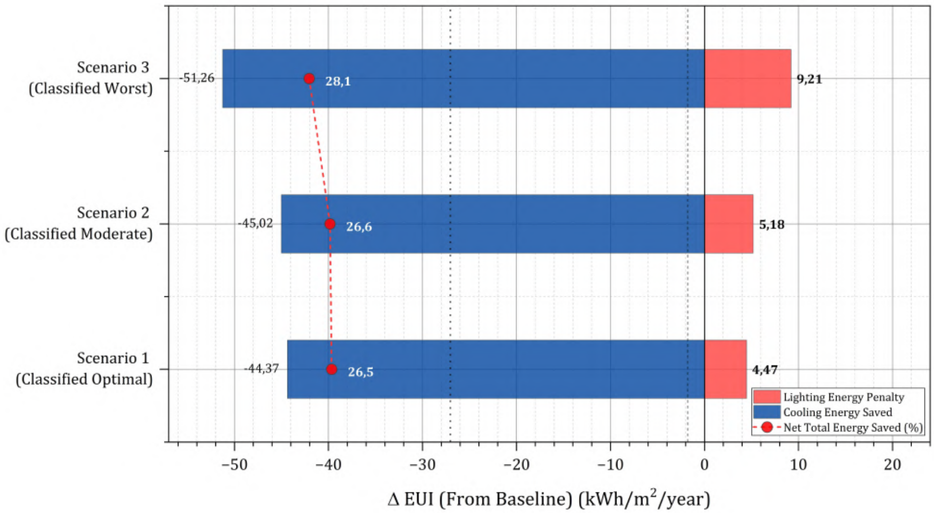


Figure 59. Diverging trade-off: sensible cooling savings vs. lighting penalties

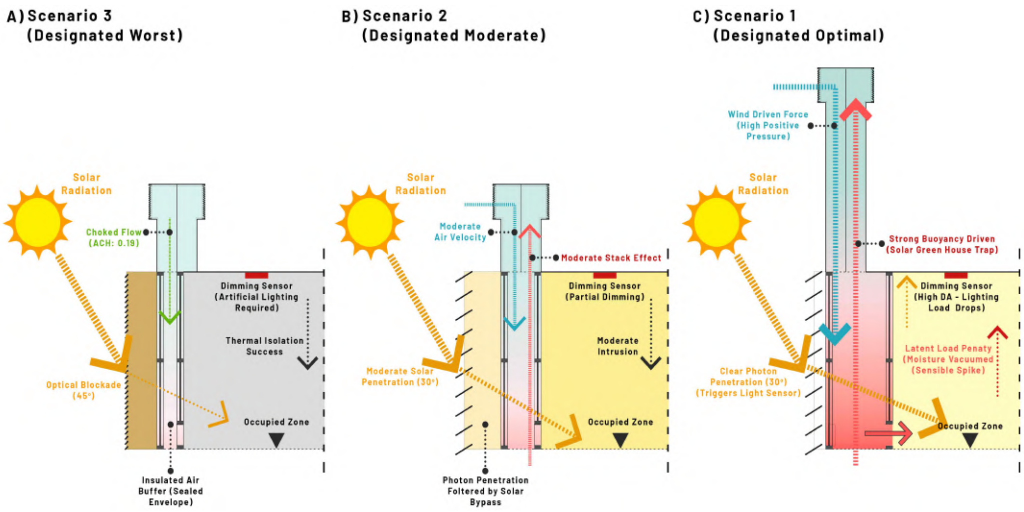


Figure 60. Passive envelope mechanics in phase 2: hygrothermal, aerodynamic, and dayligting

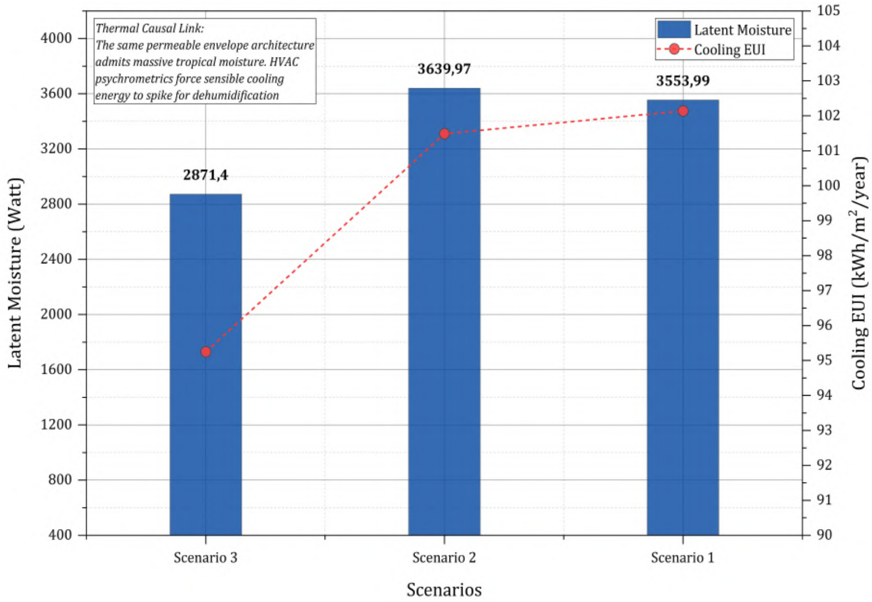


Figure 61. Thermal mechanism (moisture intrusion vs. sensible cooling)

B. Lighting Load Trendline: The Optical Penalty of Over-Shading

The reduction in cooling load was accompanied by a parasitic increase in artificial lighting demand. The baseline maintained a minimal lighting load of 3.29 kWh/m^2/year owing to unobstructed glazing, albeit suffering from intense thermal glare. In stark contrast, scenario 3 recorded the highest lighting demand at 12.50 kWh/m^2/year. The rigid external louver geometry was found to be responsible for the aggressive blocking of photon transmission, thereby suppressing the DA to a mere 38.58% of the operational hours. This severe visual deficit triggered continuous artificial illumination. In the scenario 1, the louver geometry was strategically widened, thereby driving the DA300 to 57.45% and effectively eradicating the lighting penalty down to 7.76 kWh/m^2/year. The necessity of daylighting is corroborated by Preto and Gomes (2019), who challenge the traditional physiological adequacy of static, low-illuminance workplaces (e.g., 200-500 lux). The argument is made that dynamic daylighting frameworks, which permit natural illuminance fluctuations throughout the day, are fundamentally superior in synchronising human circadian rhythms and promoting long-term organizational well-being compared to rigid artificial lighting constraints. Scenario 1 (see Figure 62) aligns with this dynamic paradigm by securing a DA300 of 57.45% (sDA300/50%: 50.54%), thus approaching – though not formally satisfying – the LEED v4 sDA300/50% credit threshold of 55%, whilst avoiding excessive glare. In contrast, scenario 3 collapsed to a visually starved 38.58%.

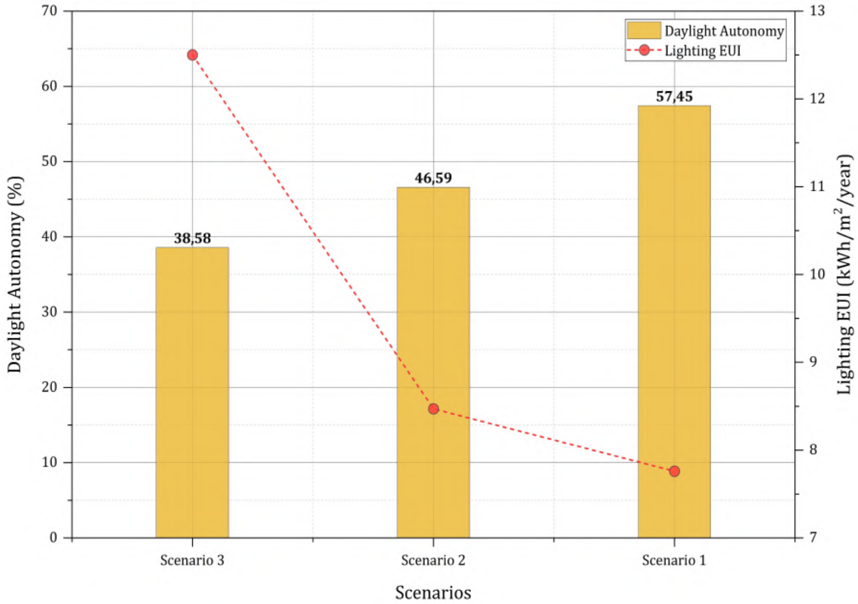


Figure 62. Optical mechanism (daylight availability vs. lighting penalty)

4.1.2. The Thermodynamic Saturation Point

The synthesis of these parameters results in the total EUI profile. The transition from the baseline (149.80 kWh/m^2/year) to passive frameworks resulted in a substantial initial energy decline. From a numerical perspective, scenario 3 was found to be most effective in terms of total energy savings, with a recorded total of 42.05 kWh/m^2/year, which corresponds to 28.07% of the total. Nevertheless, this perceived superiority is an illusion engendered by a lack of habitability, resulting in the exclusion of both daylight and fresh air. The data confirm that scenario 1 is designated as the constrained optimum, achieving a 26.47% reduction (110.14 kWh/m2/year) while maintaining the best attainable balance between thermal shielding and daylight admission — the latter approaching, though not formally meeting, the LEED v4 sDA threshold. As depicted by the pareto frontier (see Figure 63), this dynamic corroborates the fundamental thermodynamic limit of passive envelopes in Jakarta's high enthalpy climate. The data confirm that surpassing the limits of optimization in sensible heat rejection (scenario 3) precipitates a substantial parasitic penalty on visual energy, resulting in a marked deterioration in spatial visual comfort and daylight habitability.

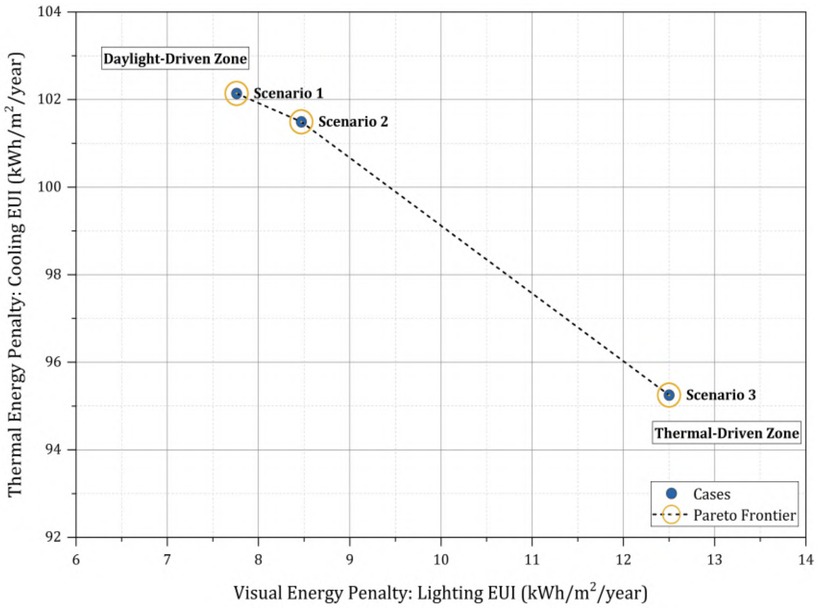
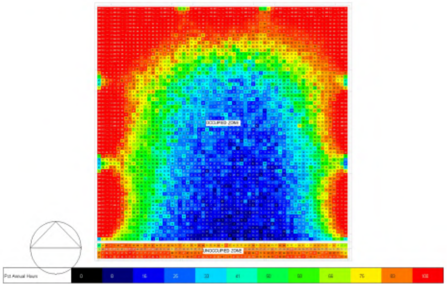
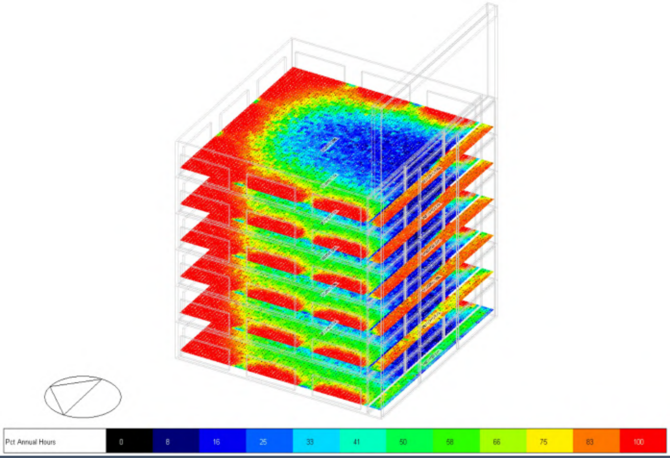
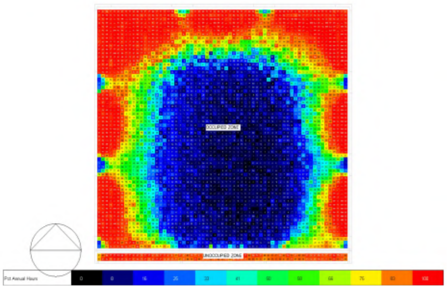
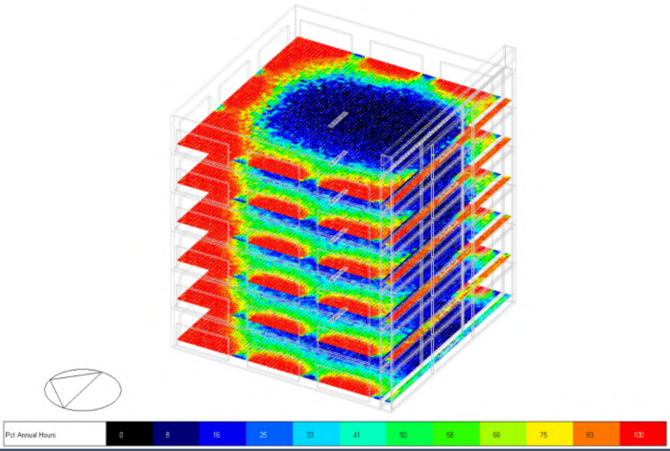


Figure 63. Energy trade-off: sensible cooling vs. daylighting admission

To rigorously validate this threshold, a multi-objective optimization analysis was performed. Figure 65 provides a visual representation of the conflicting correlation between sDA and total energy savings. The pareto frontier exhibits a steep gradient, confirming that aggressive energy minimization inherently compromises visual comfort. It is imperative to note that scenario 1 emerges as the constrained optimum of this distribution. It is only configuration that approaches the human-centric daylighting threshold (sDA300/50%: 50.54%, against the LEED v4 credit threshold of 55%) while maintaining a highly competitive 25.47% energy reduction. Critically, even this constrained optimum falls 4.46 percentage points short of the LEED v4 daylighting credit – a shortfall that does not weaken the analysis but rather substantiates it: it demonstrates that purely passive geometry cannot simultaneously satisfy thermal, hygric, and visual habitability criteria. This finding provide empirical validation of the central hypothesis that sensible thermal loads must be decoupled from latent constraints in tropical passive design. The spatial consequences of this sDA-energy trade-off are rendered directly legible through the false-colour distribution across phase 2 scenarios (Figure 64), confirming that scenario 1 (DA300: 57.45%; sDA300/50%: 50.54%) is the only integrated configuration that approaches the LEED v4 (USGBC, 2019) daylighting credit threshold of 55%, without formally attaining it. Meanwhile, scenario 3's (DA300: 38.58%) restrictive louver geometry renders the interior uniformly under-lit with sDA of 34.85% – corroborating the lighting EUI surge documented in the preceding trendline analysis (See Figure 65).

Optimal Scenario 1 DA300: 57.45% → (sDA: 50.54%)	
Floorplan	Isometry
	
Moderate Scenario 2 DA300: 46.59% → (sDA: 38.43%)	
Floorplan	Isometry
	

Worst-Case Scenario 3 DA300: 38.58% → (sDA: 34.85%)

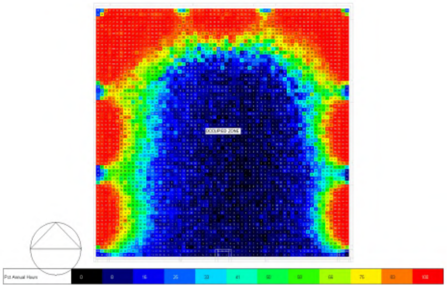
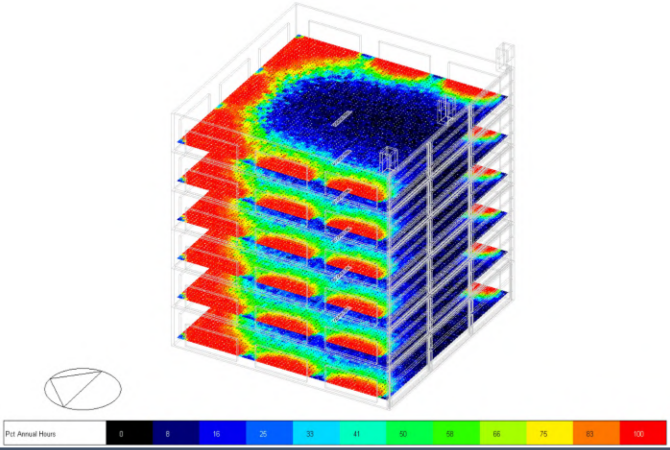
Floorplan	Isometry
	

Figure 64. sDA300/50% false-colour distribution across phase 2 integrated scenarios

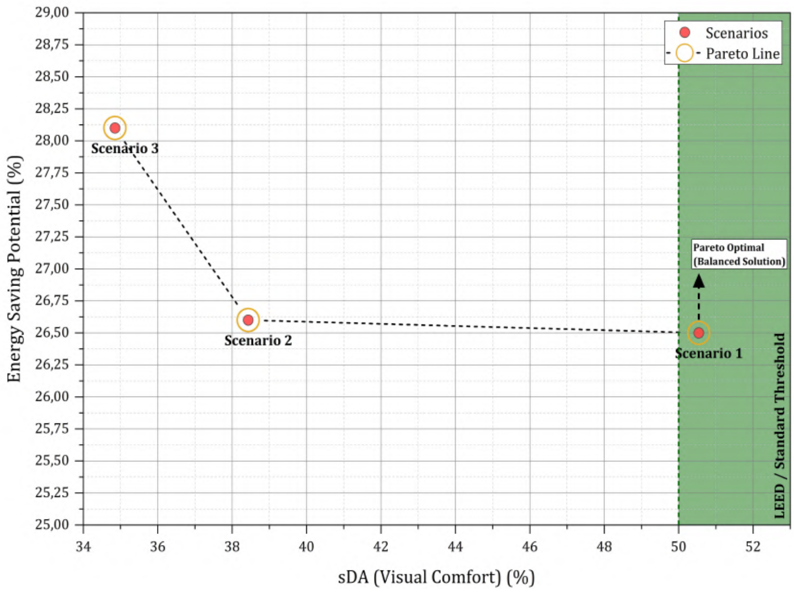


Figure 65. Multi-objective pareto frontier: Visual comfort vs. energy efficiency

4.2. Latent Energy Penalty and Hygric Boundary Diagnosis

The aggregate energy trajectory established in the preceding section demonstrates that, while sensible cooling suppression is substantial, it represents only one dimension of the total thermodynamic burden imposed on the mechanical system. A secondary constraint manifests at the hygric boundary, a problem that passive geometry alone cannot resolve.

This section examines how moisture dynamics within the integrated envelope translate into quantifiable dehumidification penalties, tracing the causal pathway from buoyancy-driven air exchange through psychrometric shift to mechanical energy overload.

4.2.1. Sensible Thermal Buffering Efficacy

The thermodynamic diagnostic of the integrated envelope confirmed a structural bifurcation between sensible buffering success and latent energy failure. In all scenarios examined, the passive skin clearly absorbed external solar heat, thereby reducing the mean OT from the initial baseline of 27.06˚C to 25.82˚C in scenario 1, 25.73˚C in scenario 2, and 26.19˚C in scenario 3, as measured by the cumulative distribution function (see Figure 66). The baseline threshold of 27.06˚C is independently corroborated by PMV-based thermal comfort analysis (ASHRAE Standard 55 appendix B; M = 1.1 met, Icl = 0.5 clo, V_a = 0.1 m/s), which identifies an OT of 27.05˚C at the PMV = +0.5 comfort zone boundary – a deviation of only 0.01˚C – confirming its thermodynamic validity as a psychrometric reference point. However, the apparent thermal superiority of scenario 3 – the most geometrically restrictive configuration – demands critical interpretation. The lower OT does not necessarily indicate a superior passive design; rather, it is a thermodynamic artefact of choked flow. The sealed geometry of scenario 3 has been demonstrated to suppress both external solar heat gain and natural ventilation simultaneously, thereby effectively transforming the envelope into a static insulation barrier. The direct energy consequence of this stagnation is a progressive escalation in the latent load fraction of total cooling demand. In the absence of convective flushing, internally generated moisture accumulates unchecked, forcing the mechanical system into continuous dehumidification under static air conditions. In contrast, scenario 1 maintains active aerodynamic breathability through the TST, albeit with a marginal sensible cooling penalty (102.14 kWh/m^2/year). This is achieved by means of buoyancy-driven heat expulsion, which prevents the structural heat accumulation observed in scenario 3.

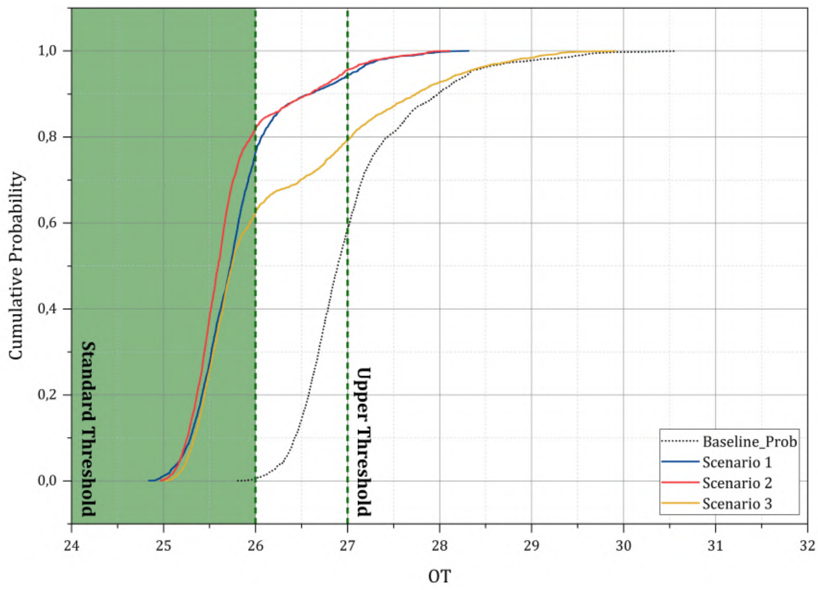


Figure 66. Thermal autonomy cumulative distribution function

4.2.2. Latent Load Thermal Diagnostic

The residual thermal indices are a quantitative measure of the dehumidification mandate imposed on the mechanical system. This mandate is the dimension of energy demand that passive geometry is unable to address. The unmitigated latent moisture introduced via the TST stack (3,553.99 W in scenario 1) sustains a thermodynamic condition in which the mechanical cooling system must operate beyond sensible heat removal, expending additional energy on active moisture extraction. This is reflected in the residual PPD, which, despite successful MRT suppression, remains at 15.04% in scenario 1. While not a comfort metric per se, it is a thermodynamic signal that the mechanical system continues to operate under latent stress despite successful sensible buffering. The persistence of this residual above the ISO 7730 (2005) Category B boundary (PPD < 10%) confirms that passive ventilation introduces tropical moisture at a rate that exceeds the envelope's passive dehumidification capacity. Table 19 provides a quantitative analysis of the hygric boundary across all configurations, with RH and latent gain identified as the primary diagnostic indicators of this mechanical energy mandate.

Table 19. Boundary diagnostic – latent energy penalty assessment (Operational hours: Monday – Friday; 09.00 – 17.00)

Configuration	Mean RH (%)	Latent Gain (W)	Latent Energy Implication
Baseline	50.68	3,740.18	Low altent penalty; severe sensible load dominates
Scenario 1 (Designated Optimal)	55.86	3,553.99	Active ventilation drives dehumidification demand
Scenario 2 (Designated Moderate)	55.75	3,639.97[9)]	Moderate latent penalty from partial ventilation
Scenario 3 (Designated Worst-Case)	56.47	2,871.40	Reduced external latent gain offset by internal moisture trapping

Note: Latent gain values represent mean operational-hour mechanical dehumidification load (W) derived from EnergyPlus zone moisture balance outputs. RH values represent mean operational-hour indoor RH.

A PMV/PPD diagnostic across 2,340 annual occupied hours confirms that scenario 1 achieves 54.66% comfortable hours (-0.5 ≤ PMV ≤ +0.5), representing a 2.96-fold improvement over the baseline (18.50%), with mean PMV reducing from 1.261 to 0.606, as summarized in Table 20 and illustrated in Figure 67. However, mean PPD of 17.76% persists above the ISO 7730 threshold (PPD < 10%) – directly consistent with the residual latent load of 3,553.99 W documented in Table 19, confirming that the hygric energy penalty constitutes a measurable thermal comfort deficit beyond the reach of passive sensible optimisation. Scenario 3 registers the lowest mean PPD (10.60%) and highest comfortable hours (70.51%), yet this outcome is thermodynamically artefactual – attributable to the sealed geometry's simultaneous suppression of both solar heat gain and natural ventilation, as corroborated by the latent moisture trapping mechanism identified.

9) Scenario 2 records a marginally higher latent gain than scenario 1 (3,639.97 W vs. 3,553.99 W) – difference of 2.4% - despite lower TST ACH. This reflects the multi-component nature of the zone moisture balance in EnergyPlus, wherein latent gain is governed by the aggregate interaction of TST ventilation, envelope infiltration, and DSF cavity thermal behavior across all three passive subsystems simultaneously, rather than by any single component in isolation.

Table 20. PMV/PPD thermal comfort diagnostic – occupant comfort performance across baseline and integrated passive envelope scenarios (Monday – Friday; 09:00 – 17:00)

Metric	Baseline	Scenario 1	Scenario 2	Scenario 3
Occupied hours (hour/yr)	2,340	2,340	2,340	2,340
Mean PMV	1.261	0.606	0.569	0.380
Median PMV	0.662	0.319	0.284	0.204
Comfortable hours (PMV ±0.5)	433	1,279	1,253	1,650
Comfortable (%)	18.50	54.66	53.55	70.51
Warm hours (PMV > +0.5)	1,907	1,061	1,087	690
Warm (%)	81.50	43.34	46.45	29.49
Mean PPD	39.91	17.76	16.49	10.60
PPD > 10% (hours)	1,958	1,064	1,091	694
PPD > 10% (%)	83.68	45.47	46.62	29.66
Mean OT – Occupied hours (˚C)	29.38	27.08	26.98	26.52
Mean RH – Occupied hours (%)	44.73	46.53	56.14	53.08
OT at PMV = +0.5 boundary (˚C)	27.05	26.68	26.75	26.78

***Note:** Cool hours (PMV < -0.5) = 0 across all scenarios, confirming the absence of overcooling risk under Jakarta's equatorial boundary condition. OT at PMV = +0.5 boundary represents the mean OT recorded during hours where PMV falls within the range 0.45 – 0.55, identifying the empirical upper comfort threshold.*

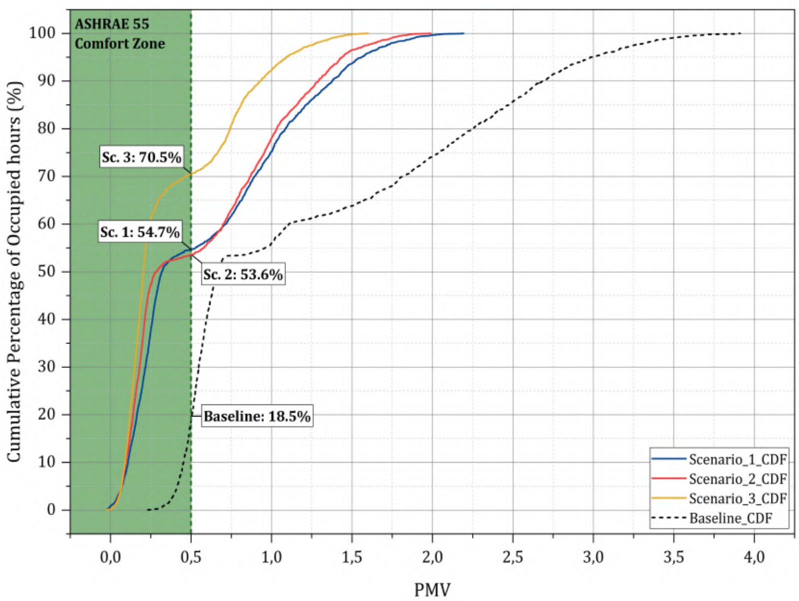


Figure 67. PMV thermal comfort diagnostic: Baseline vs. integrated passive scenarios

4.2.3. Hygric Energy Constraint: The Dehumidification Mandate

The hygric data reveals the binding energy constraint that passive geometry alone cannot overcome. As demonstrated in Figure 68, the suppression of OT towards the thermally stable range (25 – 26 ˚C) triggers a corresponding vertical shift in RH, approaching the 60 – 65% upper threshold stipulated by ASHRAE Standard 62.1 for the prevention of microbial proliferation. This hygric trajectory is not an accident; it is a direct thermodynamic consequence of the inverse relationship between sensible cooling and moisture retention capacity. As dry-bulb temperature falls under passive intervention, the saturation vapour pressure of the air decreases, leading to the concentration of residual moisture at progressively higher RH values. The latent gain data (Table 19) provides precise quantification of this boundary. In scenario 1, 3,553.99 W of latent load is introduced through active TST ventilation, representing the thermodynamic cost of maintaining buoyancy-driven convective flushing. In scenario 3, the envelope is sealed, reducing external latent intrusion to 2,871.40 W. however, this apparent improvement is counterbalanced by the accumulation of internally generated moisture from occupancy under stagnant air conditions, resulting in a mean RH of 56.47% - marginally exceeding the 55.86% mean RH achieved in scenario 1. The simulation thus corroborates a fundamental thermodynamic deadlock; it is not possible for any purely passive configuration to simultaneously minimise both sensible and latent energy loads in Jakarta's hot-humid climate. This is not a design failure; rather, it is a climate-imposed boundary condition that defines the limit of passive envelope performance and establishes the necessary scope for mechanical supplementation.

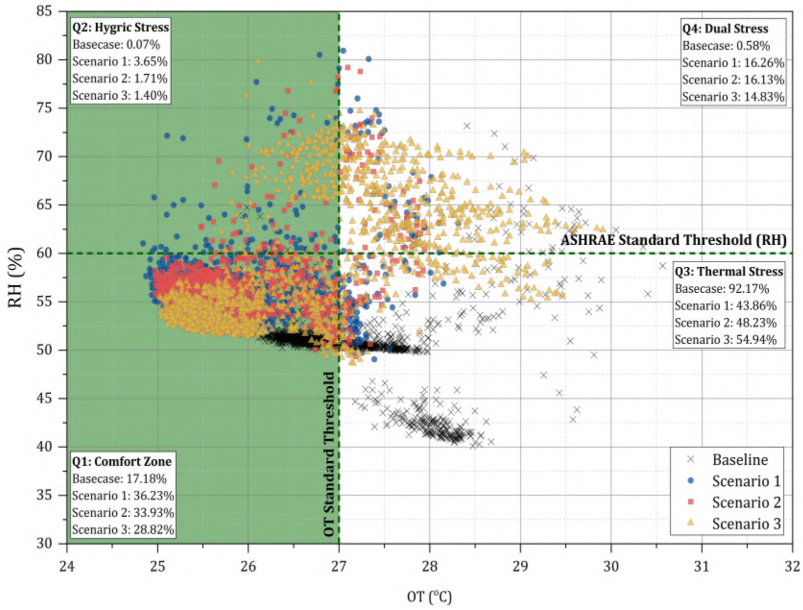


Figure 68. OT-RH quadrant distribution: sensible-latent performance boundary across baseline and integrated passive scenarios

To further substantiate this thermodynamic deadlock with temporal precision, a quadrant cluster analysis was conducted. This analysis mapped the full annual hourly distribution of OT and RH across all scenarios against the dual-axis thresholds of ASHRAE Standard 62.1 (RH ≤ 60%) and the upper bound of the warm comfort zone category as prescribed by SNI (OT ≤ 27.1˚C) which corresponds to the initial baseline OT of 27.06˚C established in Section 4.2.1. As illustrated in Figure 63, the analysis divides the psychrometric space into four quadrants: Q1 (comfort zone: OT ≤ 27.06˚C and RH ≤ 60%), Q2 (thermally adequate but hygric stress: OT ≤ 27.06˚C and RH > 60%), Q3 (thermal stress but hygric adequate: OT > 27.06˚C and RH ≤ 60%), and Q4 (dual stress: OT > 27.06˚C and RH > 60%). The baseline condition is found to be predominantly concentrated in Q3 (92.17% of annual hours; 8,074 hours), thereby confirming that the unmitigated building operates in persistent thermal overload whilst maintaining relatively low RH – a direct consequence of the absence of any ventilative moisture intrusion. Scenario 1 achieves the most substantial redistribution towards Q1 (36.23%; 3,174 hours), representing a fivefold improvement over the baseline, yet simultaneously generates the highest Q2 exposure (3.65%; 320 hours).

This is the quantitative fingerprint of the hygrothermal paradox, wherein effective sensible suppression inevitably elevates RH towards the microbial proliferation threshold. It is crucial to note that all active scenarios demonstrate significantly elevated Q4 occupancy (scenario 1: 16.26%; scenario 2: 16.13%; scenario 3: 14.83%) relative to the baseline (0.58%). This finding serves to corroborated the assertion that TST-driven ventilation, whilst thermally beneficial, simultaneously imports tropical moisture that passive geometry is unable to expel. This is a finding that directly and empirically quantifies the necessary scope for active decoupled mechanical supplementation that is prescribed by the Thermal-Hygric Adaptive Decoupled Envelope System (THADES) framework. Table 21 presents the complete annual quadrant distribution, quantifying the temporal displacement of each scenario across the four psychrometric zones and the corresponding deviation from the baseline condition.

Table 21. Annual OT-RH quadrant distribution: Psychrometric cluster analysis across passive envelope scenarios

Quadrant	Condition	Baseline	Scenario 1	Scenario 2	Scenario 3
Q1 – Comfort Zone	OT ≤ 27.06˚C; RH ≤ 60%	629 hours; 7.18%	3,174 hours; 36.23%	2,972 hours; 33.93%	2,525 hours; 28.82%
Q2 – Hygric Stress	OT ≤ 27.06˚C; RH > 60%	6 hours; 0.07%	320 hours; 3.65%	150 hours; 1.71%	123 hours; 1.40%
Q3 – Thermal Stress	OT > 27.06˚C; RH ≤ 60%	8,074 hours; 92.17%	3,842 hours; 43.86%	4,225 hours; 48.23%	4,813 hours; 54.94%
Q4 – Dual Stress	OT > 27.06˚C; RH > 60%	51 hours; 0.58%	1,424 hours; 16.26%	1,413 hours; 16.13%	1,299 hours; 14.83%
Total		8,760 hours; 100%	8,760 hours; 100%	8,760 hours; 100%	8,760 hours; 100%
ΔQ1 vs. Baseline	Comfort gain	-	+29.75 pp	+26.75 pp	+21.64 pp
ΔQ4 vs. Baseline	Dual stress penalty	-	+15.68 pp	+15.55 pp	+14.25 pp

The ΔQ1 and ΔQ4 rows reveals the fundamental trade-off that defines passive envelope performance in humid equatorial climate: every percentage point of comfort gain (ΔQ1) is accompanied by a disproportionate dual stress penalty (ΔQ4). Scenario 1 achieves the highest comfort distribution (+29.05 pp), yet simultaneously incurs the largest dual stress exposure (+15.68 pp) – a ratio of 1.85:1. This quantitative result confirms that passive sensible optimisation and latent load containment are mutually exclusive objectives within the constraints of purely passive geometry. This ratio constitutes the empirical basis for the THADES decoupled hybrid framework: passive elements are designated to maximise ΔQ1 gains, whilst active mechanical supplementation is mandate to suppress the ΔQ4 penalty that passive geometry structurally cannot resolve.

4.3. Aerodynamic Stagnation Diagnostic and Ventilation Energy Deficit

Aerodynamic diagnostics of the integrated passive system reveal a critical structural limitation that operates independently of thermal performance: the progressive decoupling of total internal air movement from the delivery of actual outdoor fresh air. This phenomenon, termed aerodynamic short-circuiting, constitutes the boundary condition that determines the minimum mechanical energy input required to sustain the system's operational viability. To isolate this mechanism, the airflow data from the AFN diagnostic framework is decomposed into two distinct flow components: the TST-delivered outdoor airflow (TST ACH), representing genuine outdoor air intake; and the DSF cavity-driven internal circulation (DSF ACH), representing buoyancy-driven air movement that is predominantly recirculated indoor air. As demonstrated in Figure 69, these two components exhibit a fundamental divergence across all scenarios. The TST's outdoor air delivery is severely constrained, registering 0.073 ach in scenario 1, 0.065 ach in scenario 2, and 0.020 ach in scenario 3. These figures represent critical deficits against the ASHRAE Standard 62.1 minimum outdoor airflow requirement of 0.68 ach for the 2,400 m^2 office floor plate. The corresponding ventilation deficits are 89.3%, 90.5%, and 97.0%, respectively, thereby demonstrating that no purely passive scenario fulfils the minimum fresh air mandate.

However, this severe deficit in fresh air intake does not necessarily imply a stagnant indoor environment. The building instead exhibits a deceptively active internal airflow regime – driven overwhelmingly by DSF cavity recirculation rather than genuine outdoor air induction – as consolidated in Figure 69. This discrepancy highlights a significant thermodynamic paradox: while air is moving vigorously within the building's core, it is primarily driven by the internal redistribution of existing air rather than the induction of new outdoor air. This verifies the hypothesis that the stack-induced suction from the DSF acts as a parasitic force, thereby impeding the TST's efficacy by prioritising internal upward flows over downward fresh air displacement.

Concurrently, the DSF cavity generates substantially higher internal air circulation rates – 0.392 ach in scenario 1, 0.225 ach in scenario 2, and 0.191 ach in scenario 3. The aerodynamic short-circuit ratio, defined as the proportion of total internal circulation attributable to recirculated rather than fresh outdoor air, quantifies the severity of this decoupling: 81.4% in scenario 1, 71.2% in scenario 2, and 89.4% in scenario 3 (see Figure 69). This metric reveals a counterintuitive hierarchy – scenario 3, with the most restrictive geometry, paradoxically exhibits the highest recirculation fraction despite its lowest total airflow. This phenomenon can be attributed to the sealed envelope configuration of scenario 3 (case B6 TST: ACH – 0.020 ach), which virtually eliminates outdoor air intake, while the DSF cavity continues to drive a residual internal recirculation of 0.191 ach, yielding an effective fresh air fraction of merely 10.6% of total air movement.

The physical mechanism underlying this short-circuiting is elucidated by the DSF cavity directionality data. Across all three scenarios, the daytime net flow within the DSF cavity is uniformly negative, registering – 1.5 m^3/s in scenario 1, -0.851 m^3/s in scenario 2, and -0.696 m^3/s in scenario 3. This confirms that the cavity operates in dominant exhaust mode during operational hours. Instead of functioning as a passive air supply conduit, the deep cavity generates an upward buoyancy-driven draft that siphons and recirculates indoor air through the bottom ventilation inlets. This competes directly with, rather than supplements, the TST's outdoor air intake. The TST's geometric optimisation for maximum aerodynamic capture – embodied in scenario 1's 9 m long-rectangular tower – is thus systematically undermined by the cavity's dominant exhaust pull, which stalls the TST's net fresh air delivery at 0.073 ach despite its design capacity of 0.084 ach whole-building capacity established in Phase 1. This 81.4% parasitic recirculation confirms an aerodynamic short-circuit; demonstrating that increasing TST height in deep-cavity system does not improve fresh air delivery, but rather causes the envelope to monopolize airflow at the severe expense of building-wide ventilation efficiency.

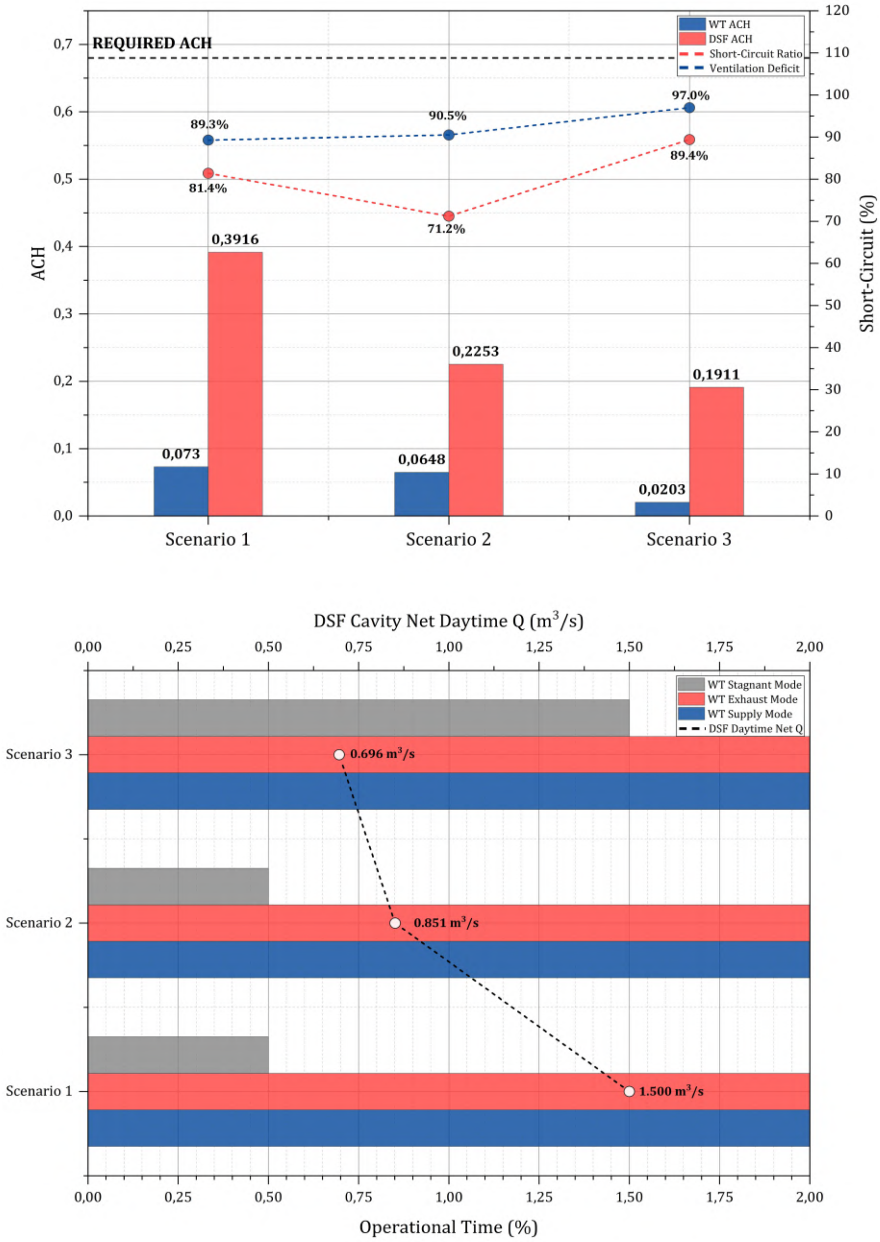


Figure 69. Aerodynamic short-circuiting diagnostic: Fresh air delivery vs. recirculated internal flow across integrated passive scenarios

The TST's operational mode distribution further corroborates this stagnation diagnosis. In scenario 1, the system operates in supply mode for 62.5% of the operational time, with a mean daytime volumetric flow of 0.277 m^3/s. However, the net outdoor air delivered to the occupied zone remains critically insufficient. As demonstrated in scenario 3, the stagnation rate is 1.5%, which is three times that of scenarios 1 and 2, indicating that the restrictive configuration is experiencing progressive aerodynamic paralysis. It is imperative to note that the nocturnal TST data reveals a further divergence: in scenario 1, the mean night-time flow was recorded at -0.028 m^3/s (thermal buoyancy-driven exhaust – active heat purging). In scenario 3, the approach to near-complete aerodynamic stagnation was -0.002 m^3/s, confirming that the sealed envelope forfeits even the residual nocturnal ventilation benefit.

The complete directionality and magnitude of this decoupling are consolidated in Figure 70 and Table 22. The diverging flow chart presents the signed daytime mean flow of the TST and DSF cavity simultaneously, making the parasitic dominance visually explicit: the DSF exhausts 5.41x more air than the TST supplies in scenario 1, 3.52x in scenario 2, and 9.64x in scenario 3. The scenario 3 condition is particularly diagnostic – its geometric restriction produces the highest DSF-to-TST ratio despite its lowest flows, confirming that envelope airtightness amplifies rather than mitigates the short-circuit mechanism. The consolidated AFN diagnostic in Table 22 synthesizes the complete aerodynamic and energy profile, adding three metrics not reported in the preceding analysis: the TST's supply mode stability across scenarios (62.5%, 62.4%, and 65.7%), which demonstrates that the TST's aerodynamic regime is governed by its own geometry rather than the integration scenario; the DSF's sustained exhaust dominance (70.1%, 63.6%, and 73.4%), and the total EUI trajectory (110.14, 109.96, and 107.75 $kWh/m^2/year$), which reveals that the apparent energy gain in more restrictive scenarios is achieved through airtightness that simultaneously forfeits outdoor air delivery. The aggregate implication of this aerodynamic short-circuiting is a necessary mechanical supplementation energy cost that cannot be resolved through further passive geometric optimisation. The 89.3% fresh air deficit (see Figure 69) in the optimal scenario 1 represents the aerodynamic boundary of the integrated passive system – the precise gap that defines the minimum active energy input required to achieve compliance with occupancy-driven ventilation requirements without compromising the 26.5% total energy reduction achieved by the passive envelope.

The three diagnostic assessments presented above collectively expose a structural incompatibility within the integrated passive system: whilst Sections 4.1 and 4.2 confirm that the passive envelope successfully manages the sensible and hygric dimensions of the thermal burden, Section 4.3 reveals that aerodynamic short-circuiting renders the system incapable of fulfilling the outdoor air delivery mandate that underpins both of those achievements. These concurrent failures are not merely independent anomalies; rather, they are manifestations of a single underlying thermodynamic incompatibility between sensible optimisation and ventilation sufficiency in high-enthalpy equatorial climates. Sections 4.4 and 4.5 advance beyond the diagnostic quantification of these deficits to define their physical boundaries: the precise aerodynamic and hygric thresholds at which passive intervention is thermodynamically exhausted and active mechanical decoupling becomes the only viable resolution pathway.

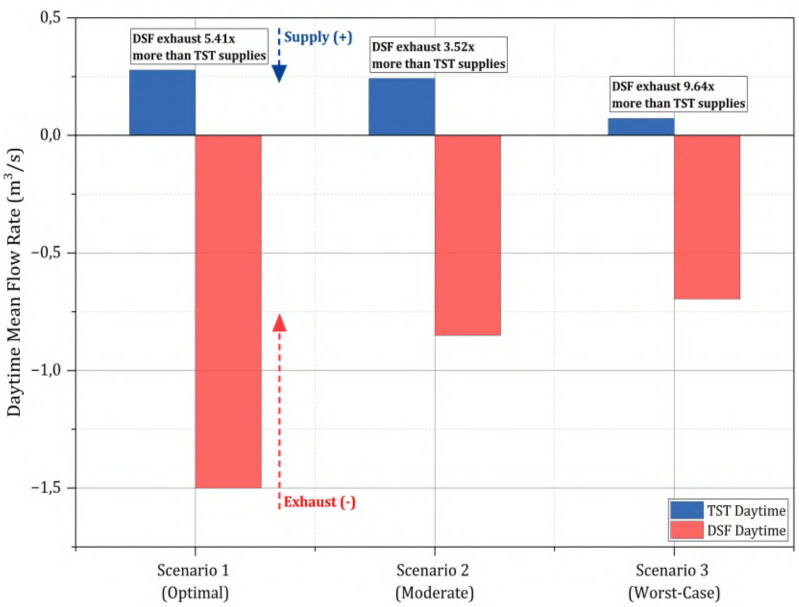


Figure 70. Signed daytime mean flow - TST tower supply vs. DSF cavity exhaust: aerodynamic short-circuit evidence across integration scenarios

Table 22. Consolidated AFN aerodynamic and energy diagnostic all phase 2 integration scenarios – key metric summary

Metric	Scenario 1 (Optimal)	Scenario 2 (Moderate)	Scenario 3 (Worst-case)	Unit
TSTMean ACH	0.0730	0.0648	0.0203	ach
DSF Wind Chimney Mean	0.3916	0.2253	0.1911	ach
ASR	81.4	71.2	89.4	%
TST Daytime Mean Flow	+0.2774	+0.2419	+0.0722	m^3/s
DSF Daytime Mean Flow	-1.5000	-0.8507	-0.6959	m^3/s
TST Daytime	5.41	3.52	9.64	x

Ratio				
TST Night-time Mean Flow	-0.0276	-0.0230	-0.0023	m^3/s
DSF Night-time Mean Flow	+0.0114	+0.0616	-0.0523	m^3/s
TST Supply Mode	62.5	62.4	65.7	%
TST Exhaust Mode	36.9	37.1	32.9	%
DSF Exhaust Mode	70.1	63.6	73.4	%
TST vs. ASHRAE Deficit	-89.3	-90.4	-97.1	%
Mean Indoor RH	55.86	55.75	56.47	%
Latent Gain	3,554	3,640	2,871	W
Sensible Cooling	102.14	101.49	95.25	$kWh/m^2/year$
Lighting EUI	7.76	8.47	12.50	$kWh/m^2/year$
Total EUI	110.14	109.96	107.75	$kWh/m^2/year$

4.4. Integrated Performance Summary – Phase 2 Consolidated Diagnostic

To synthesize the multi-dimensional performance profile established across Section 4.1 to 4.3, Table 23 consolidates the complete diagnostic output for all phase 2 configurations against the baseline. This unified view makes the thermodynamic trade-offs across scenarios explicitly legible – in particular the inverse relationship between sensible performance and aerodynamic fresh air sufficiency, which constitutes the central empirical argument of this chapter.

Table 23. Summary of phase 2 simulation process

Metric	Unit	Baseline	Scenario 1 (Optimal)	Scenario 2 (Moderate)	Scenario 3 (Worst-Case)
Energy Performance					
Sensible Cooling EUI	$kWh/m^2/year$	146.51	102.14	101.49	95.25
Lighting EUI	$kWh/m^2/year$	3.29	7.76	8.47	12.50
Total EUI	$kWh/m^2/year$	149.80	110.14	109.96	107.75
Thermal Comfort					
Mean OT (Occupied Hours)	˚C	27.06	25.82	25.73	26.19
Mean PMV	-	1.261	0.606	0.569	0.380
Comfortable	%	18.50	54.66	53.55	70.51

Hours (PMV ±0.5)					
Mean PPD	%	39.91	17.76	16.49	10.60
Hygric Performance					
Mean RH (Occupied Hours)	%	50.68	55.86	55.75	56.47
Latent Gain	W	3,740.18	3,553.99	3,639.98	2,871.40
Daylighting					
Daylight Autonomy (DA)	%	-	57.45	46.59	38.58
Spatial Daylight Autonomy (sDA)	%	-	50.54	38.43	34.85
Aerodynamic Performance					
Mean ACH (TST Tower)	ach	-	0.073	0.065	0.020
DSF Internal Recirculation	ach	-	0.392	0.225	0.191
ASR	%	-	81.4	71.2	89.4
TST vs. ASHRAE 62.1 Deficit	%	-	-89.3	-90.4	-97.1

The consolidated data reveals the central paradox of phase 2: while scenario 3 records the lowest EUI (107.75 kWh/m^2/year) and the lowest latent gain (2,871.40 W), it simultaneously delivers the most severe aerodynamic deficit (ASR: 89.4%) and the poorest daylighting performance (sDA: 38.58%) – confirming that its numerical superiority is achieved through habitability exclusion rather than genuine thermodynamic optimization. Scenario 1 remains the only configuration that satisfies the thermal and comfort thresholds (OT ≤ 27.06˚C, PPD trajectory toward ISO 7730), whilst approaching but not attaining the LEED v4 daylighting credit (sDA300/50%: 50.54% against the 55% threshold) while maintaining competitive energy performance.

4.5. Phase 2: The "Choked Flow" Phenomenon and Aerodynamic Boundaries

The empirical evidence synthesized in phase 2 elucidates a profound paradox within tropical building physics: architectural optimization that successfully mitigates sensible cooling loads invariably generates a secondary thermodynamic penalty – the escalation of latent dehumidification demand and the depletion of aerodynamic fresh air delivery capacity. Section 4.4 and 4.5 go beyond the conventional framework of parametric maximization by establishing the thermodynamic boundaries of passive integrated envelopes in equatorial climates. The argument advanced herein is that the identification of aerodynamic stagnation threshold – quantified through AFN component-level diagnostics – provides the empirical justification for shifting the architectural paradigm from purely passive reliance towards a hybrid decoupled operational framework.

The central premise of integrating a deep-cavity DSF with a TST was to harness localized buoyancy to drive continuous indoor flushing. However, the diagnostic evaluation revealed a critical failure in this mechanism, geometrically defined herein as the "choked flow" phenomenon. Although the optimal configuration (scenario 1) was successful in suppressing the OT to near-neutral levels, it did so by placing considerable reliance on the envelope's physical airtightness and the cavity's thermal exhaust. The diagnostic process conducted on the AFN analysis revealed that the thermodynamic draft generated within the 1.5 m DSF cavity functions as a predominant parasitic vacuum. Instead of drawing in uncontaminated outdoor air from the ambient TST intake (which stalled at a mere 0.073 ach), the cavity's stack effect aggressively siphoned and recirculated existing indoor air through the bottom ventilation inlets (see Figure 71). The directionality and magnitude of this parasitic dominance are made visually explicit in Figure 65, wherein the DSF exhausts 5.41x more air than the TST supplies in scenario 1 – a ratio that escalates to 9.64x in scenario 3, confirming that envelope airtightness amplifies rather than mitigates the short-circuit mechanism.

This aerodynamic short-circuiting delineates the physical boundary of natural ventilation in high-enthalpy microclimates. The requirement of 0.68 ach – which is derived from the ventilation rate procedure of ASHRAE Standard 62.1 for a standard 2400 m^2

office floorplate – represents a fixed physiological baseline. The passive envelope's inability to deliver more than 0.073 ach of actual fresh outdoor air translates to a critical 89.3% ventilation deficit. This deficit is not scenario-specific but constitutes a structural ceiling: between 0.020 and 0.073 ach – a range that represents 3.0% to 10.7% of the ASHRAE mandate respectively, as consolidated in Table 22 and Figure 70. Consequently, the premise that an integrated passive envelope can autonomously satisfy both sensible heat rejection and outdoor fresh air delivery requirements in a tropical context is thermodynamically invalid. The severity of this boundary is further quantified by the aerodynamic short-circuit ratio: 81.4% of total inherent air movement in scenario 1 constitutes recirculated indoor air rather than genuine outdoor exchange – confirming that geometric optimization of the TST cannot overcome the parasitic vacuum generated by the DSF cavity's dominant exhaust pull. This constraint is consistent with the finding of Li et al. (2025), whose review of climate-adaptive windcatcher systems established that buoyancy-driven mechanisms in high-enthalpy tropical environments are inherently bounded by competing pressure gradients that geometric scaling alone cannot resolve – a finding that directly corroborates the TST's empirically confirmed aerodynamic ceiling in this study. The aerodynamic threshold dictates that once the exterior skin has been sufficiently optimized to block external solar enthalpy, it inherently creates a frictional drag that depletes the dynamic pressure required to overcome internal hydrostatic resistance, rendering active mechanical supplementation the only viable pathway to bridge the resulting fresh air deficit.

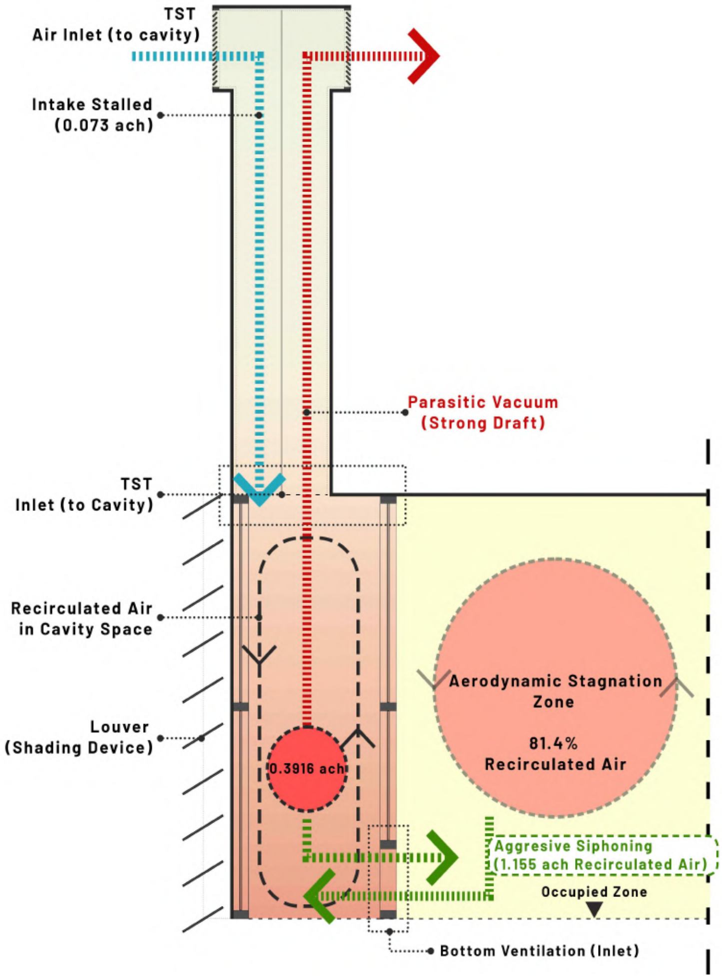


Figure 71. Aerodynamic short-circuiting: DSF cavity parasitic vacuum undermining TST outdoor air delivery

4.6. The latent Paradox: Decoupling Sensible from Hygric Constraints

The establishment of the choked flow boundary directly precipitates the second thermodynamic limit: the latent heat and humidity paradox. The continuous entrapment of air, while beneficial for stabilizing the MRT, allowed the persistent accumulation of metabolic moisture, rendering the system fragile against hygric loads. As demonstrated by the performance metrics, the optimized passive geometries exhibited a significant degree of robustness, exclusively within the domain of sensible cooling. These geometries effectively reduced the cooling intensity to 102.14 kWh/m^2/year. However, as illustrated in Figure 72, the indoor environment sustained persistent hygric failure, characterized by RH levels consistently around 55.86% and a residual latent dehumidification demand of 3,553.99 W that escalates the mechanical system's total operational energy beyond the sensible cooling fraction alone. This finding confirms the hypothesis that purely architectural interventions – such as cavity depth expansion or louver angular deflection – possess no physical capacity to execute active dehumidification. Consequently, resolving the total energy burden in such high-enthalpy environments strictly requires decoupling the latent load through the implementation of independent active dehumidification systems. This necessity is corroborated by the finding of Ding et al. (2022).

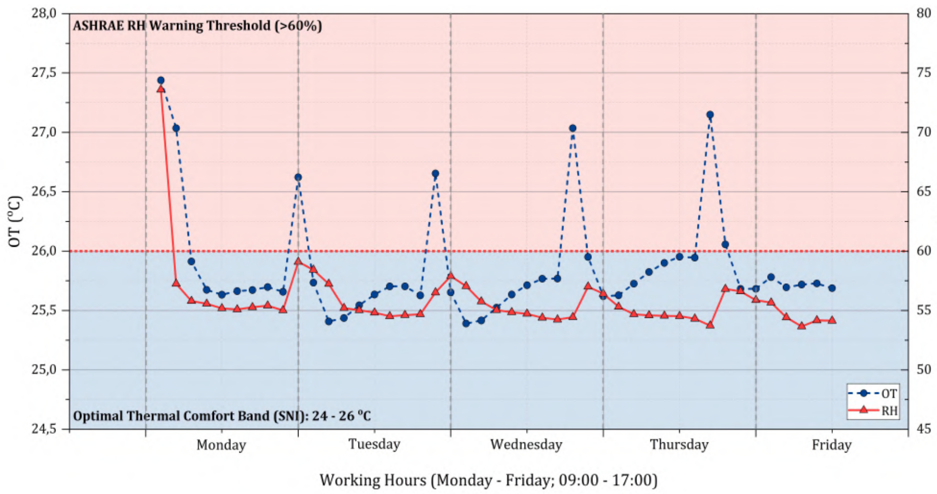


Figure 72. Temporal profiling of scenario 1: sensible robustness vs. hygric failure

Therefore, the thermodynamic limit defined by this research posits that sensible and latent cooling demands in high-enthalpy climates cannot be simultaneously resolved through singular passive mechanics (as illustrated in Figure 73). This necessitates a fundamental "decoupling imperative." To address the 89.3% deficit in fresh air and to manage latent accumulation without neutralising the sensible energy savings achieved in scenario 1, it is essential that the architectural envelope is dedicated exclusively to solar shielding and sensible buffering. Concurrently, the introduction of fresh outdoor air and moisture removal must be decoupled and aggressively managed via targeted mechanical supplementation.

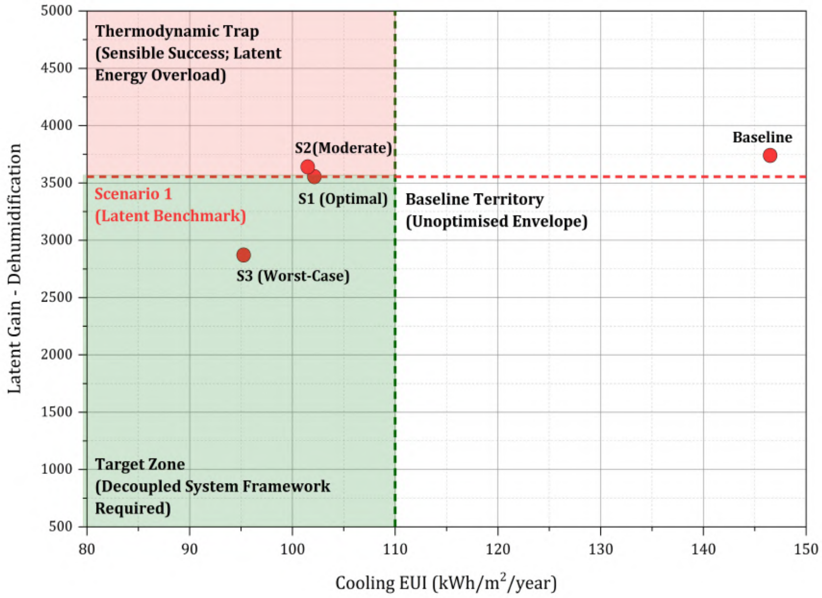


Figure 73. Thermodynamic limits: the sensible vs. latent paradox

5. Conclusion

The fundamental purpose of this research was to undertake a critical investigation into the thermodynamic performance boundaries of an integrated passive envelope system. This system comprised a deep-cavity DSF, an aerodynamic TST, and an optimized louver system. The study focused on mid-rise office buildings in the high-enthalpy tropical climate of Jakarta. The present study employed a rigorous two-phase parametric optimization coupled with AFN diagnostic to quantify the extent to which passive architectural interventions can mitigate cooling loads before reaching their hygric and aerodynamic limits.

5.1. Empirical Conclusions

The findings of this study contradict the prevailing assumption that purely passive architectural interventions can independently resolve the complete thermodynamic energy burden of mid-rise office buildings in equatorial climates – encompassing both sensible heat rejection and the aerodynamic fresh air delivery mandate imposed by occupancy-driven ventilation requirements. The synthesis of the simulations yielded the following key outcomes:

5.1.1. The Efficacy of Sensible Load Mitigation

The initial phase of this research endeavour confirmed that geometrical optimization of the passive envelope yielded substantial thermal shielding capacity. MCDA was employed to determine the optimal configuration (scenario 1), which was found to be a 1.5 m deep-cavity, a 9 m long-rectangular TST, and a highly permeable louver array (270 mm gap, 340 mm slats depth, 30° tilt, and 30 mm distance from external glazing material). This configuration successfully intercepted external solar enthalpy and suppressed the MRT, functioning as a robust sensible thermal buffer. Consequently, the system suppressed the OT to 25.82°C – confirming effective MRT interception and radiant heat decoupling from the occupied zone – whilst facilitating a substantial 30.3% reduction in sensible cooling energy demand, driving the cooling EUI from the baseline building of 146.51 kWh/m^2/year to an optimized 102.14 kWh/m^2/year; however, the concurrent visual penalty – arising from the shading geometry's suppression of daylight transmission – elevated the total EUI to 110.14 kWh/m^2/year, representing a net 26.47% reduction from the baseline total EUI of 149.80 kWh/m^2/year and confirming that sensible cooling optimisation in hot-humid climates is thermodynamically inseparable from the visual-thermal trade-offs. Consequently, scenario 1 is established not as a flawless solution, but as a delicate equilibrium – the most precise balance between passive free-cooling gains and unavoidable latent penalties in Jakarta's climate.

5.1.2. The Aerodynamic Paradox and Thermodynamic Limit

Notwithstanding the marked success achieved in the field of sensible energy conservation, the AFN diagnostic framework in phase 2 revealed a severe thermodynamic dichotomy. The operational requirement to seal for solar and thermal buffering had the inherent effect of paralysing the building's aerodynamic capacity to dilute metabolic pollutants. The system encountered a physical boundary characterized by the "choked flow" phenomenon and "aerodynamic short-circuiting." The data provided unequivocal evidence that the stack effect generated within the 1.5 m DSF cavity functioned as a parasitic vacuum. Instead of drawing in fresh outdoor air, the cavity's stack effect aggressively siphoned and recirculated the localized cavity-room air mixture through the bottom ventilation inlets. Concurrently, the actual fresh outdoor air captured by the TST stalled at a critical 0.073 ach – yielding an 89.3% deficit against the ASHRAE Standard 62.1 minimum requirement of 0.68 ach – whilst the DSF cavity continued to drive 0.392 ach of recirculated internal air, confirming that 81.4% of total air movement constitutes parasitic recirculation rather than genuine outdoor air exchange. As a direct consequence of this aerodynamic short-circuiting, the integrated envelope sustained a persistent latent dehumidification demand of 3,553.99 W (mean RH: 55.86%), confirming that passive geometry possesses no physical mechanism for active moisture removal. Therefore, this research establishes a thermodynamic limit: in hot-humid equatorial climates, the optimization of passive envelopes for sensible energy efficiency inversely correlates with aerodynamic fresh air delivery capacity and latent dehumidification demand – confirming that passive thermal optimization and active ventilation sufficiency are thermodynamically irreconcilable within a purely passive framework.

5.2. Design Guidelines: The "Hybrid Decoupled" Framework

To overcome the identified thermodynamic limitations, future architectural paradigms for tropical mid-rise offices must decisively move away from the pursuit of purely passive autonomy, which is often idealized. This study proposes the necessary adoption of a hybrid decoupled design framework (see Figure 74). Within this framework, the architectural envelope (comprising the DSF and louver system) must be strictly designated to function solely as a solar interceptor and sensible heat buffer. The expectation that the façade can serve as the primary driver for deep-plan ventilation must be abandoned. Concurrently, the critical tasks of deep-plan atmospheric flushing and latent load management (dehumidification) must be physically decoupled and delegated to active mechanical supplementation. The integration of a dedicated outdoor air system or the installation of low-energy mechanical exhaust fans within the TST shaft is imperative to artificially disrupt the buoyancy-driven stagnation, thereby ensuring the continuous extraction of metabolic moisture without neutralising the sensible cooling savings achieved by the passive skin (see Figure 75). The integrated system – comprising the deep-cavity DSF, TST, and the optimised louver array – is collectively designated the Thermal-Hygric Adaptive Decoupled Envelope System (THADES). This designation reflects the system's four empirically established operative characteristics: thermal sensible buffering by the DSF, hygric latent load as the unresolved passive ceiling, adaptive dual-mode operation of the TST, and the decoupled hybrid framework as the necessary design prescription. THADES is not synonymous with the scenario 1 passive optimum; rather, it constitutes the proposed design paradigm that prescribes active mechanical supplementation – specifically DOAS or low-energy exhaust integration within the TST shaft – to resolve the hygric and aerodynamic deficits that passive geometry alone cannot address.

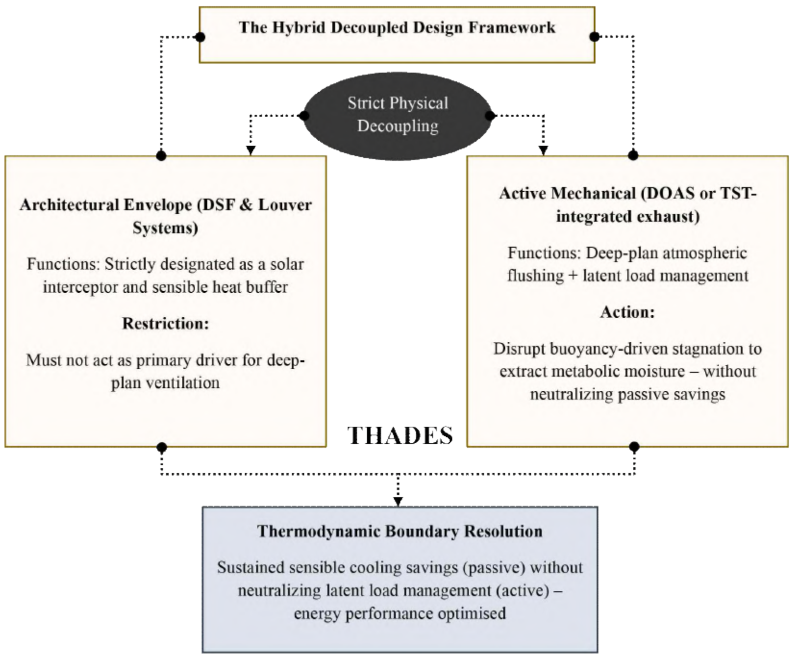


Figure 74. The conceptual architecture of the hybrid decoupled framework

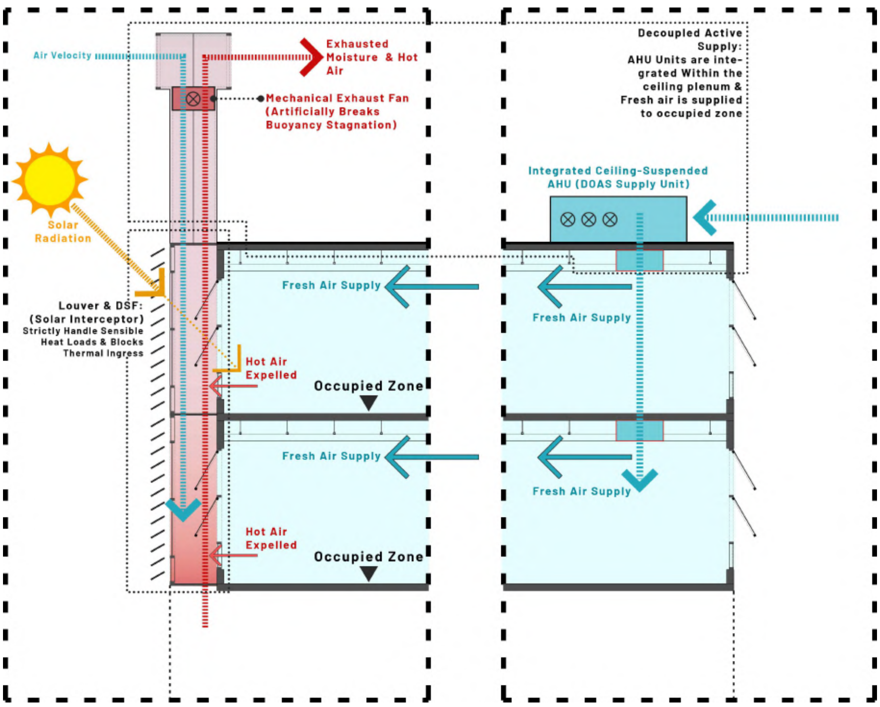


Figure 75. Schematic mechanism of the hybrid decoupled framework

5.3. Policy Implication and Recommendation for Regional Standards

The prevailing regional regulatory paradigms – notably the Indonesian National Standard (SNI 03-6572-2001) and the GBCI GREENSHIP assessment framework (2013) – exhibit a fundamental aerodynamic blind spot. The prevailing guidelines emphasise the promotion of natural ventilation through the prescription of openable WWR ratios (typically ranging from 5% to 10%), while concurrently placing a disproportionate emphasis on the reduction of gross cooling energy. However, there is an absence of enforcement mechanisms to ensure the verification of the actual delivery of outdoor air. The static geometric metrics in question are predicated on the erroneous assumption that envelope permeability is equivalent to aerodynamic flow effectiveness, a premise that this study has empirically disproven. The aerodynamic short-circuiting evidence demonstrates that a geometrically permeable envelope can simultaneously achieve high internal air circulation (0.392 ach) whilst delivering only 0.073 ach of genuine outdoor fresh air — an 89.3% shortfall that prescriptive window-to-wall ratio targets are structurally incapable of detecting.

Consequently, it is imperative that subsequent iterations of SNI and GBCI guidelines stipulate component-level AFN diagnostic modelling for any commercial building asserting natural ventilation credits. This would serve to verify not only the existence of openings, but also to ascertain that the delivered outdoor air exchange rate complies with the occupancy-driven ventilation mandate. Concurrently, a regulatory baseline for active dehumidification systems must be established for all tropical passive buildings, acknowledging that latent load management constitutes an independent energy obligation that passive geometry cannot discharge. A fundamental shift in the approach to geometry, moving from prescriptive guidelines to performance-verified aerodynamic delivery, is essential to prevent the aggressive pursuit of sensible energy reduction from generating a latent energy penalty that escapes regulatory scrutiny.

5.4. Limitations and Future Directions

Whilst the present research provides a comprehensive thermodynamic boundary definition, it is subject to certain limitations that warrant further investigation. It is further acknowledged that the thermodynamic boundaries established herein are scoped to the specific parametric domain investigated: the Jakarta-Kemayoran climatic context and the geometric ranges defined in Table 1. Quantitative thresholds reported – including the 89.3% ventilation deficit and 55.86% mean indoor RH – are constants without independent recalibration. Notwithstanding this parametric specificity, the thermodynamic boundaries identified herein are transferable to office typologies satisfying condition: a climate characterised by concurrent sensible and latent cooling demand for more than 85% of annual occupied hours, as established by the psychrometric analysis of the Jakarta-Kemayoran EPW dataset in Section 3.2. The sensitivity of the choked flow threshold and the 1.85:1 constraint ratio to variations in building scale, plan geometry, and façade-to-volume ratio beyond the present archetype is identified as a primary direction for future parametric extension.

The present simulation framework evaluated the building performance under a steady-state operational schedule that is characteristic of standard office environment. It is recommended that subsequent studies encompass transient occupancy profiles and dynamic spatial utilisation patterns, with the objective of observing real-time fluctuations in the choked flow phenomenon. Furthermore, the integration of a comprehensive life-cycle cost analysis would prove to be of significant benefit in evaluating the economic viability and return on investment of transitioning from a conventional HVAC setup to the proposed hybrid decoupled framework. The evidence shows that the application of this coupled thermal AFN diagnostic methodology to other high-density building typologies, including vertical residential towers and educational facilities, would serve to further validate the universality of these aerodynamic constraints across the tropical built environment. Ultimately, the establishment of the Decoupled Hybrid Framework constitutes the primary original contribution of this study, asserting that pure passive strategies have reached their thermodynamic limits in tropical offices and necessitate a targeted separation of sensible and latent load management.

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열대 기후 중층 오피스 빌딩을 위한 통합 패시브 외피의 에너지 성능 최적화

디마스 마울라나 파샤

부산대학교 대학원 건축학과

요약

열대 지역, 특히 자카르타의 급격한 도시화는 능동적 냉방 의존에서 패시브 건축적 방어 메커니즘으로의 근본적인 패러다임 전환을 요구한다. 패시브 설계는 지속 가능한 해결책의 정점으로 자주 거론되지만, 극한의 주변 습도 환경에서 순수 패시브 전략의 열역학적 실현 가능성은 여전히 비판적 논쟁의 대상으로 남아 있다. 본 연구는 중층 사무소 건물에 적용된 통합 패시브 외피 시스템의 에너지 성능 한계를 규명한다. 제안된 시스템은 태양열을 차단하는 심층 공기층 이중 외피(DSF), {본 연구에서 부력 구동 배기 메커니즘의 실증적 확인에 근거하여 열 스택 타워(TST)로 재명명된 공력학적 윈드캐처}, 그리고 복사 열 획득을 조절하는 최적화된 외부 루버의 세 가지 기술을 통합한다. 본 연구의 주요 목적은 단순히 냉방 부하 경감을 정량화하는 것을 넘어, 패시브 현열 냉방 최적화가 의도치 않게 잠열(제습) 에너지 패널티 및 시각-열적 트레이드오프를 유발하는 엄격한 열역학적 한계를 규명하는 것이다.

방법론적으로, 본 연구는 기류 네트워크(AFN) 진단 모델과 결합된 엄격한 2단계 매개변수 최적화 프레임워크를 적용하였다. 이 접근법은 부력 구동 환기와 동적 수분(잠열) 침입 간의 복잡한 상호관계를 다룸으로써 기존의 열적 시뮬레이션을 넘어서는 방법론적 진보를 나타낸다. 본 연구는 공기층 깊이 및 루버 각도와 같은 기하학적 변수가 현열 및 잠열 열전달 메커니즘에 미치는 영향을 분리하기 위해 최적 구성에서 최악 구성에 이르는 다양한 설계 시나리오를 평가하였다. 나아가, 새로운 3경로 열 획득 분해 분석을 통해, 루버 시스템(C2)이 태양 복사 차단 측면에서 소폭 우위를 보임에도 불구하고(18.07 대 19.23 kWh/m²/년), DSF(A10)가 전도 열 유속 경로에서 결정적 우위를 발휘함(21.25 대 32.75 kWh/m²/년)을 규명하였다. 이는 습공기선도적 태양 차양 요건과 DSF의 총괄적 냉방 우위 간의 반직관적 역설을 해소하는 핵심 근거이다. 이러한 열적 우위는, 그러나 습열적 역설에 의해 제약된다. 시뮬레이션은 현열 냉방을 극대화하기 위해 탑 높이가 9 m(사례 B5)로 증가함에 따라, 결과적인 습공기선도적 변

화가 유입 상대습도를 포화 수준에 가까운 73.73%까지 상승시킴으로써 상당한 잠열 에너지 부담을 필요로 함을 보여준다 결과적인 습공기선도적 변화가 유입 상대습도를 포화 수준에 가까운 73.73%까지 상승시킴으로써 상당한 잠열 에너지 부담을 필요로 함을 보여준다.

도출된 결과는 명확한 열역학적 한계를 규정한다. 1.5 m 공기층 깊이와 고성능 유리 열 특성으로 구성된 최적 구성(시나리오 1)은 태양 복사를 효과적으로 차단하였다. 이로 인해 연간 현열 냉방 에너지 수요가 30.3% 대폭 감소하였으며, 냉방 에너지 사용 집약도(EUI)는 기준 모델의 146.51 kWh/m²/년에서 102.14 kWh/m²/년으로 낮아졌다. 인공 조명 패널티를 반영한 총 EUI는 110.14 kWh/m²/년으로, 기준 총 EUI(149.80 kWh/m²/년) 대비 26.47%의 절대적 감소에 해당한다. 반면, 최악 구성(시나리오 3)은 역설적으로 95.25 kWh/m²/년의 소폭 낮은 냉방 EUI를 달성하였는데, 이는 과도한 차양이 건물의 인공 조명 에너지 수요를 심각하게 악화시키는 방식으로 외견상의 냉방 절약이 이루어지는 심각한 시각-열적 트레이드오프를 부각시킨다. 더불어, 시나리오 1의 최적 냉방 효율은 임계적인 잠열 에너지 역설을 드러내었다. 자연 환기를 촉진하기 위한 다공성 기하 구조가 대량의 열대 수분을 재실 공간으로 흡입하여 기계 시스템이 제습을 위한 추가적인 에너지를 소비하도록 강제하였다.

나아가, 가열된 외피에 의해 발생한 강력한 굴뚝 효과(−1.50 m³/s의 막대한 배기율로 작동)가 의도된 풍력 구동 환기 효율을 저하시키는 심각한 "공력학적 단락" 및 "열역학적 불균형" 현상이 발생하였다. 시나리오 1의 DSF 공동은 0.392 ACH의 기만적인 내부 순환율을 생성하였으나, TST에 의해 실제로 포집된 신선 외기는 0.073 ACH에 불과하였으며, 이는 기준 건물 모델 (0.0439 ACH) 대비 단 0.029 ACH의 미미한 공력학적 개선만을 나타낸다. 이러한 공력학적 단락은 ASHRAE Standard 62.1 최소 요구량 대비 89.3%의 신선 외기 공급 부족으로 정량화되며, 전체 내부 기류의 81.4%가 실질적인 외기 교환이 아닌 기생적 재순환으로 구성됨을 의미한다. 이러한 공력학적 단락은 심층 공기층 통합에서 외피가 건물의 근본적인 환기 기능을 심각하게 희생하면서 자체적인 열 조절을 우선시함을 입증한다. 본 연구의 결과는 고온 다습한 적도 기후에서 최적화가 인공 조명 및 기계식 제습으로부터의 잠재적 에너지 패널티를 역으로 악화시키는 "경성(hard)" 열역학적 한계에 도달한다는 실증적 근거를 제공한다. 본 연구는 집계적 현열 냉방 절약에서 총 에너지 평형의 세밀한 진단으로 담론을 전환함으로써 열대 건물 과학 분야의 중요한 연구 공백을 해소한다. 이는 순수 패시브 설계가 열대 기후에 대한 보편적 해결책을 구성한다는 일반적으로 수용되는 개념의 재고를 촉구한다. 따라서 본 연구의 결과는 패시브 외피가 현열 부하를 제어하고 표적화된 기계 시스템이 잠열 부하 및 주간 환기를 전담하는 '디커플링된 하이브리드(decoupled hybrid)' 프레임워크가 열대 기후에서 현열 배출과 총 에너지 효율 간의 충돌을 해결하기 위한 유일한 실행 가능한 경로임을 입증한다.